TAMIL NADU AGRICULTURAL UNIVERSITY

B.Sc. (Hons) Agriculture - Third Semester

AGR 202

Principles and Practices of Natural Farming

Complete Lecture-Wise Theory Study Material

As per the 2025 Syllabus - All 17 Lectures

About this study material

- One numbered chapter per syllabus lecture, in the exact order of the course outline.
- Every sub-topic is expanded with headed sections, plain-English definitions and bullet points.
- Comparison tables, worked numerical examples and text diagrams are placed inside the relevant lecture.
- Each lecture closes with an Exam focus section: key points, short-answer questions with model answers, and long-answer questions with answer outlines.
- Examples are drawn from Indian and Tamil Nadu conditions wherever possible.

How to use this material

- Read the lecture notes first, then work through the tables and diagrams - they carry the comparison points that appear in examinations.
- Attempt the short-answer questions without looking at the model answers, then check.
- For long-answer questions, write the outline from memory before expanding it into full paragraphs.
- Use the glossary at the end to revise definitions quickly before the examination.

Mounish Raj · MouRa · Agri AI

Syllabus Map - Lecture Order

The course is organised into seventeen lectures. Lectures 1 to 8 and 10 to 17 carry the theory of natural farming; Lecture 9 is the mid-semester examination. The table below is the master index of this material.

Lecture	Topic
1	Indian Heritage of Ancient Agriculture
2	History and Evolution of Natural Farming
3	Concept, Definition and Objectives of Natural Farming
4	Essential Characteristics and Principles of Natural Farming
5	Main Pillars of Natural Farming
6	Scope and Importance of Natural Farming
7	Methods and Schools of Natural Farming
8	Characteristics and Design of a Natural Farm
9	Mid Semester Examination
10	Ecological Balance, Ecological Engineering and Community Responsibility
11	Integration of Crops, Trees and Animals in Natural Farming
12	Biodiversity, Indigenous Seeds and Farm Waste Recycling
13	Water Conservation, Renewable Energy and Nutrient Management
14	Insect, Pest, Disease and Weed Management in Natural Farming
15	Mechanization and Farm Operations in Natural Farming
16	Processing, Labelling and Economic Viability of Natural Farming
17	Certification, Standards, Export Potential, Government Initiatives and Entrepreneurship

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Lecture 1. Indian Heritage of Ancient Agriculture

Learning objectives

- Trace the origins of farming in the Indian subcontinent and the civilisations that shaped it.
- Identify the indigenous wisdom behind traditional soil, water and crop management.
- Explain how ancient practices link to modern sustainable and natural farming.
- Recognise the sources - Vedas, Krishi-Parashara, Kautilya's Arthashastra and regional traditions.

1.1 Introduction

India is one of the oldest agricultural civilisations in the world. Farming began here about ten thousand years ago in the fertile plains of the Indus and the Ganga, and the knowledge that grew around it was not merely a means of survival - it was a complete system of living with nature. The farmer of ancient India observed soils, seasons, rainfall, insects, cattle and crops as one connected whole. Natural farming, which is studied in this course, is therefore not a new invention. It is a return to a way of farming that Indian communities practised for centuries before chemical inputs arrived.

The phrase used in the syllabus is 'Indian heritage of ancient agriculture'. Heritage means the knowledge, skills and practices handed down from earlier generations. The task of this lecture is to see what that heritage contained, and why the modern natural farming movement treats it as its root.

1.2 Early farming civilisations of India

Indian agriculture evolved in several distinct regions, each with its own crops and methods. Together they form the base of the country's farming tradition.

- **Indus Valley Civilisation (about 3300-1300 BCE):** one of the earliest farming cultures, with evidence of wheat, barley, cotton, peas, sesame and dates. Well-planned granaries, brick-lined storage and canal irrigation show advanced crop management.
- **Vedic period (about 1500-500 BCE):** agriculture became central to settled life. The Rigveda mentions ploughing, irrigation channels (kulya) and the use of cattle manure. Terms such as 'krishi' (agriculture), 'karshaka' (farmer) and 'sita' (furrow) come from this period.
- **Mauryan period (about 320-185 BCE):** Kautilya's Arthashastra records a state department of agriculture, rules on land classification, irrigation, cattle care and crop rotation, and even the protection of forests and wildlife.
- **Sangam and later South Indian period:** Tamil literature describes tank irrigation (eri), land classification into five 'thinai', cattle rearing and rain-fed dryland cropping, all of which shaped Tamil Nadu farming.
- **Medieval period:** Persian and Mughal influence introduced new crops and gardens, but the village-level system of manure, mixed cropping and water harvesting continued unchanged.

1.3 Indigenous wisdom in soil management

Ancient Indian farmers understood that the soil is a living body that must be fed with organic matter and protected from harm. Several practices reflect this idea.

1.3.1 Feeding the soil, not the crop

The traditional principle was to keep the soil fertile and let the crop draw its nutrition from it. Farmers applied cattle dung, urine, compost, green leaf manure and ash. The Vedic 'gomaya' (cow dung) and 'gomutra' (cow urine) were regarded as purifiers of the soil - the same materials that reappear today in Jivamrit and Bijamrit. The modern scientific explanation is that these materials add organic carbon, humus and beneficial microorganisms.

1.3.2 Crop rotation and mixed cropping

Cereals were rotated with pulses so that the nitrogen removed by one crop was replaced by the next. Mixtures of cereals, pulses and oilseeds were grown together - a practice still seen in Tamil Nadu as 'varagu-thuvarai' and similar mixed stands. Rotation broke pest cycles and maintained soil fertility without external inputs.

1.3.3 Conservation of soil and moisture

Contour bunding, field bunds, dead furrows, mulching with crop residues and the protection of trees on field boundaries all reduced erosion and conserved moisture. The ancient practice of covering the soil surface with straw or leaves is exactly the 'mulching' pillar of natural farming.

1.4 Indigenous wisdom in water management

Water harvesting was a highly developed skill in ancient India, especially in the semi-arid south where rainfall is seasonal.

Traditional water system	Region	Principle and modern relevance
Eri (tank) system	Tamil Nadu	Cascading village tanks stored monsoon runoff and recharged wells; the classic example of watershed-based water management.
Kudimaramathu	Tamil Nadu	Community labour to desilt and repair tanks and channels before the monsoon; a model of participatory water governance.
Kuhl / kulya	Himalayan and northern plains	Gravity channels carried stream water to terraced fields; efficient use of slope and gravity.
Stepwells (baoli / vav)	Gujarat and Rajasthan	Deep groundwater access with minimal evaporation loss.
Phad system	Rajasthan	Community-managed diversion of river water over a chain of fields.
Zabo and bamboo drip	Nagaland and Meghalaya	Hill runoff harvesting and gravity drip irrigation from streams.

1.5 Indigenous wisdom in crop and pest management

Traditional farmers managed pests and diseases through diversity and natural materials rather than poisons.

- **Seed treatment:** seeds were soaked in cow urine, dung slurry, buttermilk or beeja-amrit-like preparations before sowing to improve germination and vigour.
- **Botanical pest control:** neem (Azadirachta indica), nochi (Vitex negundo), erukku (Calotropis), adathoda (Adhatoda vasica), garlic, chilli and tobacco were used as sprays and dusts.
- **Cultural control:** border rows of trap crops, mixed cropping, timely sowing, hand-picking of pests and burning of infested material.
- **Sacred groves and refuges:** patches of vegetation were left undisturbed to shelter natural enemies such as birds, spiders and predatory insects.

- **Astronomical and seasonal calendars:** sowing and harvesting were timed by the position of stars (nakshatra) and the monsoon, a practice encoded in the Tamil 'panchangam' and in the proverbs of 'pattam' and 'paruvam'.

1.6 Knowledge sources of ancient Indian agriculture

The heritage survives in a body of texts and in living tradition. The important written sources are listed below.

Source	Period / author	Contribution
Rigveda and later Vedas	c. 1500-500 BCE	Earliest references to ploughing, irrigation and cattle-based fertility.
Krishi-Parashara	c. 4th-6th century CE (attributed to Parashara)	First systematic Indian text on agriculture: soil types, seed selection, sowing, manuring, rainfall prediction.
Kautilya's Arthashastra	c. 4th century BCE	State policy on land, irrigation, cattle, crop protection and forest conservation.
Vrikshayurveda	Surapala, c. 10th century CE	Treatise on trees, orchards, plant health and plant nutrition.
Agni Purana and Brihat Samhita	c. 6th-9th century CE	Rainfall forecasting, agronomy and plant protection.
Sangam Tamil literature	c. 300 BCE-300 CE	Descriptions of dryland farming, tank irrigation and cattle culture in the Tamil country.
Village tradition and proverbs	Continuous	Living knowledge of local seeds, soil types, manures and pest control.

1.7 From heritage to natural farming

The table below links each ancient practice to the modern natural farming principle it has become.

Indigenous practice	Modern scientific basis / natural farming link
Cow dung and cow urine application (gomaya, gomutra)	Source of organic carbon, nitrogen, auxins and a rich microbial load; basis of Jivamrit and Bijamrit.
Covering soil with crop residues and leaves	Reduces evaporation, moderates soil temperature and suppresses weeds; now called mulching.
Cereal-pulse rotation and mixed cropping	Biological nitrogen fixation, break in pest and disease cycles, better use of soil layers.
Green leaf manuring (e.g. Glyricidia, neem, Pongamia)	Adds nitrogen and lignin, builds humus, feeds soil organisms.
Neem and botanical sprays	Azadirachtin and other compounds act as repellents, antifeedants and growth regulators.
Tank and community water harvesting	Watershed recharge, groundwater improvement, drought buffer.
Seed selection from the farmer's own harvest	Local adaptation, genetic diversity, seed sovereignty; basis of indigenous seed conservation.
Sacred groves and field bunds with trees	Habitat for pollinators and natural enemies; ecological engineering.

WORKED EXAMPLE - Heritage practice, modern reading

- A Tamil Nadu farmer soaks paddy seed in cow urine solution before sowing. Traditionally this is done to 'protect the seed'.

- Modern reading: cow urine is mildly alkaline and carries nitrogen, potassium and microbes. The soak improves germination percentage, hardens the seed and reduces seed-borne fungal infection - an early form of biological seed treatment.

- Link to the course: the same idea is formalised today in the Bijamrit seed treatment of natural farming.

EXAM FOCUS

Key points to remember

- Indian agriculture is about ten thousand years old; the Indus Valley, Vedic and Mauryan periods each contributed practices still in use.
- The heritage rested on feeding the soil with cattle dung, urine, compost and green manure rather than feeding the crop with chemicals.
- Crop rotation, mixed cropping and mulching maintained fertility and broke pest cycles.
- Tank irrigation, the eri system and Kudimaramathu are Tamil Nadu's contribution to community water management.
- Botanical pest control used neem, nochi, erukku and adathoda; cultural control used diversity and timing.
- Krishi-Parashara, the Arthashastra and Vrikshayurveda are the key written sources.
- Every pillar of modern natural farming can be traced to an indigenous practice.

Short-answer questions with model answers

Q1. Name any four traditional water-harvesting systems of India.

Model answer: Eri (tank) system of Tamil Nadu, kuhl/kulya gravity channels of the hills and northern plains, stepwells (baoli/vav) of Gujarat and Rajasthan, and the phad system of Rajasthan. The zabo system of Nagaland and bamboo drip irrigation of Meghalaya are further examples.

Q2. What is Krishi-Parashara and why is it important?

Model answer: It is an ancient Sanskrit treatise on agriculture attributed to the sage Parashara. It is important because it is the first systematic Indian text on farming, covering soil types, seed selection, sowing, manuring and rainfall prediction, and it shows that scientific agronomy existed in India long before modern times.

Q3. How did ancient farmers manage soil fertility without chemical fertilisers?

Model answer: They applied cattle dung and urine, compost, ash and green-leaf manure to feed the soil; rotated cereals with nitrogen-fixing pulses; grew mixed crops; and returned crop residues to the field. These practices built organic matter and kept the soil biologically active.

Long-answer questions with model answer outlines

Q1. Describe the indigenous wisdom of ancient India in soil, water and crop management, and show how it forms the root of natural farming.

Outline:

- Introduction: India as one of the oldest agricultural civilisations; heritage as transmitted knowledge.
- Soil management: cattle dung and urine, compost, green manure, rotation with pulses, mulching with residues.
- Water management: eri tanks, Kudimaramathu, kuhl channels, stepwells, phad, zabo; contour and bunding.
- Crop and pest management: seed treatment, botanical sprays (neem, nochi, erukku), mixed cropping, trap crops, sacred groves.
- Written sources: Krishi-Parashara, Arthashastra, Vrikshayurveda, Sangam literature.
- Linkage: each practice maps to a modern natural farming principle - Jivamrit, mulching, rotation, botanical pest control, water harvesting.
- Conclusion: natural farming is a revival, not an invention.

Q2. Explain the traditional water-harvesting systems of India with special reference to Tamil Nadu and their relevance today.

Outline:

- Introduction: seasonal monsoon and the need to store runoff.

- Eri tank system: cascading tanks, catchment, sluice, recharge of wells.
- Kudimaramathu: community desilting and repair as participatory governance.
- Other systems: kuhl, stepwells, phad, zabo, bamboo drip.
- Relevance: groundwater recharge, drought buffer, reduced dependence on borewells, revival under state programmes.
- Conclusion: traditional systems are the template for modern watershed management.

1.8 Traditional soil classification and cattle wealth

Ancient Indian farmers classified soils by colour, texture, water-holding capacity and suitability, and matched crops to each. This was practical soil science long before laboratories existed.

Traditional class	Character	Suitability
Urvara (fertile) and Anurvara (barren)	General fertility classification	Choice of field for the main crop
Krushna (black)	Heavy, moisture-retentive	Cotton and deep-rooted crops
Pandu (pale / light)	Light texture, quick to warm	Millets and pulses
Lohita (red)	Iron-rich, well drained	Groundnut and pulses
Tamil 'thinai' classification	Five ecological zones - kurinji, mullai, marutham, neydal, palai	Matching crops, livestock and livelihood to the landscape

The Tamil 'thinai' system is a striking early example of ecological land classification, dividing the landscape into hill, forest, agricultural, coastal and dry zones, each with its own crops and practices. Cattle were central to this system: the indigenous cow and bullock provided dung, urine, draught power, milk and leather, and native breeds such as Kangayam, Umbalachery, Pulikulam, Alambadi and Bargur were selected for local conditions. Modern natural farming draws directly on this cattle wealth.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Indian agriculture is about ____ years old.

2 The Indore method of composting was developed by ____.

3 The ancient Tamil water-harvesting system of village tanks is called ____.

4 Krishi-Parashara is attributed to the sage ____.

5 Community labour for desilting village tanks is known as ____.

6 The traditional Tamil land classification of five ecological zones is the ____ system.

B. State whether true or false

1 The Rigveda contains references to ploughing and irrigation channels. (True)

2 Ancient Indian farmers relied mainly on chemical fertilisers. (False - on cattle dung, urine and compost)

3 Neem was used as a botanical pesticide in ancient India. (True)

4 The Arthashastra has no reference to agriculture. (False - it records a state department of agriculture)

C. Match the following

Column A	Column B
Eri	Tamil Nadu village tank system
Kudimaramathu	Community tank and channel maintenance
Vrikshayurveda	Ancient treatise on trees
Gomutra	Cow urine used as soil purifier
Phad	Community-managed water diversion in Rajasthan

D. Answer in one word or one line

1 Name two ancient Indian texts on agriculture.

2 Which traditional practice is the ancestor of modern mulching?

3 Name two botanical pest-control plants used in ancient India.

4 What was the role of cattle in traditional Indian farming?

Practical application in Tamil Nadu - heritage practices still in use

- Eri tanks and Kudimaramathu are revived under state watershed and tank restoration programmes.
- Indigenous paddy varieties such as Kallurundai Samba and Mappillai Samba are conserved by farmer groups and seed banks.
- Cow-urine seed soaking and neem sprays are common in natural farming across the state.

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Lecture 2. History and Evolution of Natural Farming

Learning objectives

- Trace the global evolution of natural and organic farming thought.
- Describe the Indian journey from traditional farming to Zero Budget Natural Farming.
- Explain Fukuoka's 'Do-nothing' farming and its influence.
- List the milestones and movements that shaped natural farming in India and Tamil Nadu.

2.1 Introduction

Natural farming did not appear suddenly. It grew out of a century of concern about the damage caused by chemical-intensive agriculture - soil erosion, falling fertility, pesticide poisoning, loss of biodiversity and the rising cost of cultivation. This lecture follows that journey across the world and within India, so that the student can see natural farming as the latest stage of a long movement rather than an isolated idea.

2.2 The global journey

The modern organic and natural farming movement began in the early twentieth century and passed through several phases.

Period	Development	Key idea or person
1900s-1920s	Biodynamic agriculture in Europe	Rudolf Steiner (1924) - the farm as a self-contained living organism; preparations from manure and herbs.
1930s-1940s	Organic farming concept	Sir Albert Howard (UK, working in India) - 'The Waste Products of Agriculture' and 'An Agricultural Testament'; the Indore method of composting.
1940s	Organic movement formalised	Lady Eve Balfour's Haughley Experiment; Jerome Rodale in the USA popularised 'organic'.
1930s-1970s	Fukuoka's natural farming	Masanobu Fukuoka (Japan) developed 'Do-nothing' or natural farming; wrote 'The One-Straw Revolution' (1975).
1960s-1970s	Green Revolution and its costs	High-yielding varieties, fertilisers and pesticides raised output but caused soil and water degradation.
1980s-1990s	Korean Natural Farming (KNF)	Cho Han-Kyu developed Indigenous Micro-Organisms (IMO) and natural input preparations.
1990s-2000s	ZBNF in India	Subhash Palekar developed Zero Budget Natural Farming after observing natural forests.
2000s onward	Institutional and policy phase	Organic and natural farming missions, certification systems (PGS), and state-scale programmes such as Andhra Pradesh Community Managed Natural Farming.

2.3 Sir Albert Howard and the Indore method

Albert Howard worked as an agricultural adviser in India in the early twentieth century. Studying traditional Indian farmers, he concluded that the health of the soil, the plant, the animal and the human being are one chain. He developed the Indore method of composting, in which crop residues, cattle dung and urine are layered and turned to produce humus. This work became a foundation of the world organic movement and is a direct Indian contribution to it.

2.4 Fukuoka and 'Do-nothing' farming

Masanobu Fukuoka was a Japanese plant pathologist who left his job and returned to his family farm. After years of trial he arrived at what he called natural farming or 'Do-nothing' farming. The name does not mean doing nothing at all; it means removing unnecessary human intervention so that nature does the work.

- **No tillage:** the soil is never ploughed; roots, earthworms and microorganisms maintain structure.
- **No chemical fertiliser and no prepared organic fertiliser:** the soil is enriched by returning straw and by green manure crops such as clover.
- **No weeding by tillage or herbicide:** weeds are managed by ground cover, timing and disturbance of the surface layer.
- **No dependence on pesticides:** a balanced ecosystem keeps pests in check.
- His book 'The One-Straw Revolution' (1975) became a global classic and is listed in the reference list of this course.

2.5 The Indian journey

India's path to natural farming runs through the traditional practices of Lecture 1, then the Green Revolution, then a corrective phase.

2.5.1 Before the Green Revolution

Indian farming up to the 1960s was largely organic by default. Fertility came from cattle dung, compost and green manure; pests were managed with botanicals and cultural methods; and mixed cropping was common. Yields were modest but stable and the system was self-reliant.

2.5.2 The Green Revolution

From the mid-1960s India adopted high-yielding varieties of wheat and rice with assured irrigation, chemical fertilisers and pesticides. Food production rose sharply and famine was avoided, which was a genuine achievement. However, the costs appeared later: falling soil organic matter, micronutrient deficiency, groundwater depletion, pesticide residues, pest resurgence and rising input debt for farmers.

2.5.3 The corrective phase

From the 1980s awareness grew that chemical-intensive farming was not sustainable. Non-governmental organisations, farmer groups and scientists promoted organic farming. In the 1990s Subhash Palekar developed Zero Budget Natural Farming (ZBNF) in Maharashtra, and the idea spread rapidly across states. In the 2000s and 2010s, state governments, including Andhra Pradesh and Tamil Nadu, launched large programmes, and the central government introduced Paramparagat Krishi Vikas Yojana and the PGS-India certification system.

2.6 Subhash Palekar and Zero Budget Natural Farming

Subhash Palekar, a farmer from Maharashtra, observed that natural forests grow luxuriantly without any external input and concluded that the soil has its own capacity to supply nutrition if it is kept alive and covered. He distilled his approach into four pillars (Bijamrit, Jivamrit, Mulching and Waaphasa - see Lecture 5) and called it Zero Budget Natural Farming because the external cash cost of inputs is designed to be close to zero. The word 'budget' here means the cash input cost, not the total cost of cultivation.

Stage	Approximate period	Character of Indian agriculture
Traditional / organic by default	up to 1960s	Cattle-based fertility, mixed cropping, botanical pest control, self-reliant.
Green Revolution	mid-1960s to 1980s	High-yielding varieties, irrigation, chemical fertiliser and pesticides; rising output and rising cost.
Awareness and organic promotion	1980s to 1990s	NGO and farmer-led organic movements; concern over soil and water degradation.
ZBNF and natural farming spread	1990s to 2010s	Palekar's ZBNF; farmer-to-farmer training; state interest.
Policy and programme phase	2010s onward	Paramparagat Krishi Vikas Yojana, PGS-India, state natural farming missions, AP community-managed natural farming.

2.7 Evolution of natural farming in Tamil Nadu

Tamil Nadu has a long tradition of organic and natural farming rooted in its tank culture, cattle wealth and dryland cropping. The state has supported organic and natural farming through its agriculture department and the Tamil Nadu Organic Certification Department, and through schemes delivered on the Uzhavan platform. Farmer interest is strong in districts such as Erode, Dindigul, Krishnagiri, Salem and the Cauvery delta, where both organic certification and natural farming are practised. Note: specific scheme names, subsidy amounts and years change with each state budget and should be checked in the current government order; this material describes them in general terms.



Fig. 2.1 The evolution of natural farming from tradition to policy.

WORKED EXAMPLE - Budget: what 'zero budget' really means

- A natural farmer spends on seed, some labour and land preparation, but not on purchased fertiliser and pesticide.
- Suppose a conventional one-acre crop spends Rs. 9,000 on chemical fertiliser and Rs. 4,000 on pesticide per season.
- A natural farmer replaces these with Jivamrit, Bijamrit, mulching material and botanical sprays prepared on the farm.
- The cash input cost therefore falls close to zero for fertiliser and pesticide, though family labour and organic materials are still used. This is the sense in which ZBNF is 'zero budget'.

EXAM FOCUS

Key points to remember

- Natural farming evolved from a global organic movement and from India's own traditional practices.
- Albert Howard's Indore method of composting is an Indian contribution to world organic farming.
- Fukuoka's 'Do-nothing' farming rests on no tillage, no chemical inputs, no weeding by tillage and no pesticides.
- The Green Revolution raised output but caused soil, water and health problems.
- Subhash Palekar developed ZBNF in the 1990s around four pillars.
- In policy, Paramparagat Krishi Vikas Yojana, PGS-India and state missions carry the movement forward.
- Tamil Nadu's tradition of tanks, cattle and dryland cropping gives natural farming a strong local base.

Short-answer questions with model answers

Q1. Who was Masanobu Fukuoka and what is his contribution?

Model answer: Fukuoka was a Japanese plant pathologist turned farmer who developed natural or 'Do-nothing' farming. He showed that a farm can be productive without tillage, chemical fertiliser, weeding by tillage or pesticides, and wrote 'The One-Straw Revolution', which inspired the worldwide natural farming movement.

Q2. What is meant by 'zero budget' in ZBNF?

Model answer: It means that the external cash cost of inputs is brought close to zero. Fertiliser and pesticide are replaced by on-farm preparations such as Jivamrit and Bijamrit and by mulching. It does not mean there is no cost at all, because land, seed and family labour still involve cost.

Q3. State two achievements and two problems of the Green Revolution.

Model answer: Achievements: India became self-sufficient in foodgrain and famine was avoided; yields of wheat and rice rose sharply. Problems: soil organic matter declined and micronutrient deficiencies appeared; groundwater was depleted and pesticide residues and pest resurgence increased, along with farmer indebtedness.

Long-answer questions with model answer outlines

Q1. Trace the history and evolution of natural farming from traditional Indian agriculture to the present day.

Outline:

- Introduction: define natural farming as a revival of tradition shaped by global thought.
- Global phase: Steiner (biodynamic), Howard (organic, Indore method), Balfour and Rodale, Fukuoka, Korean Natural Farming.
- Indian traditional phase: organic by default with cattle-based fertility and mixed cropping.
- Green Revolution: benefits and costs.
- Corrective phase: organic movements, Palekar's ZBNF and the four pillars.
- Policy phase: PKVY, PGS-India, state missions, AP community-managed natural farming, Tamil Nadu initiatives.
- Conclusion: natural farming today is a mature, policy-supported system.

Q2. Discuss the contributions of Albert Howard, Masanobu Fukuoka and Subhash Palekar to natural farming.

Outline:

- Howard: worked in India; Indore method of composting; soil-plant-animal-human health as one chain; foundation of world organic farming.

- Fukuoka: Japan; natural/'Do-nothing' farming; four principles (no tillage, no chemical input, no weeding by tillage, no pesticides); 'The One-Straw Revolution'.
- Palekar: India; observed natural forests; ZBNF; four pillars (Bijamrit, Jivamrit, Mulching, Waaphasa); farmer-to-farmer training.
- Common thread: soil as a living system; reduced external inputs; diversity and natural processes.
- Conclusion: together they converted a local idea into a global and Indian movement.

2.8 Milestones of the organic and natural farming movement in India

Milestone	Significance
Albert Howard's work in India	Indore method of composting; foundation of world organic farming.
Green Revolution (mid-1960s)	High-yielding varieties and chemical inputs; raised output, later costs appeared.
Organic farming movements (1980s-1990s)	NGO and farmer-led promotion of organic methods.
ZBNF by Subhash Palekar (1990s)	Four-pillar natural farming; farmer-to-farmer training.
National Programme for Organic Production (NPOP)	Set the Indian organic standard and certification framework.
PGS-India	Low-cost participatory certification for small farmers.
Paramparagat Krishi Vikas Yojana (PKVY)	Cluster-based support for organic and traditional farming.
Andhra Pradesh Community Managed Natural Farming	Large state programme; a widely cited model of scale.
National Mission on Natural Farming	National push to promote natural farming through demonstration and training.

Note: the precise year of each milestone should be confirmed from official documents; the sequence and significance above are the important points.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Masanobu Fukuoka wrote the book ____.

2 ZBNF was developed by ____.

3 The Indore method of composting is associated with ____.

4 Korean Natural Farming was developed by ____.

5 The Green Revolution in India began in the mid-____.

6 Fukuoka called his method ____ farming.

B. State whether true or false

1 The Green Revolution raised yields but had no environmental cost. (False - it caused soil, water and health problems)

2 Albert Howard worked in India and developed a composting method. (True)

3 ZBNF means the total cost of cultivation is zero. (False - it means the external cash cost of inputs is near zero)

4 Fukuoka's method uses chemical fertiliser in small amounts. (False - no chemical fertiliser is used)

C. Match the following

Column A	Column B
Fukuoka	Do-nothing natural farming, Japan
Palekar	ZBNF, India
Howard	Indore method of composting
Cho Han-Kyu	Korean Natural Farming
Rodale	Popularised organic farming in the USA

D. Answer in one word or one line

1 Expand ZBNF.

2 Name the four pillars of ZBNF.

3 What is a seed ball?

4 In which decade did the Green Revolution begin in India?

Practical application in Tamil Nadu - the movement in the state

- Natural farming and organic farming are promoted through the state agriculture department and its extension network.
- The Tamil Nadu Organic Certification Department supports certification of organic produce.
- Farmer interest is strong in Erode, Dindigul, Krishnagiri, Salem and the Cauvery delta districts.

Lecture 3. Concept, Definition and Objectives of Natural Farming

Learning objectives

- Define natural farming clearly and distinguish it from organic and conventional farming.
- State the ecological and socio-economic objectives of natural farming.
- Explain its importance for climate change, food security and rural sustainability.

3.1 What natural farming means

Natural farming is a farming system that works with nature rather than against it. It avoids synthetic chemical fertilisers, pesticides and growth regulators, and it avoids disturbing the soil through deep or repeated tillage. Instead it relies on the natural processes of the soil - the activity of microorganisms, earthworms, organic matter and plant roots - to supply nutrition and to keep pests and diseases in check. Its central idea is simple: if the soil is kept alive, covered and moist, it can feed the crop without purchased inputs.

Several definitions are used. A widely accepted working definition is given below.

Working definition

- Natural farming is a chemical-free, low-input, ecologically integrated farming system in which crop nutrition is met from on-farm organic resources and the soil's own biological processes, pest and disease are managed by ecological methods, and the soil is protected from disturbance and exposure, so that the farm becomes self-reliant, sustainable and in harmony with nature.

3.2 Natural farming, organic farming and conventional farming

Students often confuse natural farming with organic farming. Both avoid synthetic chemicals, but they differ in their attitude to external inputs and to soil disturbance. The comparison table below is important for the examination.

Basis	Natural farming	Organic farming	Conventional farming
Core principle	Work with nature; minimum human intervention	Avoid synthetic chemicals; use approved organic inputs	Maximise yield with external inputs
Synthetic fertiliser	Not used	Not used	Used heavily
Synthetic pesticide	Not used	Not used	Used heavily
External organic inputs (compost, oil cakes, rock phosphate)	Minimal; mostly on-farm preparations	Permitted and often purchased	Not the main input
Tillage	No tillage or minimum disturbance	Tillage usually allowed	Intensive tillage
Mulching	Central practice (crop residue / live mulch)	Practised but not mandatory	Rare
Soil cover	Kept covered at all times	Encouraged	Often bare
Source of nutrition	Soil biology and on-farm recycling	Organic amendments and manures	Synthetic NPK and micronutrients
Input cost	Very low (close to zero cash cost for nutrients)	Moderate to high	High and rising
Certification	Often community or self-declared; PGS where applicable	Mandatory certification for premium market (PGS, NPOP)	Not applicable
Typical yield in transition	May dip initially, then stabilise	May dip initially, then recover	High but input-dependent

Basis	Natural farming	Organic farming	Conventional farming
Main aim	Sustainability, soil health, self-reliance	Chemical-free produce and market	Production and profit

3.3 Ecological objectives

Natural farming has clear ecological aims, which are the reason it is promoted for sustainability.

- **Build soil health:** increase organic carbon, humus and microbial diversity, and improve soil structure and water-holding capacity.
- **Protect water:** reduce chemical runoff and leaching, and improve infiltration and groundwater recharge.
- **Conserve biodiversity:** encourage pollinators, natural enemies and soil organisms through crop diversity and habitat patches.
- **Reduce pollution:** eliminate pesticide residues in soil, water, food and the farmer's body.
- **Mitigate climate change:** sequester carbon in soil, reduce greenhouse gas emissions from fertiliser manufacture and use, and lower fossil energy use.
- **Adapt to climate change:** healthier soils resist drought and heavy rain better, giving resilience to erratic monsoons.

3.4 Socio-economic objectives

- **Reduce cost of cultivation:** remove or greatly reduce expenditure on fertiliser and pesticide.
- **Increase and stabilise net income:** lower costs can raise net returns even if gross yield is similar.
- **Free farmers from debt:** reduce dependence on input dealers and moneylenders.
- **Improve food quality:** chemical-free, more nutritious produce for the family and the market.
- **Empower rural communities:** strengthen local knowledge, seed systems and self-reliance.
- **Generate rural employment:** preparation of inputs, mulching, processing and local marketing create work.
- **Improve farmer health:** fewer pesticide-related illnesses.

3.5 Importance for climate change, food security and rural sustainability

Concern	How natural farming helps
Climate change mitigation	Soil carbon sequestration through organic matter and no tillage; lower emissions from avoided fertiliser manufacture and use; reduced fossil fuel in tillage.
Climate change adaptation	Better soil moisture retention and structure give drought and flood resilience; diverse crops spread risk.
Food security	Stable, low-cost production for the household; diversified output (cereals, pulses, vegetables, milk) improves nutrition.
Food safety	No pesticide residues; safer produce for consumers and safer work for farmers.
Rural sustainability	Lower input cost, local input production and local value addition keep money in the village.
Natural resource base	Protects soil, water and biodiversity for future generations.

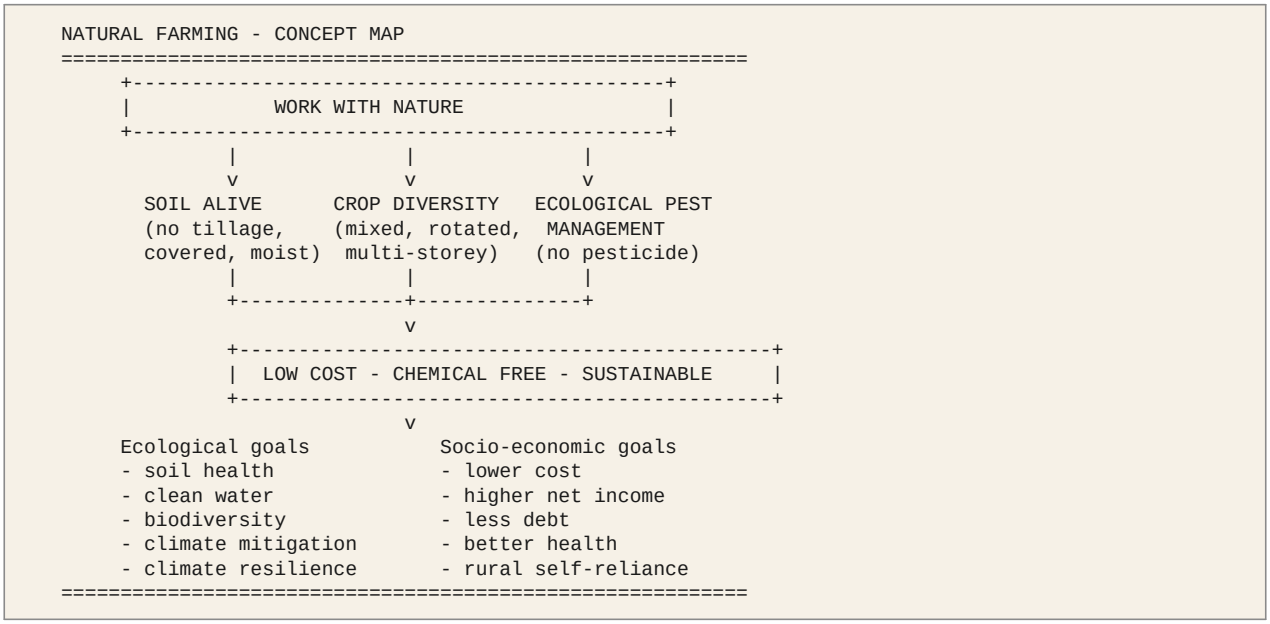


Fig. 3.1 The concept of natural farming and its twin sets of objectives.

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Key points to remember

- Natural farming works with nature, avoids synthetic inputs and avoids soil disturbance.
- It is not the same as organic farming: natural farming minimises external inputs and tillage, while organic farming permits approved external organic inputs and tillage.
- Ecological objectives: soil health, clean water, biodiversity, lower pollution, climate mitigation and adaptation.
- Socio-economic objectives: lower cost, higher net income, less debt, better health, rural self-reliance.
- For climate change it both mitigates (carbon sequestration) and helps adaptation (resilience).
- For food security it provides stable, diverse and safe food at low cost.
- For rural sustainability it keeps inputs and value local.

Short-answer questions with model answers

Q1. Define natural farming.

Model answer: Natural farming is a chemical-free, low-input, ecologically integrated farming system in which crop nutrition comes from on-farm organic resources and soil biological processes, pest and disease are managed by ecological methods, and the soil is protected from disturbance and exposure, making the farm self-reliant and sustainable.

Q2. Give any three differences between natural farming and organic farming.

Model answer: Natural farming uses no or minimum tillage while organic farming permits tillage. Natural farming relies mainly on on-farm preparations with minimal external inputs while organic farming permits purchased organic inputs such as compost, oil cakes and rock phosphate. Natural farming keeps the soil covered with mulch at all times, whereas mulching is only encouraged and not mandatory in organic farming.

Q3. How does natural farming help in climate change mitigation and adaptation?

Model answer: Mitigation: it sequesters carbon in soil through organic matter and no tillage, and it avoids the greenhouse gases and fossil energy associated with fertiliser manufacture and use. Adaptation: healthier soils hold more moisture and resist drought and heavy rain better, and crop diversity spreads risk, so the farm is more resilient to erratic weather.

Long-answer questions with model answer outlines

Q1. Define natural farming and explain its ecological and socio-economic objectives.

Outline:

- Definition and central idea: work with nature, chemical-free, low input, soil alive and covered.
- Ecological objectives: soil health, water protection, biodiversity, reduced pollution, climate mitigation and adaptation.
- Socio-economic objectives: lower cost, higher net income, less debt, better food quality and health, rural employment and self-reliance.
- Comparison table: natural vs organic vs conventional.
- Conclusion: natural farming meets production and conservation goals together.

Q2. Discuss the importance of natural farming for climate change, food security and rural sustainability.

Outline:

- Introduction: the three concerns and why they matter for India.
- Climate change: mitigation (carbon sequestration, lower emissions) and adaptation (resilience, diversity).
- Food security: stable low-cost output, household nutrition through diversification, food safety.
- Rural sustainability: lower input cost, local input production, local value addition, employment, reduced migration.
- Supporting evidence: state programmes and farmer experiences.
- Conclusion: natural farming is a triple-win system for environment, food and rural economy.

3.6 Natural farming compared with other sustainable systems

Natural farming is one of several sustainable systems. Students should be able to distinguish them briefly.

System	Key idea	Difference from natural farming
Organic farming	Avoid synthetic chemicals; use approved organic inputs	Permits external organic inputs and tillage; requires certification for the premium market
Biodynamic farming	Farm as a self-contained organism; special preparations	Uses specific fermented preparations and a cosmic calendar; natural farming does not
Permaculture	Design system mimicking natural ecosystems	Emphasis on design of the whole homestead; natural farming focuses on field practice
Regenerative agriculture	Restore soil health and sequester carbon	Often permits some external inputs and managed grazing; broader umbrella
Conservation agriculture	Minimum tillage with residue retention and rotation	May still use herbicides; natural farming does not
Zero Budget Natural Farming	Four pillars; desi cow based; zero external cash cost	A specific school within natural farming

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Natural farming avoids ____ fertilisers and pesticides.

2 Natural farming keeps the soil ____ at all times.

3 A key socio-economic objective of natural farming is to reduce the ____ of cultivation.

4 Natural farming sequesters carbon in the soil, which helps climate change ____.

5 Natural farming differs from organic farming mainly in minimising ____ inputs and tillage.

6 Food ____ is improved because natural produce is free of pesticide residues.

B. State whether true or false

1 Natural farming and organic farming are exactly the same. (False - they differ in external inputs and tillage)

2 Natural farming can help both mitigate and adapt to climate change. (True)

3 Natural farming increases the farmer's dependence on input dealers. (False - it reduces dependence)

4 Natural farming has only environmental objectives. (False - it also has socio-economic objectives)

C. Match the following

Column A	Column B
Ecological objective	Soil, water and biodiversity health
Socio-economic objective	Lower cost and higher net income
Climate mitigation	Carbon sequestration in soil
Climate adaptation	Resilience to drought and heavy rain
Food safety	No pesticide residues

D. Answer in one word or one line

1 Define natural farming in one sentence.

2 Name two ecological objectives of natural farming.

3 Name two socio-economic objectives.

4 Give one way natural farming helps climate adaptation.

Practical application in Tamil Nadu - objectives in practice

- Lower input cost is the strongest motivation for small farmers adopting natural farming.
- Chemical-free produce is sold through organic stores, farmers' markets and online platforms at a premium.
- Improved soil moisture retention matters greatly in the state's dryland districts.

Lecture 4. Essential Characteristics and Principles of Natural Farming

Learning objectives

- State the essential characteristics that define a natural farm.
- Explain the principles of no tillage, no chemical inputs and minimum disturbance.
- Describe reliance on natural processes for fertility and pest control.

4.1 Essential characteristics

The characteristics below distinguish a natural farm from any other farming system. A farm need not satisfy every point perfectly at the start, but these are the direction of travel.

- **No synthetic chemical input:** no urea, DAP, complex fertiliser, synthetic pesticide, herbicide or growth regulator.
- **No tillage or minimum tillage:** the soil is not inverted or pulverised; structure built by roots and soil life is preserved.
- **Soil always covered:** crop residues, straw or live mulch keep the surface shaded and moist.
- **Soil always moist (Waaphasa):** a balance of air and moisture is maintained in the soil pores.
- **Reliance on soil biology:** microorganisms, earthworms and mycorrhiza supply and mobilise nutrients.
- **Crop diversity:** mixed cropping, rotation and multi-storey systems instead of monoculture.
- **On-farm input production:** inputs such as Jivamrit and botanical sprays are prepared from farm and local materials.
- **Integration of crops, trees and livestock:** the farm recycles nutrients within its own boundary.
- **Ecological pest management:** prevention through diversity and natural enemies, with botanicals as a last resort.
- **Seed from the farm and indigenous varieties:** local seed adapted to the soil and climate.
- **Low external dependence and self-reliance:** minimal purchase, minimal debt.
- **Long-term soil and ecosystem health as the primary goal.**

4.2 Principle of no tillage

Tillage is the mechanical disturbance of soil. Conventional farming ploughs and pulverises the soil to prepare a seedbed and control weeds. Natural farming rejects this because tillage destroys soil structure, breaks fungal networks, oxidises organic matter, exposes soil to erosion and kills earthworms and other soil life.

- Soil is a living structure of aggregates, pores and channels built over years by roots and organisms.
- Tillage collapses aggregates, seals pores and reduces infiltration.
- Tillage accelerates the loss of organic carbon to the atmosphere.
- No tillage keeps root channels and earthworm burrows intact, improving water movement and aeration.
- Seeds are sown by dibbling, direct seeding or transplanting into a mulch layer without ploughing.

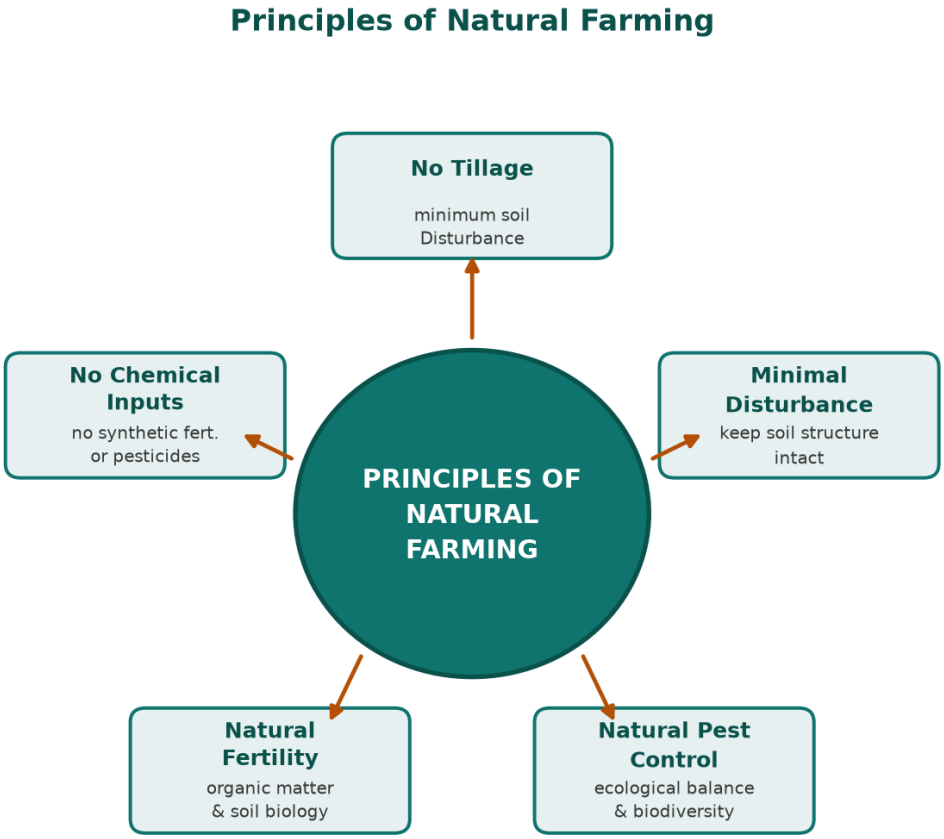


Figure 4.1 Principles of natural farming

No tillage, no chemical inputs, minimal disturbance, natural fertility and natural pest control.

4.3 Principle of no chemical inputs

Synthetic fertilisers and pesticides give a quick response but harm the soil ecosystem and create dependence.

- Chemical fertilisers supply only a few nutrients and do not build organic matter.
- Excess nitrogen acidifies soil and leaches into groundwater as nitrate.
- Pesticides kill beneficial insects, pollinators and natural enemies, causing pest resurgence.
- Residues enter the food chain and harm farmer and consumer health.
- Natural farming substitutes Jivamrit, compost, green manure and botanical preparations.

4.4 Principle of minimum disturbance and ecological balance

Besides tillage, other disturbances are also minimised - unnecessary weeding, burning of residues, removal of all biomass and excessive grazing. The aim is to keep the farm's living communities intact so that they regulate themselves. A balanced ecosystem has predators that control pests, decomposers that recycle nutrients and pollinators that set seed and fruit.

4.5 Reliance on natural processes for fertility

In natural farming, fertility is a product of biological activity rather than of applied chemicals. The main natural processes are given below.

Natural process	How it supplies nutrition
Organic matter decomposition	Soil microbes break down mulch and residues into humus, releasing nitrogen, phosphorus, potassium and micronutrients slowly.
Biological nitrogen fixation	Rhizobia in legume root nodules and free-living bacteria fix atmospheric nitrogen.
Mycorrhizal association	Fungal networks extend root reach and help absorb phosphorus, zinc and water.
Nutrient mobilisation	Phosphate-solubilising and potassium-mobilising bacteria release locked nutrients from soil minerals.
Earthworm activity	Earthworms mix organic matter with soil and produce nutrient-rich casts.
Green manuring	Leguminous green manure crops (sunhemp, daincha) add nitrogen and organic matter.
Mulching	Residue mulch feeds soil life and conserves moisture, driving the whole process.

4.6 Reliance on natural processes for pest control

Natural process	How it controls pests
Natural enemies	Predators and parasitoids (ladybird beetles, spiders, lacewings, Trichogramma) keep pest populations low.
Crop diversity	Mixed cropping confuses pests, hosts alternative prey for predators and slows spread.
Botanical preparations	Neem, nochi, erukku and adathoda act as repellents, antifeedants and growth regulators.
Healthy soil	Balanced plant nutrition from organic matter makes plants less attractive to pests and more tolerant of damage.
Cultural methods	Timely sowing, trap crops, border rows, sanitation and hand-picking.
Habitat for enemies	Bund vegetation, flowering plants and trees shelter beneficial insects and birds.

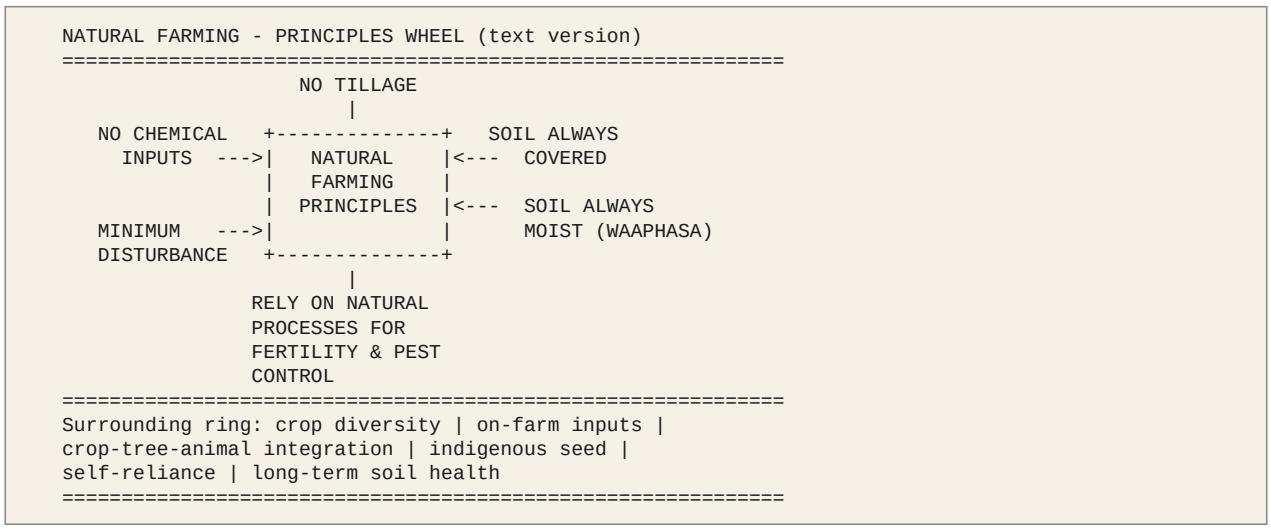


Fig. 4.1 The principles wheel - five core principles with supporting ideas around them.

WORKED EXAMPLE - Applying the principles on one acre

- No tillage: sow with a dibble or seed drill directly into the previous crop's mulch; do not plough.
- No chemical input: replace the usual urea top-dress with Jivamrit; replace the usual pesticide with a neem-based spray only if the pest crosses the threshold.
- Soil covered: spread 2-3 tonnes of straw or crop residue per acre as mulch.
- Soil moist: keep the mulch layer intact and irrigate only when the soil below the mulch begins to dry.
- Diversity: sow a cereal with a pulse and a border row of marigold or castor to shelter natural enemies.

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Key points to remember

- The defining characteristics are no chemical input, no or minimum tillage, permanent soil cover, moist soil (Waaphasa), reliance on soil biology, diversity and self-reliance.
- No tillage protects soil structure, organic matter and soil life.
- No chemical input avoids soil acidification, pollution and pest resurgence.
- Minimum disturbance extends to weeding, residue burning and grazing.
- Fertility comes from decomposition, nitrogen fixation, mycorrhiza, nutrient mobilisation, earthworms, green manuring and mulch.
- Pest control comes from natural enemies, diversity, botanicals, healthy plants and cultural methods.
- The five principles together make the farm self-regulating.

Short-answer questions with model answers

Q1. Why does natural farming avoid tillage?

Model answer: Tillage destroys soil structure, breaks fungal networks, oxidises organic carbon, seals pores, increases erosion and kills earthworms and soil microorganisms. No tillage preserves root channels and burrows, keeps organic matter in place and maintains the biological activity on which natural farming depends for nutrition.

Q2. Name four natural processes that supply nutrition in a natural farm.

Model answer: Decomposition of organic matter and mulch into humus; biological nitrogen fixation by legumes and free-living bacteria; mycorrhizal association helping uptake of phosphorus, zinc and water; and nutrient mobilisation by phosphate-solubilising and potassium-mobilising bacteria. Earthworm activity and green manuring are further examples.

Q3. What is minimum disturbance in natural farming?

Model answer: It means reducing all unnecessary interference with the farm ecosystem beyond tillage - avoiding excessive weeding, not burning crop residues, leaving biomass on the field and controlling grazing - so that the living communities of soil, plants and beneficial insects remain intact and regulate themselves.

Long-answer questions with model answer outlines

Q1. State the essential characteristics and principles of natural farming and explain any four of them.

Outline:

- List the characteristics: no chemical input, no tillage, permanent cover, Waaphasa, soil biology, diversity, on-farm inputs, integration, ecological pest management, indigenous seed, self-reliance.
- Explain no tillage: soil structure, organic carbon, soil life.
- Explain no chemical input: pollution, resurgence, health.
- Explain reliance on soil biology for fertility: decomposition, fixation, mycorrhiza.
- Explain ecological pest management: enemies, diversity, botanicals.
- Conclusion: together they make the farm self-regulating and sustainable.

Q2. Explain how natural farming relies on natural processes for fertility and pest control instead of external inputs.

Outline:

- Fertility: organic matter decomposition, nitrogen fixation, mycorrhiza, nutrient mobilisation, earthworms, green manuring, mulching.
- Pest control: natural enemies, crop diversity, botanical preparations, healthy plants, cultural methods, habitat for enemies.
- Why it works: a balanced ecosystem regulates itself; diversity and soil health are the foundation.
- Contrast with conventional reliance on fertiliser and pesticide.
- Conclusion: natural processes replace purchased inputs.

4.7 Common mistakes in applying the principles

Knowing the mistakes helps the student apply the principles correctly.

- **Stopping all inputs at once:** an abrupt change can cause a severe yield drop; transition gradually, especially reducing fertiliser over a season or two.
- **Neglecting mulch:** without mulch, Waaphasa cannot be maintained and the soil dries out; mulch is not optional.
- **Over-irrigation:** flooding destroys Waaphasa and drives out air from the soil.
- **Poorly prepared Jivamrit:** wrong proportions or too long a fermentation reduce its microbial value.
- **Using non-desi cattle products:** the microbial richness of desi cow dung and urine is considered essential in ZBNF.
- **Expecting immediate results:** soil biology takes time to rebuild; the first year or two may be a learning period.
- **Ignoring diversity:** monoculture under natural farming is fragile; diversity is part of the system.
- **Spraying botanicals routinely:** even botanicals can harm beneficial insects; spray only above the action threshold and in the evening.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 The five core principles are no tillage, no chemical input, minimum disturbance, permanent ____ and ____.

2 Tillage accelerates the loss of soil organic ____.

3 Biological nitrogen fixation is carried out by ____ in legume root nodules.

4 Fungal networks that help roots absorb phosphorus are called ____.

5 Earthworms produce nutrient-rich ____.

6 A balanced ecosystem controls pests through natural ____.

B. State whether true or false

1 Natural farming uses deep ploughing to prepare the seedbed. (False - no tillage)

2 Tillage improves soil structure in the long run. (False - it destroys structure)

3 Mycorrhiza help plants absorb phosphorus and water. (True)

4 Natural farming sprays pesticide on a fixed calendar. (False - only above the action threshold)

C. Match the following

Column A	Column B
No tillage	Preserves soil structure and soil life
No chemical input	Avoids pollution and pest resurgence
Waaphasa	Balance of air and moisture in soil pores
Rhizobium	Nitrogen fixation in legumes
Green manuring	Adds nitrogen and organic matter

D. Answer in one word or one line

1 Name the five core principles of natural farming.

2 Why is tillage harmful to soil?

3 What is mycorrhiza?

4 Name two natural processes that supply nutrition.

Practical application in Tamil Nadu - principles on the ground

- No-till sowing with a dibbler or no-till drill is used for rain-fed crops such as millets and pulses.
- Mulching with crop residues is common in coconut and banana gardens to conserve moisture.
- Botanical sprays made from neem and nochi are widely used instead of synthetic pesticides.

Lecture 5. Main Pillars of Natural Farming

Learning objectives

- Name the four pillars of natural farming and explain each.
- Give the composition and function of Bijamrit and Jivamrit.
- Explain mulching and Waaphasa and their scientific basis.
- Describe supporting practices such as Whapasa management and botanical inputs.

5.1 The four pillars

Subhash Palekar's ZBNF rests on four pillars: Bijamrit (seed treatment), Jivamrit (soil inoculum), Mulching (soil cover) and Waaphasa (soil moisture-air balance). Together they keep the soil alive, covered, moist and biologically active, so that the crop is nourished without external inputs.

Pillar	Meaning	Main composition	Function
Bijamrit	Seed treatment / seed inoculum	Cow dung (desi cow), cow urine, lime (calcium), a little soil from the farm bund, water; often jaggery	Coats the seed with beneficial microbes; protects against seed-borne disease; improves germination and early vigour.
Jivamrit	Living microbial culture for the soil	Desi cow dung, desi cow urine, jaggery or jaggery powder, pulse flour (besan), and soil from the farm bund, mixed in water	Inoculates the soil with a large population of beneficial microorganisms; accelerates decomposition of mulch; mobilises nutrients.
Mulching	Covering the soil surface	Crop residues, straw, dry leaves, weeds, green biomass, or a live mulch crop	Conserves moisture, moderates temperature, suppresses weeds, feeds soil life and adds organic matter.
Waaphasa	Soil moisture-air balance in the pores	Achieved through mulching and judicious irrigation, not a material	Maintains the ideal proportion of air and water in soil pores so that aerobic microbes and roots thrive.

5.2 Bijamrit - seed treatment

Bijamrit means 'nectar of the seed'. It is a coating applied to seeds before sowing. The purpose is to surround the seed with beneficial microorganisms and a mild protective layer so that it germinates strongly and resists early disease.

5.2.1 Preparation

A common preparation uses the following materials. Quantities are for treating seed for about one acre; adjust in proportion.

Ingredient	Quantity (typical)	Role
Desi cow dung (fresh)	5 kg	Carries a large microbial population; provides organic matter.
Desi cow urine	5 litre	Mildly alkaline; nitrogen, potassium and microbes; disinfecting action.
Lime (calcium) or chuna	50-100 g	Raises pH slightly, provides calcium, helps coat the seed.
Farm soil from the bund	a handful (about 100 g)	Introduces locally adapted microbes.
Water	sufficient to make a slurry	Medium for mixing and coating.
Jaggery (optional)	50-100 g	Food for microbes in the slurry.

Method: Mix the cow dung and cow urine in water to form a thin slurry. Add lime, farm soil and jaggery and stir well. Soak the seed in the slurry for a short time, then either sow immediately or dry the seed in shade and sow the same day. The seed should be evenly coated but not left wet for long.

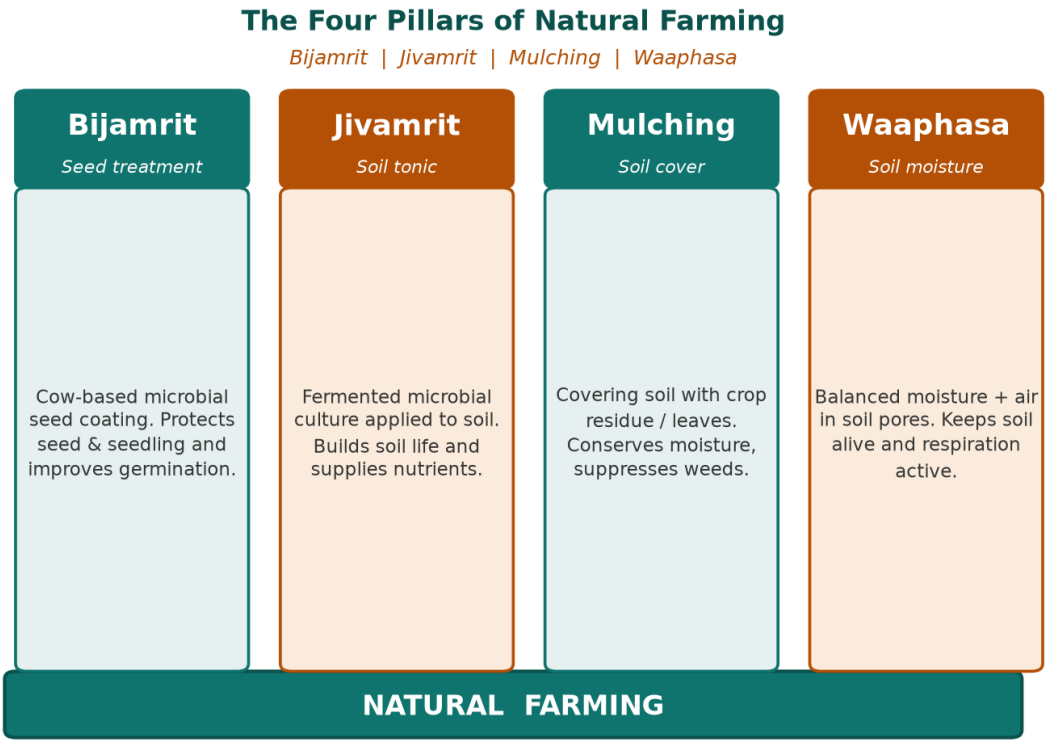


Figure 5.1 The four pillars of natural farming

Bijamrit, Jivamrit, Mulching and Waaphasa - what each provides to the farm.

Preparation of Bijamrit and Jivamrit

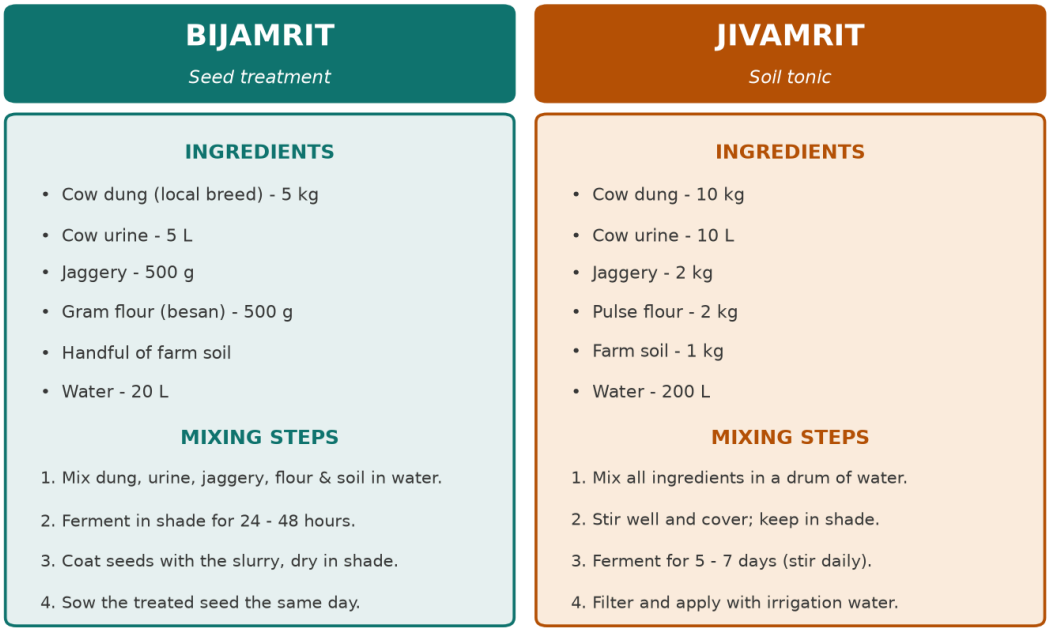


Figure 5.2 Bijamrit and Jivamrit preparation

Ingredients and mixing steps for Bijamrit seed treatment and Jivamrit soil tonic.

5.3 Jivamrit - the living soil inoculum

Jivamrit means 'nectar of life'. It is a microbial culture that is applied to the soil or through irrigation to multiply beneficial microorganisms. These organisms decompose mulch, release nutrients and improve soil structure.

5.3.1 Preparation

Ingredient	Quantity (for one acre)	Role
Desi cow dung (fresh)	10 kg	Base microbial source and organic matter.
Desi cow urine	10 litre	Nitrogen, potassium, microbes, mild disinfectant.
Jaggery or jaggery powder	2 kg	Energy source for rapid microbial multiplication.
Pulse flour (besan or any pulse)	2 kg	Protein and nutrient source for microbes.
Farm soil from the bund	a handful (about 100 g)	Locally adapted microbial inoculum.
Water	200 litre (in a drum)	Medium for the culture.

Method: Dissolve jaggery in about 200 litres of water in a drum. Add cow dung, cow urine, pulse flour and farm soil. Stir well and allow to ferment for about two to three days in shade, stirring twice a day. The culture is ready when a mild fermenting smell develops. Apply through irrigation water or as a soil drench. Note: the exact ratio and fermentation time vary with climate and source material; the quantities above are the standard ZBNF range.

5.4 Mulching

Mulching is the covering of the soil surface with organic material. In natural farming it is not optional - it is a pillar, because it drives the other three.

- **Types:** straw mulch, crop residue mulch, dry leaf mulch, green mulch and live mulch (a cover crop grown in place).
- **Moisture conservation:** mulch cuts evaporation from the soil surface, so the same irrigation goes further.
- **Temperature moderation:** it keeps the soil cooler in summer and warmer in winter, protecting roots and soil life.
- **Weed suppression:** a thick mulch layer blocks light and reduces weed germination.
- **Soil life:** as mulch decomposes it feeds earthworms and microorganisms and builds humus.
- **Erosion control:** mulch absorbs the impact of raindrops and reduces runoff.

5.5 Waaphasa - the soil moisture-air balance

Waaphasa (also written Whapasa) is the state in which soil pores hold both air and moisture in the right proportion for plant roots and aerobic microorganisms. It is not merely 'wet soil'. Saturated soil has no air and anaerobic conditions develop; dry soil has no moisture. Waaphasa is the middle, balanced condition.

- Achieved mainly through mulching, which slows moisture loss and prevents surface crusting.
- Judicious, light irrigation rather than flooding helps maintain the balance.
- In a well-aggregated soil with good organic matter, pores of different sizes hold air and water together.
- Aerobic microbes need oxygen; Waaphasa keeps them active and prevents putrefaction.

- Over-irrigation destroys Waaphasa by filling all pores with water and driving out air.

5.6 Supporting practices

Beyond the four pillars, natural farming uses several supporting practices.

Practice	Purpose
Achhadana / straw mulching of paths and bunds	Keep the whole farm covered, not only the cropped area.
Beejamrit and Jivamrit through drip	Deliver microbes efficiently with irrigation water.
Ghana Jivamrit (solid form)	A concentrated solid Jivamrit for dryland and plantation crops.
Botanical sprays (neem, nochi, erukku, adathoda)	Ecological pest and disease management.
Dravajivamrit / liquid organic preparations	Quick nutrition and microbial boost as a foliar or soil drench.
Green manuring and cover cropping	Add nitrogen and organic matter and provide live mulch.
Crop-tree-animal integration	Internal nutrient recycling and risk reduction.
Indigenous seed conservation	Local adaptation and seed sovereignty.

WORKED EXAMPLE - Preparing Bijamrit and Jivamrit for 1 acre

- Bijamrit: 5 kg desi cow dung + 5 L cow urine + 50-100 g lime + a handful of farm soil, mixed in water to a slurry; coat the seed before sowing.

- Jivamrit: 10 kg desi cow dung + 10 L cow urine + 2 kg jaggery + 2 kg pulse flour + a handful of farm soil in 200 L water; ferment 2-3 days; apply to the soil.

- For 2 acres, double every quantity: Bijamrit 10 kg dung + 10 L urine; Jivamrit 20 kg dung + 20 L urine + 4 kg jaggery + 4 kg pulse flour in about 400 L water.

- Rule of thumb: the ratio of dung, urine, jaggery and pulse flour is kept in a fixed proportion and the water is adjusted to the area.

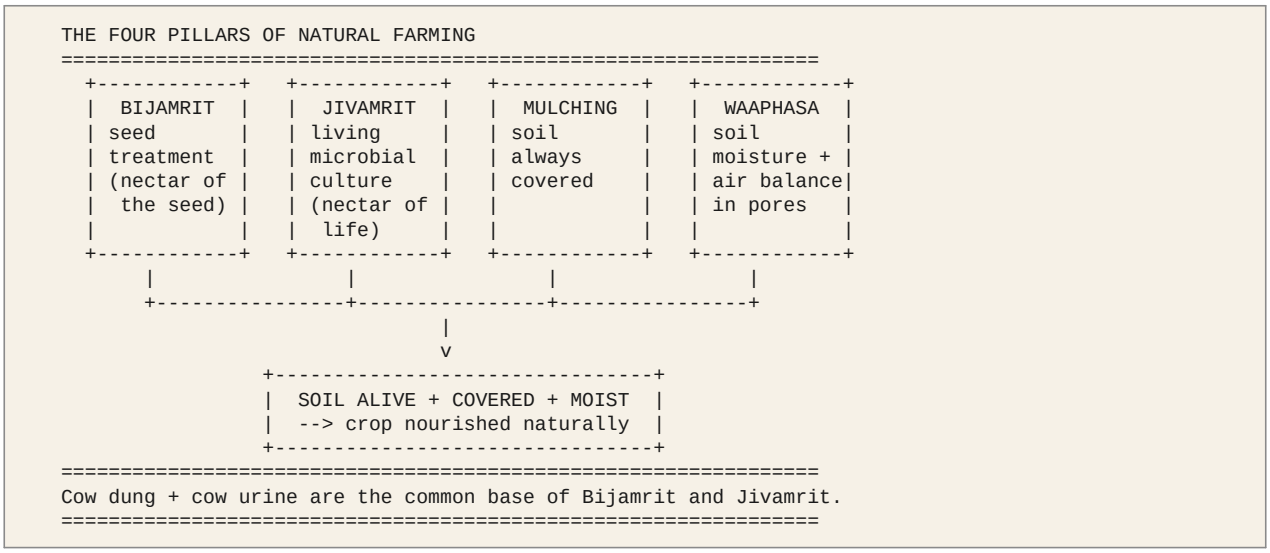


Fig. 5.1 The four pillars and how they combine to nourish the crop.

EXAM FOCUS

Key points to remember

- The four pillars are Bijamrit (seed treatment), Jivamrit (soil microbial culture), Mulching (soil cover) and Waaphasa (moisture-air balance).
- Bijamrit coats the seed with beneficial microbes; typical base is desi cow dung, cow urine and lime with farm soil.
- Jivamrit is a fermented microbial culture of cow dung, cow urine, jaggery, pulse flour and farm soil in water.

- Mulching conserves moisture, moderates temperature, suppresses weeds and feeds soil life.
- Waaphasa is the balanced state of air and water in soil pores; over-irrigation destroys it.
- Cow dung and cow urine are the common base of the preparations.
- Supporting practices include Ghana Jivamrit, botanical sprays, green manuring and drip application.

Short-answer questions with model answers

Q1. Name the four pillars of natural farming.

Model answer: Bijamrit (seed treatment), Jivamrit (living microbial culture applied to the soil), Mulching (permanent soil cover) and Waaphasa (the balance of moisture and air in soil pores).

Q2. What is Jivamrit and how is it prepared?

Model answer: Jivamrit is a living microbial culture applied to the soil. It is prepared by dissolving jaggery in about 200 litres of water and adding desi cow dung, desi cow urine, pulse flour and a handful of farm soil, then fermenting for two to three days in shade with stirring. It multiplies beneficial microorganisms that decompose mulch and mobilise nutrients.

Q3. Define Waaphasa and state why it is important.

Model answer: Waaphasa is the state in which soil pores hold air and moisture in the correct proportion for roots and aerobic microbes. It is important because saturated soil becomes anaerobic and dry soil starves the plant; the balanced state keeps aerobic decomposition and root activity at their best. Mulching and light irrigation maintain Waaphasa.

Long-answer questions with model answer outlines

Q1. Explain the four pillars of natural farming with their composition and function.

Outline:

- Bijamrit: meaning, composition (cow dung, cow urine, lime, farm soil), function (seed protection, germination).
- Jivamrit: meaning, composition (cow dung, cow urine, jaggery, pulse flour, farm soil, water), function (microbial multiplication, decomposition, nutrient mobilisation).
- Mulching: types and functions (moisture, temperature, weeds, soil life, erosion).
- Waaphasa: definition and importance, role of mulching and irrigation.
- How the four work together.
- Conclusion: they keep the soil alive, covered and moist, replacing external inputs.

Q2. Describe the preparation of Bijamrit and Jivamrit for one acre and explain their role in soil fertility.

Outline:

- Bijamrit method: mix cow dung and urine with water, add lime, farm soil and jaggery, coat the seed.
- Jivamrit method: dissolve jaggery in 200 L water, add dung, urine, pulse flour, farm soil, ferment 2-3 days, apply.
- Role in fertility: microbial inoculation, decomposition of mulch, nitrogen fixation and nutrient mobilisation, improved soil structure.
- Scaling to larger areas by proportion.
- Precautions: use desi cow products, shade fermentation, apply fresh.
- Conclusion: these preparations convert on-farm materials into a substitute for purchased fertiliser.

5.7 Precautions and quality checks for preparations

- **Use fresh desi cow dung and urine:** stale material loses microbial activity.

- **Ferment in shade:** direct sunlight can kill microorganisms and dry the culture.
- **Stir twice daily:** stirring aerates the culture and promotes aerobic microbes.
- **Do not over-ferment:** excessive fermentation produces a foul smell and reduces quality.
- **Use clean water:** chlorinated water can harm the microbes; allow it to stand before use.
- **Apply fresh:** use Jivamrit within a short period of preparation; its value declines with storage.
- **Filter before spraying:** particles clog sprayer nozzles.
- **Scale by proportion:** keep the ratio of dung, urine, jaggery and pulse flour fixed and adjust the water to the area.

A simple quality check: a well-prepared Jivamrit has a mild sweet-fermenting smell, a slightly frothy surface and no foul odour. If it smells bad, discard and prepare again.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 The four pillars are Bijamrit, Jivamrit, Mulching and ____.

2 Jivamrit contains cow dung, cow urine, jaggery and ____ flour.

3 Bijamrit is a treatment applied to the ____.

4 Mulching reduces ____ from the soil surface.

5 Waaphasa is the balance of air and ____ in soil pores.

6 Over-irrigation destroys ____ by filling all pores with water.

B. State whether true or false

1 Jivamrit is a chemical fertiliser. (False - it is a microbial culture)

2 Mulching suppresses weeds by blocking light. (True)

3 Bijamrit is applied to the soil, not the seed. (False - it is a seed treatment)

4 Waaphasa means fully saturated soil. (False - it is a balance of air and moisture)

C. Match the following

Column A	Column B
Bijamrit	Seed treatment
Jivamrit	Living microbial soil culture
Mulching	Permanent soil cover
Waaphasa	Moisture-air balance
Ghana Jivamrit	Solid form of Jivamrit

D. Answer in one word or one line

1 Name the four pillars of natural farming.

2 Name four ingredients of Jivamrit.

3 What does Bijamrit mean?

4 Give two benefits of mulching.

Practical application in Tamil Nadu - preparing inputs locally
- Jivamrit drums are commonly set up near the cattle shed in natural farms across the state.
- Desi cow breeds such as Kangayam, Umbalachery and Pulikulam are valued for their dung and urine.
- Rice straw from the delta is used as mulch and in Jivamrit preparation.

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Lecture 6. Scope and Importance of Natural Farming

Learning objectives

- Explain the economic, environmental and social significance of natural farming.
- Assess its scope in Indian agriculture and its contribution to the Sustainable Development Goals.
- Identify the conditions under which natural farming is most suitable.

6.1 Introduction

Scope means the range of situations in which a practice can be applied and the benefits it can deliver. Importance means why it matters. Natural farming has a wide scope in India because most Indian farms are small, rain-fed or partially irrigated, and run by families who cannot absorb the rising cost of purchased inputs. It also matters globally because agriculture is both a victim of and a contributor to environmental degradation.

6.2 Economic significance

- **Lower cost of cultivation:** fertiliser and pesticide, usually the largest cash cost, are replaced by on-farm preparations.
- **Higher net income:** even where gross yield is similar, the reduction in cost raises net return per acre.
- **Price premium:** chemical-free and certified natural produce often fetches a higher price in niche and export markets.
- **Reduced risk:** the farmer is less exposed to input price shocks and to crop failure from input misuse.
- **Less debt:** reduced borrowing for inputs and reduced dependence on moneylenders.
- **Diversified income:** integration of crops, livestock, trees, honey, mushrooms and value-added products gives several income streams.
- **Employment:** input preparation, mulching, processing and local marketing create rural jobs.
- **Foreign exchange:** certified organic and natural produce has export potential.

6.3 Environmental significance

- **Soil health:** rising organic carbon, better structure, higher microbial and earthworm activity.
- **Water:** less chemical contamination of groundwater and surface water; better infiltration and recharge.
- **Biodiversity:** more pollinators, natural enemies and soil organisms; conservation of indigenous varieties.
- **Climate mitigation:** carbon sequestration in soil and lower greenhouse gas emissions.
- **Climate resilience:** soils with more organic matter withstand drought and heavy rain better.
- **Reduced pollution:** no pesticide drift, no nitrate leaching, no residue in food.

6.4 Social significance

- **Farmer health:** fewer pesticide-poisoning incidents and lower exposure to agrochemicals.
- **Consumer health:** safe, residue-free, often more nutritious food.

- **Food sovereignty:** farmers control their own seed and inputs instead of depending on the market.
- **Knowledge revival:** traditional and local knowledge becomes valuable again.
- **Women's participation:** input preparation, seed conservation and home gardens involve and empower women.
- **Community strength:** group farming, community seed banks and participatory certification build social capital.

6.5 Scope in India and Tamil Nadu

Situation	Suitability of natural farming
Small and marginal holdings	High - low cash cost suits limited capital.
Rain-fed and dryland areas	High - mulching and moisture conservation reduce drought risk.
Tribal and hilly regions	High - long tradition of organic practices and rich local seed.
Intensive irrigated paddy and sugarcane	Moderate - possible, but needs careful water and mulch management in the transition.
Plantation crops (coconut, mango, cashew)	High - mulching, cover crops and intercropping work well.
Horticulture and vegetables	High - premium market and quick returns from value addition.
Peri-urban and market-linked farms	High - direct and niche markets reward chemical-free produce.

6.6 Natural farming and the Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) are seventeen global goals for 2030. Natural farming contributes directly to several of them.

SDG	How natural farming contributes
SDG 1 - No poverty	Lower input cost and diversified income raise net farm income.
SDG 2 - Zero hunger	Stable, diverse, low-cost food production and better household nutrition.
SDG 3 - Good health	Residue-free food and reduced farmer exposure to pesticides.
SDG 6 - Clean water	Less chemical runoff and leaching; better infiltration and recharge.
SDG 8 - Decent work	Rural employment in input preparation, processing and marketing.
SDG 12 - Responsible consumption	Sustainable production and reduced chemical use.
SDG 13 - Climate action	Carbon sequestration and lower emissions; climate resilience.
SDG 15 - Life on land	Soil health, biodiversity and conservation of indigenous species.

6.7 Limitations and the honest picture

A balanced answer also notes the difficulties, which examiners appreciate.

- Yield may fall in the first one to three years of transition while the soil rebuilds its biology.
- The system is knowledge-intensive; wrong preparation or timing of inputs can reduce results.
- Mulch material may be scarce where crop residues are used for cattle feed or fuel.
- Certification and market linkage are still developing for natural farming produce.
- Large-scale evidence on long-term yield is still being built; claims should be treated with care.

WORKED EXAMPLE - Economics of scope: net return per acre

- Conventional paddy, one acre, one season: gross value of produce Rs. 45,000; input cost (fertiliser, pesticide, seed, labour) about Rs. 26,000; net return about Rs. 19,000.
- Natural farming paddy, one acre: gross value of produce Rs. 43,000 (slightly lower yield, small price premium); input cost (seed, some labour, on-farm inputs) about Rs. 11,000; net return about Rs. 32,000.
- Difference: net return is higher by about Rs. 13,000 per acre per season despite a lower gross value, because the input cost is much lower.
- Note: figures are illustrative for a teaching example; actual values depend on season, variety, price and location.

EXAM FOCUS

Key points to remember

- Scope means where natural farming can be applied; importance means why it matters.
- Economic: lower cost, higher net income, premium price, less debt, diversified income.
- Environmental: soil health, clean water, biodiversity, climate mitigation and resilience.
- Social: farmer and consumer health, food sovereignty, knowledge revival, women's participation.
- It fits small, rain-fed, tribal, plantation and horticulture farms best.
- It contributes to SDGs 1, 2, 3, 6, 8, 12, 13 and 15.
- Limitations include a transition yield dip, knowledge intensity, mulch scarcity and weak market linkage.

Short-answer questions with model answers

Q1. List any five Sustainable Development Goals supported by natural farming.

Model answer: SDG 1 No poverty, SDG 2 Zero hunger, SDG 3 Good health, SDG 6 Clean water, SDG 13 Climate action. SDG 8, 12 and 15 are also supported. Natural farming raises net income, produces safe food, protects water, and sequesters carbon in soil.

Q2. Why is natural farming especially suited to small and marginal farmers?

Model answer: Because its cash input cost is very low, which suits farmers with limited capital; it does not require borrowing for fertiliser and pesticide; it can use family labour and local materials; and it gives diversified output for home consumption and local sale, reducing risk.

Q3. State any three limitations of natural farming.

Model answer: Yield may decline for one to three years during the transition while soil biology is rebuilt; the system is knowledge-intensive and errors in preparing inputs reduce results; and mulch material and market or certification linkage may be limited. Long-term large-scale yield evidence is still being built.

Long-answer questions with model answer outlines

Q1. Discuss the economic, environmental and social significance of natural farming.

Outline:

- Introduction: define scope and importance.
- Economic significance: cost, net income, premium, debt, diversification, employment.
- Environmental significance: soil, water, biodiversity, climate mitigation and resilience.
- Social significance: health, food sovereignty, knowledge, women, community.
- Scope in India and Tamil Nadu with examples of suitable situations.
- Limitations for balance.
- Conclusion: natural farming is significant on all three fronts.

Q2. Explain the contribution of natural farming to the Sustainable Development Goals and its scope in India.

Outline:

- Introduction: the SDGs and agriculture's role.
- Goal-wise contribution: 1, 2, 3, 6, 8, 12, 13, 15 with one line each.
- Scope: small holdings, rain-fed areas, tribal regions, plantations, horticulture, peri-urban farms.
- Conditions needed: knowledge, mulch availability, market linkage.
- Conclusion: natural farming aligns production with the SDG agenda.

6.8 Natural farming and the Indian farmer's economy

India has a very large number of small and marginal holdings, and a large share of farm households carry debt. The economics of natural farming must be judged against this reality.

- Small and marginal holdings dominate Indian agriculture, so low-cost methods have wide relevance.
- Input cost, especially fertiliser and pesticide, is a large share of the cash cost for many crops.
- Removing this cost reduces the need to borrow at the start of the season.
- Diversified output (grain, milk, vegetables, fruit) improves household food security.
- A price premium, where available, further improves net return.
- The main constraint is the knowledge and labour needed, and the market linkage for the produce.

Note: precise national figures on indebtedness and holding size change with each agricultural census; the general pattern above is what matters for the examination.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Natural farming reduces the ____ of cultivation.

2 Chemical-free produce often earns a price ____.

3 Natural farming contributes to SDG ____ (zero hunger).

4 Natural farming is well suited to ____ and marginal holdings.

5 A limitation of natural farming is a possible ____ dip during transition.

6 Natural farming protects ____ by reducing chemical runoff.

B. State whether true or false

1 Natural farming always gives higher yield than conventional farming in the first year. (False - yield may dip initially)

2 Natural farming can earn a premium price in niche markets. (True)

3 Natural farming has no social benefits. (False - it improves health and food sovereignty)

4 Natural farming reduces the farmer's debt burden. (True)

C. Match the following

Column A	Column B
SDG 2	Zero hunger
SDG 6	Clean water
SDG 13	Climate action
SDG 15	Life on land
SDG 3	Good health

D. Answer in one word or one line

1 Name three economic benefits of natural farming.

2 Name two environmental benefits.

3 Which farms are best suited to natural farming?

4 State one limitation of natural farming.

Practical application in Tamil Nadu - scope in the state

- Rain-fed millet and pulse areas of the southern districts suit natural farming well.
- Hill and tribal areas of the Western and Eastern Ghats have a long organic tradition.
- Plantation crops such as coconut, mango and cashew benefit from mulching and intercropping.

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Lecture 7. Methods and Schools of Natural Farming

Learning objectives

- Describe the major schools of natural farming and their unique principles.
- Compare Fukuoka natural farming, ZBNF, Rishi Krishi and Korean Natural Farming.
- Recognise the practices shared by all schools and those that differ.

7.1 Introduction

Several schools of natural farming have developed in different countries. They share the core idea of working with nature and avoiding synthetic chemicals, but they differ in their specific techniques, their attitude to external organic inputs and their philosophical base. This lecture compares the four main schools named in the syllabus.

7.2 Fukuoka natural farming (Japan)

Developed by Masanobu Fukuoka on his farm in Japan, this is the most radical of the schools. It is also called 'Do-nothing' farming or the natural way of farming.

- **No tillage:** the soil is never ploughed.
- **No chemical fertiliser and no prepared organic fertiliser:** fertility comes from returning straw and from a green manure crop such as clover grown with the main crop.
- **No weeding by tillage or herbicide:** weeds are managed by timing, a clover ground cover and temporary flooding in paddy.
- **No pesticides:** the balanced ecosystem keeps pests in check.
- **Seed balls:** seeds are enclosed in clay balls and broadcast, so no nursery or transplanting is needed.
- **Philosophy:** the farm should be as close to nature as possible; human effort is reduced to observation and minimal timely intervention.

7.3 Subhash Palekar's Zero Budget Natural Farming (India)

ZBNF is the most widely adopted school in India. It grew from Palekar's observation that natural forests are fertile without external inputs.

- **Four pillars:** Bijamrit (seed treatment), Jivamrit (microbial soil inoculum), Mulching and Waaphasa (soil moisture-air balance).
- **Desi cow based:** the dung and urine of the indigenous cow are considered essential, because of their rich microbial load.
- **Zero external cash input:** the aim is to eliminate purchased fertiliser and pesticide.
- **No tillage and permanent mulch:** soil is kept covered and undisturbed.
- **Crop diversity and multi-storey systems:** mixed and layered cropping is encouraged.
- **Farmer-to-farmer training:** knowledge spreads through farmer training camps rather than formal institutions.

7.4 Rishi Krishi (India)

Rishi Krishi, or 'farming of the sages', is a school based on ancient Indian texts and the traditional practices of rishis. It emphasises the sacred and holistic dimension of farming.

- **Rooted in Vedic and Puranic texts:** practices are drawn from Krishi-Parashara, Vrikshayurveda and traditional wisdom.
- **Amritpani and Amritjal:** liquid preparations of cow dung, cow urine, jaggery and gram flour used as soil and foliar inputs.
- **Cow-centred:** the indigenous cow is central; panchagavya and other preparations are used.
- **Spiritual and ritual element:** the farm is treated as sacred; timings may follow traditional calendars.
- **No chemical inputs and no deep tillage;** emphasis on soil health and the farmer's ethical relationship with the land.
- **Relevance:** it is closest to India's heritage and is often grouped with natural farming under traditional practices.

7.5 Korean Natural Farming (KNF)

Korean Natural Farming was developed by Cho Han-Kyu in South Korea. It is a highly technical system built around cultivating and using local microorganisms.

- **Indigenous Micro-Organisms (IMO):** collected from a forest floor, multiplied in cooked rice and then in a larger substrate, and applied to the soil.
- **Oriented to local organisms:** the aim is to strengthen the microbes already adapted to the local environment.
- **Fermented plant and fruit extracts:** preparations such as FPJ (Fermented Plant Juice) and FFJ (Fermented Fruit Juice) supply nutrition.
- **Mineral and fish preparations:** e.g. Oriental Herbal Nutrient (OHN), water-soluble calcium and phosphorus preparations.
- **Lactic acid bacteria and yeast:** used in preparations for decomposition and soil conditioning.
- **Systematic and measurable:** it uses specific recipes and stages, unlike the more philosophical Fukuoka approach.
- **No synthetic chemicals:** the entire input set is made from local natural materials.

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7.6 Comparison of the schools

Basis	Fukuoka natural	ZBNF (Palekar)	Rishi Krishi	Korean Natural Farming
Origin	Japan	India (Maharashtra)	India (traditional/Vedic)	South Korea
Founder	Masanobu Fukuoka	Subhash Palekar	Based on ancient texts and gurus	Cho Han-Kyu
Central idea	Do-nothing; let nature work	Four pillars; desi cow based	Farming of the sages; sacred cow	Cultivate local microorganisms
Tillage	None	None	None to minimal	Minimal
Main inputs	Straw, clover, seed balls	Bijamrit, Jivamrit, mulch, Waaphasa	Amritpani, Amritjal, panchagavya	IMO, FPJ, FFJ, OHN, LAB
External organic input	Discouraged	Minimised	Minimised	Minimised; local materials only
Cow dependency	Not central	Essential (desi cow)	Essential (desi cow)	Not central; uses local resources
Signature practice	Seed balls, clover ground cover	Jivamrit and mulching	Amritpani and ritual calendar	IMO collection from forest
Philosophical base	Very strong; minimal intervention	Practical; farmer training	Spiritual and Vedic	Technical and systematic

7.7 Common ground among the schools

Despite their differences, all four schools agree on the essentials, and these are the points to write in an examination.

- No synthetic chemical fertiliser and no synthetic pesticide.
- Soil life is the foundation of fertility; keep it alive.
- Soil should be covered and protected, not left bare.
- Crop diversity is preferred over monoculture.
- Inputs should be made from local, on-farm materials.
- The farmer should be self-reliant rather than dependent on the market.
- Long-term soil and ecosystem health is the goal.

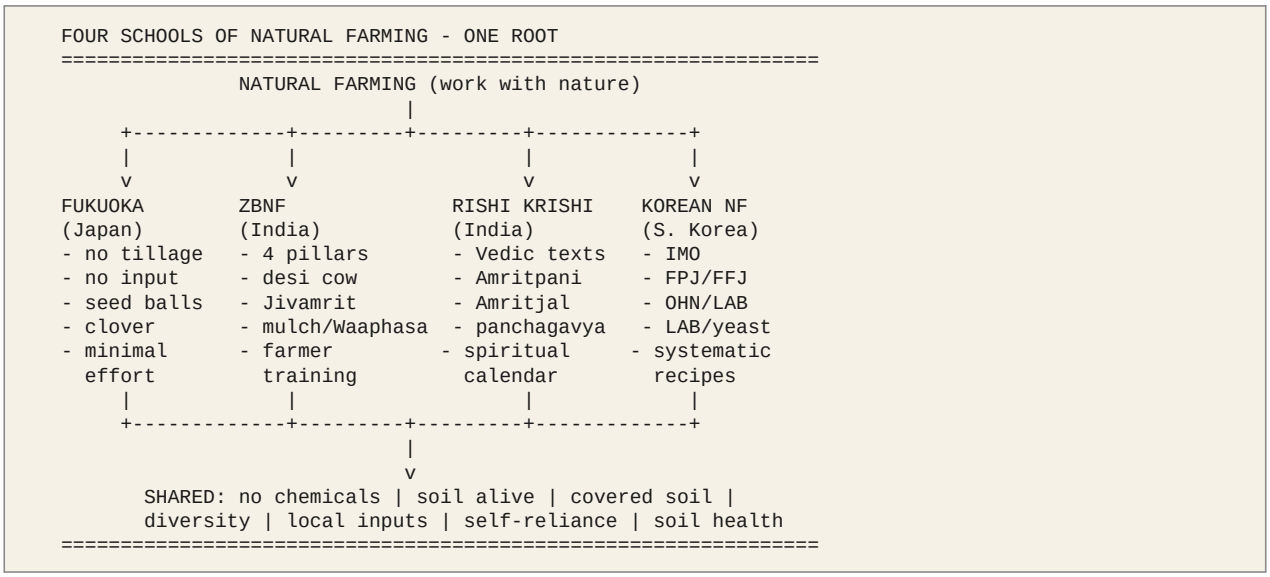


Fig. 7.1 The four schools and their shared principles.

WORKED EXAMPLE - Choosing a school for a Tamil Nadu farm

- A dryland farmer in Virudhunagar with a few desi cows and limited capital: ZBNF suits best - Jivamrit and mulching use the cattle and crop residues available on the farm.
- A traditional paddy grower near Kumbakonam interested in the least possible labour: Fukuoka-style no-till paddy with clover and straw return is an option, though it needs careful water management.
- A farmer with a strong cultural and religious orientation and access to panchagavya: Rishi Krishi fits the farm's ethos.
- A horticulture grower willing to learn precise recipes: Korean Natural Farming's IMO and FPJ preparations give a structured input programme.
- Note: most Indian farmers mix methods; the four pillars of ZBNF are the most common practical base.

EXAM FOCUS

Key points to remember

- Four main schools: Fukuoka natural farming, ZBNF, Rishi Krishi and Korean Natural Farming.
- Fukuoka: no tillage, no input, no weeding by tillage, no pesticides, seed balls, clover cover, minimal intervention.
- ZBNF: four pillars, desi cow based, zero external cash input, farmer-to-farmer training.
- Rishi Krishi: Vedic base, Amritpani and Amritjal, panchagavya, sacred-cow-centred, spiritual calendar.
- Korean Natural Farming: Indigenous Micro-Organisms, FPJ, FFJ, OHN, LAB, systematic recipes.
- All schools share no chemicals, live soil, covered soil, diversity, local inputs and self-reliance.
- Indian farmers often combine methods, usually around the ZBNF pillars.

Short-answer questions with model answers

Q1. What is Korean Natural Farming and what is its signature technique?

Model answer: Korean Natural Farming, developed by Cho Han-Kyu in South Korea, is a technical system built on cultivating and using local microorganisms. Its signature technique is the collection and multiplication of Indigenous Micro-Organisms (IMO) from a forest floor, along with fermented plant and fruit juice preparations, Oriental Herbal Nutrient and lactic acid bacteria.

Q2. State the four principles of Fukuoka's natural farming.

Model answer: No tillage; no chemical fertiliser and no prepared organic fertiliser; no weeding by tillage or herbicide; and no dependence on pesticides. Fertility comes from returning straw and a clover ground cover, and weeds are managed by timing and ground cover rather than by cultivation.

Q3. Name two preparations used in Rishi Krishi.

Model answer: Amritpani and Amritjal - liquid preparations of cow dung, cow urine, jaggery and gram flour used as soil and foliar inputs. Rishi Krishi also uses panchagavya, a preparation of five cow products.

Long-answer questions with model answer outlines

Q1. Describe the four main schools of natural farming and compare their principles.

Outline:

- Fukuoka natural farming: origin, four principles, seed balls, clover, philosophy.
- ZBNF: origin, four pillars, desi cow base, zero external cash input, farmer training.
- Rishi Krishi: Vedic base, Amritpani/Amritjal/panchagavya, sacred cow, spiritual calendar.
- Korean Natural Farming: IMO, FPJ, FFJ, OHN, LAB, systematic recipes.
- Comparison table on origin, founder, tillage, inputs, cow dependency, signature practice.

- Common ground and conclusion.

Q2. Explain ZBNF and Korean Natural Farming in detail and state how they differ.

Outline:

- ZBNF: Palekar, four pillars, desi cow, mulching, Waaphasa, no tillage, zero budget.
- KNF: Cho Han-Kyu, IMO collection, FPJ/FFJ, OHN, LAB, structured recipes.
- Differences: cow dependency, philosophical vs technical, input recipes, method of microbial cultivation.
- Similarities: no chemicals, local inputs, soil biology, self-reliance.
- Conclusion: both are natural farming schools with different techniques.

7.8 Convergence of schools and hybrid practice

In practice, farmers rarely follow one school strictly. They combine methods to suit their resources, climate and market. This convergence is a strength.

- A farmer may use ZBNF's Jivamrit and mulching with Korean Natural Farming's IMO for a horticulture crop.
- Panchagavya from the Rishi Krishi tradition is used widely even by ZBNF farmers.
- Fukuoka's no-till and cover-crop ideas combine well with ZBNF's four pillars.
- Botanical sprays from traditional practice are used across all schools.
- The common thread - no chemicals, live soil, diversity and local inputs - remains the same.

For the examination, describe the schools separately but note that farmers often combine them, which shows a practical understanding of the subject.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Fukuoka's method is also called ____ farming.

2 ZBNF stands for ____.

3 Rishi Krishi uses preparations called Amritpani and ____.

4 Korean Natural Farming is built on ____ Micro-Organisms.

5 FPJ stands for Fermented ____ Juice.

6 Panchagavya is a preparation of ____ cow products.

B. State whether true or false

1 Korean Natural Farming was developed in Japan. (False - in South Korea, by Cho Han-Kyu)

2 Rishi Krishi is based on ancient Indian texts. (True)

3 Fukuoka's method uses a clover ground cover. (True)

4 ZBNF does not use cow dung. (False - desi cow dung is essential)

C. Match the following

Column A	Column B
Fukuoka	Seed balls and clover cover
ZBNF	Four pillars
Rishi Krishi	Amritpani and panchagavya
Korean Natural Farming	Indigenous Micro-Organisms
Palekar	Founder of ZBNF

D. Answer in one word or one line

1 Name the four schools of natural farming.

2 What is a seed ball?

3 Name two preparations used in Rishi Krishi.

4 What is IMO in Korean Natural Farming?

Practical application in Tamil Nadu - schools in practice

- ZBNF is the most widely practised school in the state because of its simple four-pillar method.
- Panchagavya is widely used by organic farmers in Tamil Nadu.
- Korean Natural Farming methods are adopted by some horticulture growers for precise input programmes.

Lecture 8. Characteristics and Design of a Natural Farm

Learning objectives

- Describe how a natural farm is structured and laid out.
- Explain crop diversity, livestock integration and natural resource use.
- Design a simple natural farm layout for a small holding.

8.1 Introduction

A natural farm is not simply a field where chemicals are avoided. It is designed as an ecosystem, with diversity of plants and animals, a permanent soil cover, water conservation structures and internal nutrient recycling. Good design reduces the work needed to manage the farm and increases its stability.

8.2 Characteristics of a natural farm

- **Diversity of crops:** cereals, pulses, oilseeds, vegetables, fodder and trees grown together or in rotation.
- **Integration of livestock:** cattle, goats, poultry or fish provide dung, urine, draught power and income.
- **Permanent soil cover:** mulch or live cover on the cropped area and on bunds.
- **No chemical input:** all nutrition and protection from on-farm and local materials.
- **Water conservation structures:** farm ponds, bunds, trenches, recharge pits and drip or sprinkler where feasible.
- **Tree component:** fruit trees, timber trees, fodder trees and nitrogen-fixing trees on boundaries and as intercrops.
- **Seed and input self-reliance:** own seed and own preparations.
- **Habitat for biodiversity:** flowering plants, bund vegetation and undisturbed patches for pollinators and natural enemies.
- **Internal nutrient cycling:** residues, dung and crop by-products returned to the soil.
- **Low external dependence and family-labour based operation.**

8.3 Design principles

Design principle	What it means in practice
Mimic nature	Use the structure of a natural forest: ground cover, shrubs, trees and animals together.
Maximise diversity	Grow many species and varieties to spread risk and support beneficial organisms.
Close the nutrient loop	Return all crop residues, dung and urine to the soil; minimise export of nutrients.
Use vertical space	Multi-storey cropping with tall trees, medium shrubs and ground crops.
Conserve water	Harvest rainfall, mulch heavily and use efficient irrigation.
Use local resources	Prefer local seed, local cow breeds and locally available organic material.
Plan for resilience	Spread crops and income streams so that one failure does not sink the farm.
Integrate components	Crop, livestock and tree components should support each other, not compete.

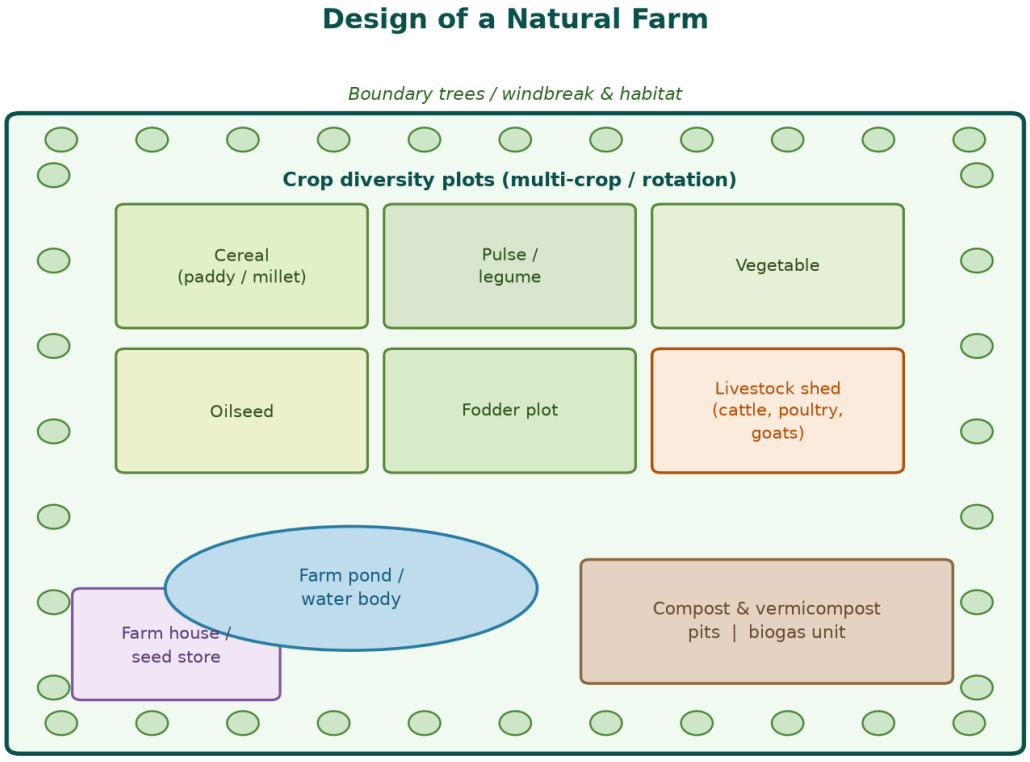


Figure 8.1 Design of a natural farm

Crop diversity, livestock, trees, water bodies and composting areas in one integrated layout.

8.4 Crop diversity and cropping systems

Diversity is the backbone of a natural farm. It is achieved through several cropping systems used at the same time.

Cropping system	Description	Benefit in a natural farm
Mixed cropping	Two or more crops grown together in the same field.	Spreads risk, uses different soil layers, confuses pests.
Intercropping	A main crop with one or more intercrops in rows.	Extra income and nitrogen fixation if a legume is included.
Crop rotation	Different crops in sequence over seasons.	Breaks pest and disease cycles; balances nutrients.
Multi-storey cropping	Layers of trees, shrubs and ground crops.	Uses vertical space; mimics a forest; high total biomass.
Strip cropping	Alternate strips of different crops.	Reduces erosion and pest spread.
Cover cropping / green manure	Legumes grown to cover soil and then incorporated or mulched.	Adds nitrogen and organic matter; live mulch.

8.5 Livestock integration

Livestock convert crop residues and fodder into dung, urine, milk and draught power, and they close the nutrient loop on the farm.

- **Desi cattle:** source of dung and urine for Bijamrit and Jivamrit, draught power and farmyard manure; their dung is preferred in natural farming for its microbial richness.
- **Goats and sheep:** convert shrubs and crop residues into manure and provide income.
- **Poultry:** eat insects and weed seeds, provide manure and eggs, and reduce pest pressure.
- **Fish and ducks:** in farm ponds, they use farm waste and add nutrients to pond silt.
- **Bee keeping:** pollination service and honey income; strongly supported by crop diversity.

8.6 Layout of a natural farm

A well-designed natural farm arranges its components so that they support one another. The text diagram below shows a typical layout for a small holding.

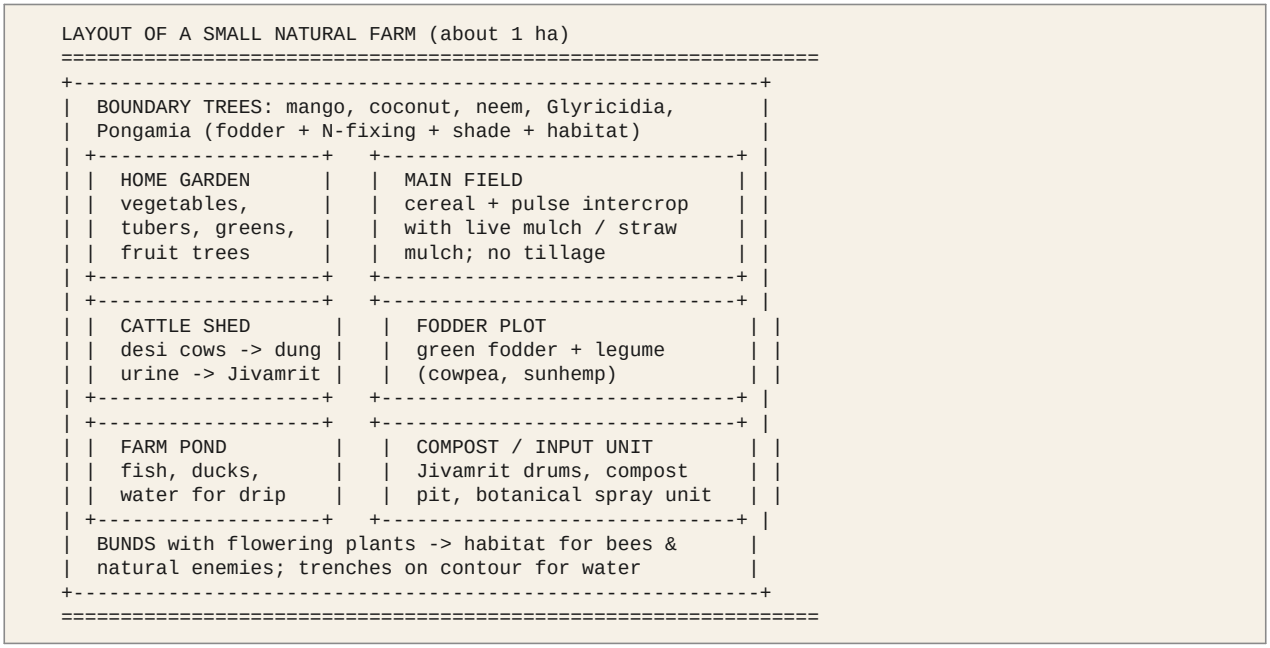


Fig. 8.1 A designed natural farm: fields, home garden, cattle, fodder, pond, input unit and boundary trees.

8.7 Natural resource use on a natural farm

Resource	How a natural farm uses it
Soil	Kept covered and undisturbed; fed with mulch and Jivamrit; protected from erosion.
Water	Rainfall harvested in ponds and trenches; mulch reduces evaporation; drip or sprinkler used sparingly.
Crop residues	Used as mulch, compost and cattle feed - not burnt.
Cattle dung and urine	Converted into Jivamrit, Bijamrit and compost.
Sunlight	Captured by multi-storey cropping and used for drying and solar equipment.
Trees	Provide fodder, fuel, fruit, shade, nitrogen fixation and habitat.
Local biodiversity	Natural enemies, pollinators and soil organisms are treated as free farm labour.

WORKED EXAMPLE - Designing a one-hectare natural farm

- Divide the holding: about 0.6 ha main field, 0.15 ha home garden, 0.1 ha fodder plot, 0.05 ha farm pond, 0.1 ha input and cattle area and boundary trees.
- Main field: cereal + legume intercrop with straw mulch; rotate with a different family each season.
- Cattle: keep two or three desi cows; their dung and urine supply all Bijamrit and Jivamrit needs.
- Water: dig a farm pond to harvest runoff; use it to irrigate the home garden and fodder plot by drip.
- Inputs: set up three or four Jivamrit drums and a compost pit near the cattle shed.
- Result: the farm produces grain, pulses, vegetables, fodder, milk and fruit, and recycles nearly all its nutrients internally.

EXAM FOCUS

Key points to remember

- A natural farm is designed as an ecosystem, not just a chemical-free field.
- Key characteristics: diversity, livestock integration, permanent cover, no chemicals, water structures, trees, seed self-reliance and biodiversity habitat.
- Design principles: mimic nature, maximise diversity, close the nutrient loop, use vertical space, conserve water, use local resources, plan for resilience.
- Cropping systems used together: mixed, inter, rotation, multi-storey, strip and cover cropping.
- Livestock - especially desi cattle - supply dung and urine for inputs and close the nutrient loop.
- A good layout arranges field, garden, cattle, fodder, pond, input unit and boundary trees to support one another.
- All natural resources are used in a way that recycles and conserves.

Short-answer questions with model answers

Q1. What are the main characteristics of a natural farm?

Model answer: Crop diversity, integration of livestock, permanent soil cover with mulch, no chemical input, water conservation structures, a tree component, seed and input self-reliance, habitat for biodiversity and internal nutrient cycling, all operated with low external dependence.

Q2. Why is livestock integration important in a natural farm?

Model answer: Livestock, especially desi cattle, supply the dung and urine needed for Bijamrit and Jivamrit, provide draught power and farmyard manure, convert crop residues and fodder into useful products, and close the nutrient loop by returning nutrients to the soil. They also add income and reduce risk.

Q3. Name any four cropping systems used in a natural farm.

Model answer: Mixed cropping, intercropping, crop rotation and multi-storey cropping. Strip cropping and cover cropping or green manuring are further systems used.

Long-answer questions with model answer outlines

Q1. Describe the characteristics and design of a natural farm with a suitable layout.

Outline:

- Introduction: a natural farm is a designed ecosystem.
- Characteristics: diversity, livestock, cover, no chemicals, water structures, trees, self-reliance, habitat, recycling.
- Design principles: mimic nature, diversity, closed nutrient loop, vertical space, water conservation, local resources, resilience.
- Layout: field, home garden, cattle shed, fodder plot, farm pond, input unit, boundary trees and bund vegetation.
- Natural resource use.
- Conclusion: good design reduces work and increases stability.

Q2. Explain crop diversity and livestock integration in the design of a natural farm.

Outline:

- Crop diversity: cropping systems (mixed, inter, rotation, multi-storey, strip, cover), benefits.
- Livestock integration: desi cattle, goats, poultry, fish, ducks, bees; roles and products.
- How they support each other: residues to cattle, dung to soil, fodder from farm, pest control by poultry.
- Nutrient loop closure and risk reduction.
- Conclusion: diversity plus livestock is the core of natural farm design.

8.8 Common mistakes in farm design

- **Too many animals for the available fodder:** the farm then has to buy feed, defeating self-reliance.
- **Monoculture despite the design:** a single crop ignores the diversity principle.
- **No water storage:** without a pond or recharge structure, the farm cannot buffer a dry spell.
- **Leaving the soil bare:** mulch must cover the soil, including bunds and paths.
- **Placing the cattle shed far from the field:** increases the labour of carrying dung and urine.
- **Ignoring boundary trees:** trees provide fodder, fuel, shade, nitrogen and habitat at little cost.
- **No input unit:** without drums and a compost pit, on-farm input production is difficult.
- **Planning for the short term only:** trees and soil health need a multi-year view.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 A natural farm integrates crops, ____ and animals.

2 Multi-storey cropping mimics the structure of a ____.

3 Desi cattle supply dung and ____ for natural inputs.

4 A farm ____ harvests runoff for supplementary irrigation.

5 Nitrogen-fixing trees such as ____ provide fodder and fix nitrogen.

6 Poultry help control ____ and weeds on the farm.

B. State whether true or false

1 A natural farm should be a monoculture for efficiency. (False - diversity is the backbone)

2 Boundary trees provide fodder, fuel and habitat. (True)

3 Livestock integration helps close the nutrient loop. (True)

4 Water conservation structures are not needed in natural farms. (False - they are important)

C. Match the following

Column A	Column B
Home garden	Vegetables, tubers and fruit trees
Cattle shed	Source of dung and urine
Farm pond	Harvests runoff and supports fish
Input unit	Jivamrit drums and compost pit
Boundary trees	Fodder, fuel, shade and habitat

D. Answer in one word or one line

1 Name three components of a natural farm layout.

2 Why is crop diversity important?

3 Name two nitrogen-fixing trees used in natural farms.

4 What is the role of a farm pond?

Practical application in Tamil Nadu - designing a natural farm

- Homesteads in the state commonly follow a multi-storey pattern with coconut, banana, spices and vegetables.
- Farm ponds are promoted under watershed development and irrigation schemes.
- Boundary planting with Glyricidia and Pongamia is common for fodder and nitrogen.

Lecture 9. Mid Semester Examination

This lecture is the mid-semester examination of the course. It is placed here in the syllabus to mark the point at which the first half of the course - Lectures 1 to 8 - is assessed. The examination covers the heritage, history, concept, principles, pillars, scope, schools and design of natural farming. The purpose of this chapter is to consolidate revision for that examination.

9.1 Coverage of the mid-semester examination

The mid-semester examination tests the following units.

Unit	Lecture	Weightage to prepare
Indian heritage of ancient agriculture	1	Definitions, examples, sources
History and evolution of natural farming	2	Timeline, key persons, movements
Concept, definition and objectives	3	Definition, comparison, objectives
Characteristics and principles	4	Principles, natural processes
Main pillars	5	Composition and function of four pillars
Scope and importance	6	Economic, environmental, social, SDGs
Methods and schools	7	Four schools and comparison
Design of a natural farm	8	Characteristics, layout, integration

9.2 Revision summary - the first half in one page

- **Heritage:** Indian agriculture is about ten thousand years old; cattle-based fertility, mixed cropping, tank irrigation and botanical pest control are its core practices; sources include Krishi-Parashara, the Arthashastra and Vrikshayurveda.
- **History:** global organic movement (Howard, Steiner, Balfour, Rodale), Fukuoka's natural farming, Korean Natural Farming, then Palekar's ZBNF; the Green Revolution raised output but degraded resources.
- **Concept:** natural farming works with nature, avoids synthetic inputs and soil disturbance, and relies on soil biology; it differs from organic farming in minimising external inputs and tillage.
- **Objectives:** ecological (soil, water, biodiversity, climate) and socio-economic (cost, income, health, self-reliance).
- **Principles:** no tillage, no chemical input, minimum disturbance, permanent cover, Waaphasa, reliance on natural processes, diversity, self-reliance.
- **Pillars:** Bijamrit (seed treatment), Jivamrit (microbial soil inoculum), Mulching (cover) and Waaphasa (moisture-air balance).
- **Scope:** best suited to small, rain-fed, tribal, plantation and horticulture farms; supports several SDGs.
- **Schools:** Fukuoka, ZBNF, Rishi Krishi and Korean Natural Farming; all avoid chemicals and rely on soil biology.
- **Design:** diversity, livestock integration, permanent cover, water structures, trees and internal nutrient recycling.

9.3 Model question paper

The following model paper follows the usual TNAU pattern for a mid-semester examination in a theory course. Marks and duration are indicative.

Model question paper (indicative)

- Time: 1.5 hours. Maximum marks: 50. Answer all questions in Part A, any two in Part B and any one in Part C.

- PART A - Objective and very short answer (10 x 1 = 10 marks):

- 1. Name the four pillars of natural farming. 2. Who wrote 'The One-Straw Revolution'? 3. Expand ZBNF. 4. What is Waaphasa? 5. Name two traditional water-harvesting systems of Tamil Nadu. 6. What is Bijamrit? 7. Name one nitrogen-fixing tree used in natural farms. 8. What is IMO in Korean Natural Farming? 9. Define mulching. 10. Name one SDG supported by natural farming.

- PART B - Short answer (2 x 5 = 10 marks):

- 11. Differentiate natural farming from organic farming. 12. Explain the role of cow dung and cow urine in natural farming. 13. Describe any two cropping systems used in a natural farm. 14. Write short notes on Rishi Krishi.

- PART C - Long answer (1 x 15 = 15 marks):

- 15. Explain the four pillars of natural farming with their composition and function. 16. Describe the history and evolution of natural farming in India. 17. Discuss the design of a natural farm with a suitable layout.

- PART D - Application (1 x 15 = 15 marks):

- 18. Prepare a plan for converting a one-hectare chemically farmed holding in your district into a natural farm.

9.4 Examination technique

- Read the whole paper first and budget time in proportion to marks.
- For comparison questions, always draw a table - it earns marks quickly and clearly.
- For 'explain with composition' questions, give ingredients, quantities and function.
- Use Indian and Tamil Nadu examples wherever possible; they show application.
- Draw a simple text diagram (pillars, principles wheel, farm layout) for long answers on those topics.
- Define the key term in the first line of every answer.
- Leave time to re-read; correct spelling of technical terms such as Jivamrit, Bijamrit and Waaphasa.

EXAM FOCUS

Key points to remember

- The mid-semester examination covers Lectures 1 to 8.
- Expect a table-based comparison of natural, organic and conventional farming.
- The four pillars and their composition are almost certain to appear.
- Be ready to name key persons: Fukuoka, Palekar, Howard, Cho Han-Kyu.
- Know the ZBNF preparations and their ingredients.
- Be able to sketch a natural farm layout and a principles wheel.
- Use Indian and Tamil Nadu examples in every long answer.

Short-answer questions with model answers

Q1. How should a comparison question be answered?

Model answer: By drawing a table with a 'basis' column and one column for each system being compared, then filling rows such as chemical input, tillage, mulching, source of nutrition, input cost and certification. A table earns marks quickly and shows clarity.

Q2. Which topics are most likely to appear as long answers in the mid-semester examination?

Model answer: The four pillars of natural farming with composition and function; the history and evolution of natural farming; the characteristics and design of a natural farm with a layout; and the comparison of natural, organic and conventional farming.

Q3. Why should Indian examples be used in answers?

Model answer: Because they demonstrate application of the theory to real conditions, which examiners reward. Examples such as Palekar's ZBNF, Andhra Pradesh community-managed natural farming, Tamil Nadu's eri tanks and PGS-India certification show that the student can connect the concept to practice.

Long-answer questions with model answer outlines

Q1. Write a complete answer on the main pillars of natural farming.

Outline:

- Introduction: name the four pillars and their origin in ZBNF.
- Bijamrit: definition, composition, method, function.
- Jivamrit: definition, composition, method, function.
- Mulching: types and functions.
- Waaphasa: definition, importance, how maintained.
- How the pillars work together; supporting practices.
- Conclusion.

Q2. Describe how you would convert a chemically farmed one-hectare holding into a natural farm.

Outline:

- Baseline survey: soil test, water source, cattle, crops, residue availability.
- Plan components: main field, home garden, fodder, pond, cattle, input unit, boundary trees.
- Year 1: introduce mulching, Jivamrit and Bijamrit; stop pesticide first, then reduce fertiliser gradually.
- Year 2-3: build diversity, add trees and livestock, strengthen water structures.
- Inputs: prepare Jivamrit, Bijamrit, botanical sprays; conserve own seed.
- Monitoring: soil organic carbon, yield, cost, net income.
- Conclusion: transition is gradual and the farm becomes self-reliant.

Lecture 10. Ecological Balance, Ecological Engineering and Community Responsibility

Learning objectives

- Explain ecological balance and how natural farming maintains it.
- Define ecological engineering and describe its tools in a natural farm.
- Describe the collective community role in sustainability and biodiversity conservation.

10.1 Ecological balance

Ecological balance is the state in which the living and non-living parts of an ecosystem are in stable relationship, so that no single species or process runs out of control. In a farm, balance means that pests do not multiply without limit because predators keep them down, that nutrients are recycled rather than lost, and that the soil, plants, animals and microorganisms support one another. Natural farming seeks this balance deliberately; conventional farming, by killing pests with broad-spectrum poisons and by adding nutrients artificially, tends to disturb it.

10.2 How natural farming maintains ecological balance

Mechanism	How balance is maintained
Predator-prey relationship	Natural enemies (ladybird beetles, spiders, lacewings, predatory bugs, birds) keep pest numbers low.
Parasitism and pathogens	Parasitoids such as Trichogramma and entomopathogens such as Beauveria and Metarhizium control pests.
Competition	Diverse plant cover competes with weeds and reduces their dominance.
Nutrient cycling	Decomposers convert residues into plant nutrients, preventing loss and buildup.
Pollination	Bees, butterflies and other pollinators sustain seed and fruit set.
Soil food web	Bacteria, fungi, protozoa, nematodes and earthworms regulate each other and release nutrients.
Habitat and refuge	Undisturbed patches, bund vegetation and trees shelter beneficial organisms.

10.3 Ecological engineering

Ecological engineering means designing the farm so that the environment itself supports natural pest control and pollination. Instead of relying on a pesticide, the farmer manipulates the habitat so that natural enemies thrive. This is the modern, scientific form of the traditional practice of leaving refuge vegetation.

10.3.1 Tools of ecological engineering

- **Habitat manipulation:** growing flowering plants on bunds and borders to provide nectar and pollen to adult parasitoids and predators.
- **Banker plants:** plants that carry alternative prey or hosts, on which natural enemies build up before the pest arrives.
- **Trap crops:** attractive plants that draw pests away from the main crop and are then destroyed.
- **Border and refuge vegetation:** strips of grass, shrubs and trees that shelter enemies and pollinators.
- **Beetle banks:** raised grassy strips that shelter ground beetles and spiders.
- **Bird perches and nest boxes:** encourage insect-eating birds.

- **Sacred groves and undisturbed patches:** preserve the local gene pool of beneficial organisms.
- **Water bodies:** farm ponds support dragonflies, frogs and other predators.

10.3.2 Examples of plants used

Plant	Role in ecological engineering
Marigold (Tagetes)	Trap crop for nematodes; nectar for beneficial insects.
Sunflower and cosmos	Nectar and pollen for parasitoids and predators.
Castor	Trap crop for Spodoptera; host for some parasitoids.
Cowpea and other legumes	Nitrogen fixation and cover; support predatory mites.
Glyricidia and Pongamia	Nitrogen fixation, fodder, shade, perching sites for birds.
Tulsi, nochi and other herbs	Botanical source and aromatic pest repellent.
Sunhemp (Crotalaria)	Green manure and trap crop for nematodes.

10.4 Community responsibility

Ecological balance cannot be maintained by one farmer alone, because pests, water and biodiversity cross farm boundaries. Community action is therefore essential.

- **Group farming and clusters:** neighbouring farmers adopting natural farming together prevent re-infestation from chemically farmed fields and enable collective marketing.
- **Community seed banks:** conserve indigenous varieties and ensure seed availability for all members.
- **Community water management:** the traditional Kudimaramathu model of joint tank and channel maintenance.
- **Participatory certification:** the PGS-India system is built on peer inspection by farmer groups.
- **Joint pest and disease monitoring:** early warning and coordinated action across a village.
- **Biodiversity conservation:** protecting sacred groves, village trees and common lands as reservoirs of beneficial species.
- **Knowledge sharing:** farmer-to-farmer training, field days and demonstration plots.
- **Collective bargaining:** groups get better prices and can meet the volume demanded by buyers.

10.5 Biodiversity conservation as a community duty

Biodiversity is the variety of living things - crops, livestock, trees, insects, microbes and wildlife. Natural farming depends on it and also protects it.

Level of biodiversity	Examples on a natural farm	Community role
Genetic diversity	Land races and indigenous varieties of paddy, millets, pulses, vegetables	Seed banks, seed exchanges, conservation of land races
Species diversity	Many crops, trees, livestock breeds, bees, natural enemies	Avoid monoculture belts; maintain common land
Ecosystem diversity	Fields, ponds, bunds, groves, homesteads	Protect tanks, groves and village forests

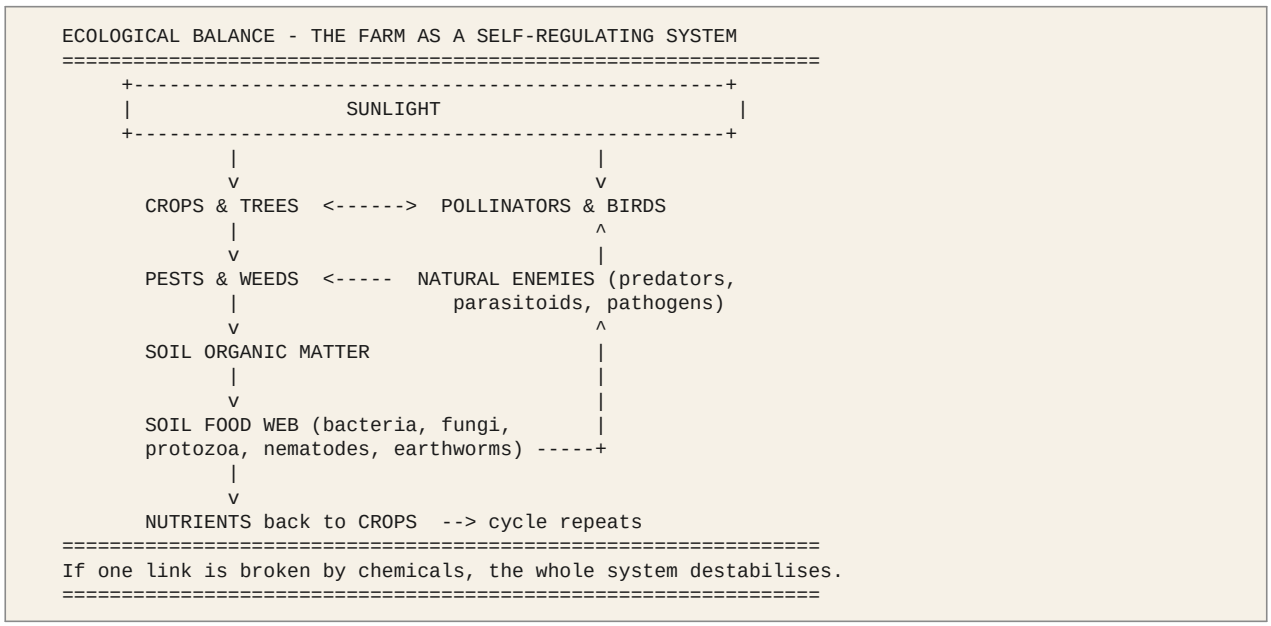


Fig. 10.1 Ecological balance as a closed, self-regulating cycle on the farm.

WORKED EXAMPLE - Ecological engineering on one acre of cotton

- Border: two rows of castor as a trap crop for Spodoptera, and marigold as a trap for nematodes and nectar source.
- Bunds: flowering plants such as sunflower and cosmos to feed adult parasitoids; grassy strips as beetle banks.
- Intercrop: cowpea as a cover and nitrogen source, supporting predatory mites.
- Bird perches: 10 perches per acre to attract insectivorous birds.
- Result: pest pressure is reduced so that a botanical spray is needed only if the pest crosses the action threshold, not on a fixed calendar.

EXAM FOCUS

Key points to remember

- Ecological balance is the stable relationship among the living and non-living parts of the farm ecosystem.
- Natural farming maintains balance through predators, parasitoids, competition, nutrient cycling, pollination and the soil food web.
- Ecological engineering designs the habitat so that natural enemies and pollinators thrive.
- Tools include habitat manipulation, banker plants, trap crops, border vegetation, beetle banks, bird perches, ponds and undisturbed patches.
- Community responsibility is essential because pests, water and biodiversity cross farm boundaries.
- Community tools: group farming, seed banks, joint water management, participatory certification and knowledge sharing.
- Biodiversity operates at genetic, species and ecosystem levels, and all three need community protection.

Short-answer questions with model answers

Q1. Define ecological engineering and give two examples.

Model answer: Ecological engineering is the design of the farm environment so that natural pest control and pollination are supported by the habitat itself. Examples: growing flowering plants such as sunflower and marigold on bunds to provide nectar and pollen to parasitoids and predators, and growing a trap crop such as castor to draw Spodoptera away from the main crop.

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Q2. Why is community responsibility important in natural farming?

Model answer: Because pests, water, pollinators and biodiversity move across farm boundaries, so one farmer acting alone cannot maintain ecological balance. Group adoption prevents re-infestation from neighbouring fields, community seed banks secure indigenous seed, joint water management maintains tanks and channels, and participatory certification and collective marketing strengthen the whole village.

Q3. Name three tools of ecological engineering.

Model answer: Habitat manipulation with flowering plants on bunds; trap crops such as marigold and castor; and banker plants that carry alternative prey for natural enemies. Beetle banks, bird perches and undisturbed refuge patches are further tools.

Long-answer questions with model answer outlines

Q1. Explain ecological balance and ecological engineering, and describe how a natural farm maintains both.

Outline:

- Ecological balance: definition and why it matters.
- Mechanisms: predators, parasitoids, competition, nutrient cycling, pollination, soil food web, habitat.
- Ecological engineering: definition and purpose.
- Tools: habitat manipulation, banker plants, trap crops, border vegetation, beetle banks, perches, ponds, groves.
- Examples of plants and their roles.
- Conclusion: design replaces chemicals.

Q2. Discuss the role of the community in maintaining ecological balance and conserving biodiversity in natural farming.

Outline:

- Introduction: ecological processes cross farm boundaries.
- Group farming and clusters to prevent re-infestation and enable marketing.
- Community seed banks and conservation of land races.
- Community water management (Kudimaramathu model).
- Participatory certification and joint pest monitoring.
- Protection of groves, tanks and common lands; knowledge sharing.
- Conclusion: natural farming is a community enterprise as much as a farm practice.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Ecological balance is maintained by natural ____ such as predators and parasitoids.

2 Designing the habitat to support natural enemies is called ecological ____.

3 A ____ crop draws pests away from the main crop.

4 A ____ plant hosts alternative prey for natural enemies.

5 Community labour for tank maintenance is called ____.

6 Biodiversity operates at genetic, species and ____ levels.

B. State whether true or false

1 Ecological engineering replaces pesticides with habitat design. (True)

2 One farmer alone can maintain ecological balance regardless of neighbours. (False - processes cross farm boundaries)

3 Trap crops protect the main crop. (True)

4 Participatory certification is part of the PGS-India system. (True)

C. Match the following

Column A	Column B
Predators	Keep pest numbers low
Trap crop	Draws pests away from main crop
Banker plant	Hosts alternative prey
Seed bank	Conserves indigenous varieties
PGS-India	Peer-based certification

D. Answer in one word or one line

1 Define ecological balance.

2 Define ecological engineering.

3 Name two tools of ecological engineering.

4 Why is community action important?

Practical application in Tamil Nadu - community initiatives

- Farmer groups and clusters adopt natural farming together to prevent re-infestation from neighbouring fields.
- Community seed banks conserve traditional paddy and millet varieties.
- Joint tank maintenance follows the traditional Kudimaramathu model.

Lecture 11. Integration of Crops, Trees and Animals in Natural Farming

Learning objectives

- Explain integrated farming systems and their components.
- Describe how crops, trees and animals recycle resources on a natural farm.
- Show how integration builds resilience and farm sustainability.

11.1 Integrated farming system

An integrated farming system (IFS) combines two or more enterprises - crops, livestock, trees, poultry, fish, bees or mushrooms - on the same farm so that the waste of one becomes the input of another. On a natural farm, integration is not just an add-on; it is how the nutrient loop is closed and how risk is spread.

11.2 Components and their interactions

Component	Provides	Receives
Crops	Grain, fodder, crop residues, mulch	Dung and urine based inputs, compost
Trees	Fodder, fuel, fruit, timber, shade, nitrogen	Compost, mulch, water
Cattle and buffaloes	Dung, urine, milk, draught power	Fodder, crop residues, water
Goats and sheep	Manure, meat, income	Fodder, shrubs, crop residues
Poultry	Manure, eggs, meat, insect control	Grain, kitchen waste, insects
Fish and ducks	Protein, pond silt as fertiliser	Farm waste, pond water
Bees	Honey, pollination	Nectar and pollen from diverse crops and trees
Mushrooms	Protein, income	Crop residues and straw

11.3 The crop-tree-animal nutrient cycle

The heart of integration is a cycle in which nutrients move from soil to plant to animal and back to soil, losing as little as possible.

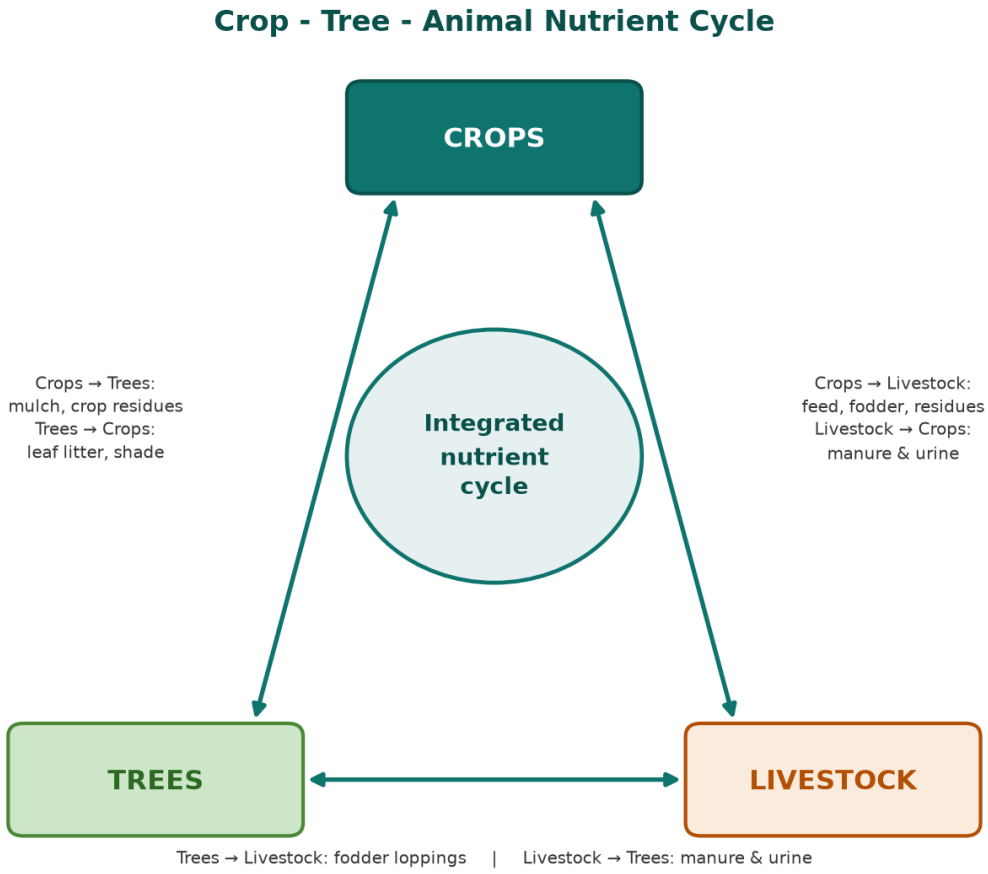


Figure 11.1 Crop-tree-animal nutrient cycle

Nutrient flows between crops, livestock and trees in an integrated natural farm.

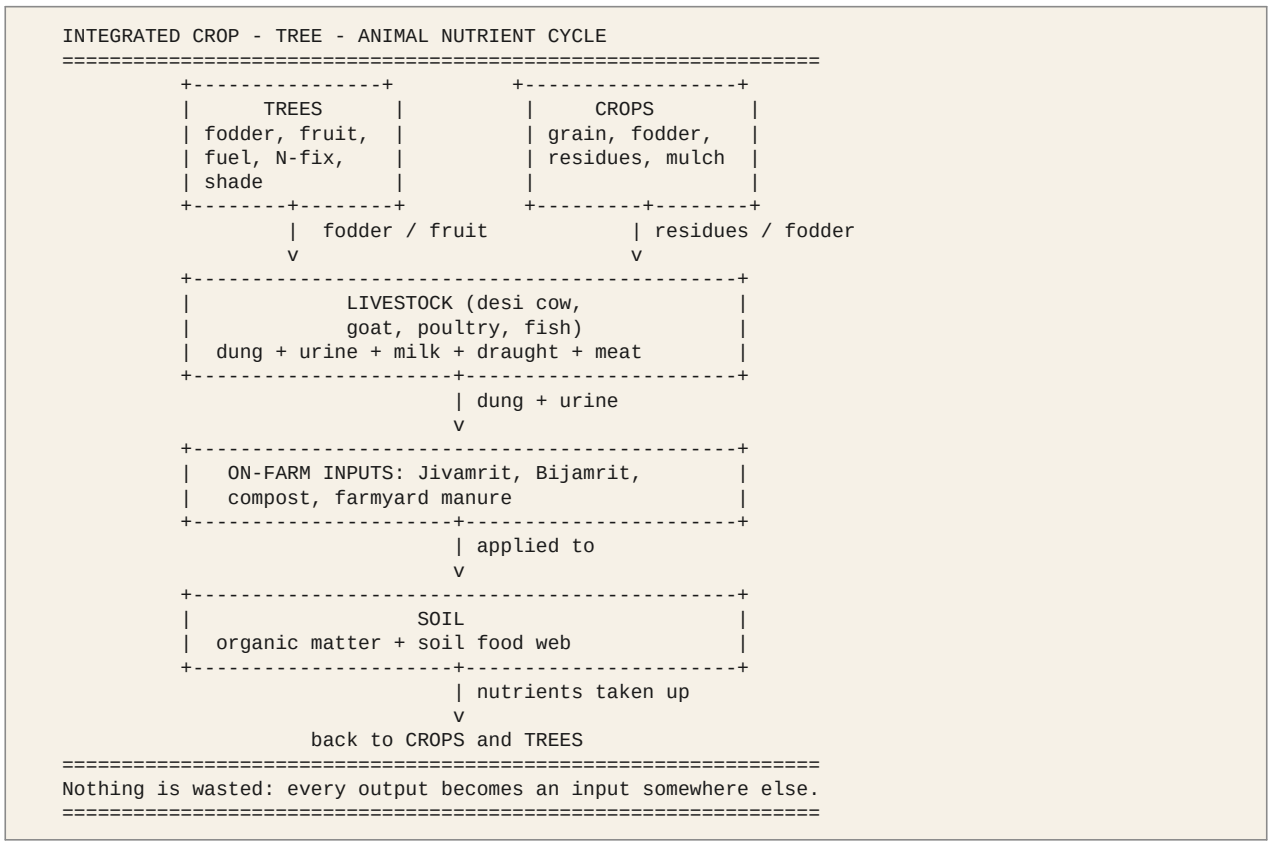


Fig. 11.1 The integrated crop-tree-animal nutrient cycle.

11.4 Common integrated models

Model	Components	Suitability
Crop + dairy	Field crops, fodder, two to five desi cows	Most small holdings; supplies dung and urine for inputs.
Crop + poultry + vegetables	Cereal, poultry, home garden	Homesteads; poultry controls insects and weeds.
Crop + goat + tree	Field crops, goats, fodder and fruit trees	Dryland and hilly areas.
Crop + fish + duck	Paddy or pond, fish, ducks	Low-lying and tank-fed areas; ducks control pests and add nutrients.
Agri-horti-forestry	Field crops with fruit and timber trees	Rain-fed and degraded lands; long-term income with short-term crops.
Crop + bees	Diverse crops and trees with bee boxes	Any diverse farm; pollination plus honey income.
Multi-storey homestead	Coconut, banana, spices, vegetables, tubers, livestock	Tamil Nadu homesteads; high output from small area.

11.5 Benefits of integration

- **Resource recycling:** residues, dung and urine are used on the farm instead of being wasted or bought outside.
- **Reduced cost:** the farm supplies its own inputs, fodder and much of its food.
- **Diversified income:** grain, milk, eggs, meat, fruit, honey and timber give several income streams.
- **Risk reduction:** if one enterprise fails, others support the household.
- **Improved soil:** continuous addition of manure and residues builds organic matter.

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- **Employment:** livestock and tree components need and provide year-round labour.
- **Nutrient balance:** livestock convert low-value residues into high-value manure.
- **Climate resilience:** trees provide shade and reduce heat stress; diversity buffers weather shocks.

11.6 Designing integration on a small farm

Integration should be planned so that components match the farm's resources and the family's needs. The steps below are a practical guide.

1. Assess resources: land, water, labour, capital, existing livestock and trees.
2. Identify the family's needs: food, fodder, fuel, cash and risk tolerance.
3. Choose a main enterprise (usually the crop) and supporting enterprises (dairy, poultry, trees).
4. Match component sizes: the number of animals should match the fodder and residue available, so that the farm does not need to buy feed.
5. Locate components for efficiency: cattle shed near the field and input unit; fodder plot close to the shed; pond downhill to harvest runoff.
6. Plan the nutrient flow: dung and urine to the input unit; residues to mulch or feed; pond silt to the field.
7. Add trees on boundaries and as intercrops for fodder, nitrogen and shade.
8. Monitor and adjust each season based on yields, costs and income.

WORKED EXAMPLE - Nutrient budget of an integrated one-hectare farm

- Cattle: 3 desi cows producing about 8-10 kg dung and 8-10 L urine per cow per day.
- Dung per year: about 3 cows x 9 kg x 365 = about 9,850 kg (say 10 tonnes) of fresh dung.
- Urine per year: about 3 x 9 L x 365 = about 9,850 L of urine - ample for the season's Jivamrit and Bijamrit needs.
- If half the dung is composted and applied, and urine is used in preparations, the farm supplies most of its nitrogen and potassium requirement from internal sources.
- Crop residues from 0.6 ha of cereals and pulses: about 2-3 tonnes of straw, used as mulch and cattle feed, which returns carbon and nutrients to the soil.
- Result: the farm's nutrient exports (grain, milk) are substantially offset by internal recycling, reducing the need for any external input.
- Note: these are indicative teaching figures; actual values depend on animal size, feed quality and crop yields.

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Key points to remember

- An integrated farming system combines crops, livestock, trees and other enterprises so that the waste of one becomes the input of another.
- Components interact: crops give residues and fodder; animals give dung, urine, milk and power; trees give fodder, fuel, fruit, nitrogen and shade.
- The crop-tree-animal nutrient cycle closes the loop from soil to plant to animal and back to soil.
- Common models: crop+dairy, crop+poultry+vegetables, crop+goat+tree, crop+fish+duck, agri-horti-forestry, crop+bees, multi-storey homestead.
- Benefits: resource recycling, lower cost, diversified income, risk reduction, better soil, employment and climate resilience.

- Design should match component sizes to available fodder and residues.
- Integration is the practical way to achieve self-reliance in natural farming.

Short-answer questions with model answers

Q1. What is an integrated farming system?

Model answer: An integrated farming system combines two or more enterprises, such as crops, livestock, trees, poultry, fish or bees, on the same farm so that the by-product or waste of one enterprise becomes the input of another. This closes the nutrient loop, reduces cost, diversifies income and spreads risk.

Q2. Name any four integrated farming models suitable for Tamil Nadu.

Model answer: Crop plus dairy, crop plus poultry plus vegetables, crop plus fish plus duck, and multi-storey homestead with coconut, banana, spices, vegetables and livestock. Agri-horti-forestry and crop plus bees are further models.

Q3. How does livestock integration close the nutrient loop?

Model answer: Livestock consume crop residues and fodder grown on the farm and convert them into dung and urine, which are used to prepare Jivamrit, Bijamrit and compost. These are applied back to the soil, so nutrients that would otherwise be exported or wasted return to the crop. Nothing leaves the farm as waste.

Long-answer questions with model answer outlines

Q1. Explain the integration of crops, trees and animals in natural farming with a nutrient-cycle diagram.

Outline:

- Introduction: definition and purpose of integration.
- Components and what each provides and receives.
- The crop-tree-animal nutrient cycle drawn as a diagram.
- Common integrated models with suitability.
- Benefits: recycling, cost, income, risk, soil, employment, resilience.
- Conclusion: integration is how a natural farm becomes self-reliant.

Q2. Describe how you would design an integrated natural farm for a one-hectare holding and estimate its nutrient budget.

Outline:

- Resource assessment and family needs.
- Choice of main and supporting enterprises.
- Matching animal numbers to fodder and residues.
- Layout: shed, fodder plot, pond, input unit, boundary trees.
- Nutrient budget: dung, urine, residues and their use.
- Monitoring and adjustment.
- Conclusion: internal recycling reduces external inputs.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 An integrated farming system combines crops, livestock and ____.

2 Livestock convert crop residues into ____ and urine.

3 Trees provide fodder, fuel, fruit and ____ fixation.

4 In crop plus fish plus duck systems, ducks control ____ and add nutrients.

5 Bee keeping provides pollination and ____ income.

6 The crop-tree-animal cycle closes the ____ loop.

B. State whether true or false

1 Integrated farming increases risk by concentrating on one enterprise. (False - it spreads risk)

2 Matching animal numbers to available fodder reduces the need to buy feed. (True)

3 Trees compete with crops and are never useful. (False - they provide many benefits)

4 Integration reduces the cost of inputs. (True)

C. Match the following

Column A	Column B
Crop + dairy	Dung and urine for inputs
Crop + poultry	Insect and weed control
Crop + fish + duck	Low-lying and tank-fed areas
Agri-horti-forestry	Short and long-term income together
Multi-storey homestead	High output from a small area

D. Answer in one word or one line

1 Define integrated farming system.

2 Name three components of an integrated farm.

3 Give two benefits of integration.

4 How does livestock close the nutrient loop?

Practical application in Tamil Nadu - integrated models

- Crop plus dairy is the most common integrated model on smallholdings in the state.
- Coconut-based multi-storey homesteads combine coconut, banana, spices, vegetables and livestock.
- Crop plus fish plus duck suits the tank-fed low-lying areas of the Cauvery delta.

Lecture 12. Biodiversity, Indigenous Seeds and Farm Waste Recycling

Learning objectives

- Explain the importance of biodiversity and the conservation of indigenous species.
- Describe indigenous seed conservation and seed treatment practices.
- Explain how farm wastes are converted into inputs for soil enrichment.

12.1 Biodiversity on the farm

Biodiversity is the variety of life - genes, species and ecosystems. On a farm it includes crops, livestock breeds, trees, insects, birds, microbes and the ecosystems of field, pond and grove. High biodiversity makes a farm stable: more species means more ways for the system to function if one is disturbed.

- **Genetic diversity:** different varieties and land races within a crop, and different breeds of livestock.
- **Species diversity:** many kinds of plants and animals on the farm.
- **Ecosystem diversity:** fields, bunds, ponds, groves, homesteads and wetlands.
- **Functional biodiversity:** the organisms that do useful work - pollinators, natural enemies, decomposers, nitrogen fixers.

12.2 Indigenous seeds

Indigenous seeds are the varieties developed and maintained by farmers over generations. They are also called land races or traditional varieties. They are adapted to local soil, climate and pests, and they carry genetic diversity that modern varieties often lack.

12.2.1 Why indigenous seeds matter

- Adapted to local conditions, so they perform well with low inputs and in variable rainfall.
- More tolerant of pests, diseases and drought than many high-yielding varieties.
- Better taste, cooking quality and nutrition in many cases.
- Genetic diversity is the raw material for future crop improvement.
- They give the farmer independence from purchased seed and protect seed sovereignty.
- Many are suited to mixed cropping and to the farmer's own cropping calendar.

12.2.2 Tamil Nadu examples

Crop	Indigenous varieties / land races (examples)
Paddy	Traditional varieties such as Kallurundai Samba, Karuppu Kavuni (black rice), Mappillai Samba, Kuzhiyadichan, Samba and Seeraga Samba.
Millets	Kambu (pearl millet), Kelvaragu (finger millet), Varagu (kodo), Samai (little millet), Thinai (foxtail) and Panivaragu (proso) land races.
Pulses	Traditional cowpea, blackgram, greengram and redgram land races.
Oilseeds	Traditional groundnut, sesame (gingelly) and castor varieties.
Vegetables	Local brinjal, chilli, okra, gourds and country tomato land races.

Note: the variety names above are widely known Tamil Nadu land races; the exact list held by any seed bank varies, and students should treat these as examples rather than a complete catalogue.

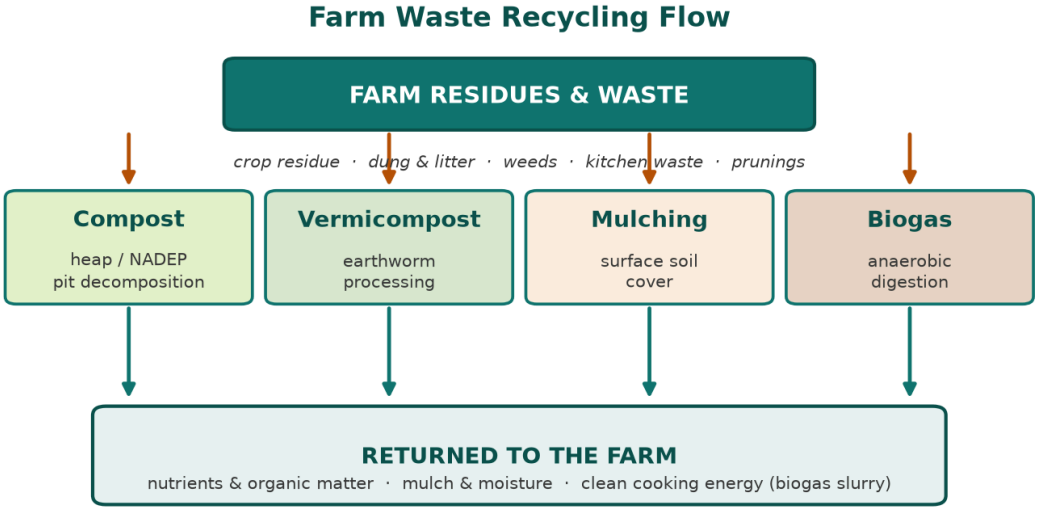


Figure 12.1 Farm waste recycling flow

Residues to compost, vermicompost, mulching and biogas - and back to the soil.

12.2.3 Seed conservation and treatment

- **Community seed banks:** farmer groups store and exchange seed of indigenous varieties.
- **On-farm conservation:** growing the variety every year keeps it alive and adapted.
- **Seed selection:** selecting the best plants and grain for seed maintains vigour.
- **Seed treatment:** Bijamrit, cow urine, dung slurry, buttermilk and botanical extracts protect the seed and improve germination.
- **Storage:** dry seed stored with neem leaves, ash or in airtight containers to prevent insect damage.
- **Seed exchange and seed festivals:** events that distribute and celebrate indigenous seed.

12.3 Farm waste recycling

A natural farm treats waste as a resource. Crop residues, animal dung, kitchen waste and weeds are all converted into inputs, so that nothing is burnt or thrown away and no nutrient is lost.

12.3.1 Types of farm waste and their use

Farm waste	Recycling method	Product and use
Crop residues and straw	Mulching, composting, cattle feed, mushroom substrate	Mulch, compost, manure, mushrooms
Cattle dung	Composting, Jivamrit, Bijamrit, vermicompost	Organic manure and microbial inoculum
Cattle urine	Jivamrit, Bijamrit, botanical sprays, pest repellent	Liquid fertiliser and repellent
Green weeds and prunings	Green manure, compost, liquid manure	Nitrogen and organic matter
Kitchen and market waste	Composting, vermicomposting	Nutrient-rich compost
Poultry litter	Composting, direct application	Nitrogen-rich manure
Pond silt	Applied to fields	Nutrient-rich silt
Banana pseudostem and other crop trash	Composting, mulch, bio-enzyme	Mulch and liquid fertiliser

12.3.2 Composting methods

- **Indore method:** layered composting of residues with dung and urine, turned periodically - the method of Albert Howard.
- **Bangalore method:** pit composting suited to drier conditions.
- **Vermicomposting:** earthworms convert organic waste into nutrient-rich vermicompost.
- **NADEP composting:** a low-cost method using biomass, dung and soil in an aerated pit.
- **Waste decomposer / microbial consortia:** cultures that speed up decomposition of residues.
- **Liquid manures:** Jivamrit, panchagavya and dung-urine extracts as quick nutrition.

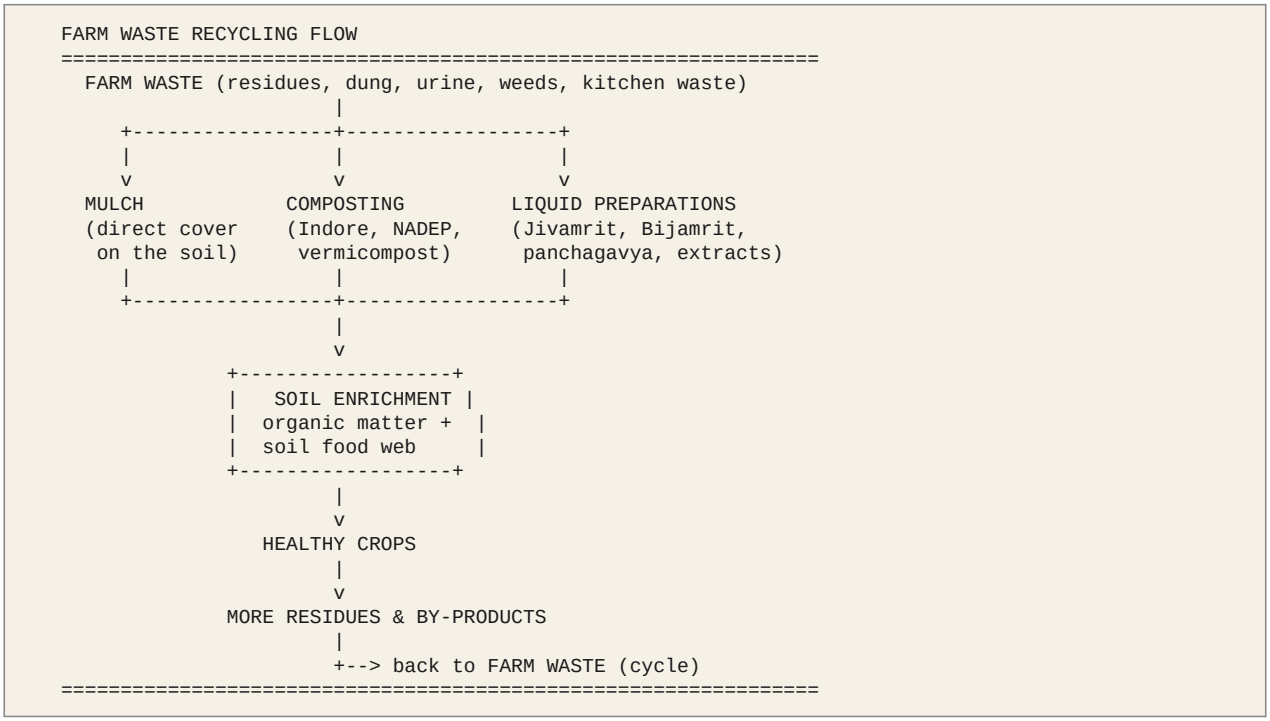


Fig. 12.1 Farm waste recycling - waste becomes input, input becomes crop, crop becomes waste again.

12.4 Nutrient value of recycled materials

Material	Approximate nutrient content (per cent, on fresh/dry basis as noted)	Note
Farmyard manure (well rotted)	N 0.5, P 0.2, K 0.5 (dry basis, indicative)	Values vary widely with source and storage.
Compost	N 0.5-1.5, P 0.3-0.8, K 0.5-1.5	Range depends on feedstock and maturity.
Vermicompost	N 1.0-2.0, P 0.5-1.0, K 0.8-1.5	Also rich in microbes and humus.
Cattle urine (fresh)	N 0.5-1.0, K 0.5-1.5	Used diluted; source of nitrogen and potassium.
Poultry manure	N 1.5-3.0, P 1.5-2.5, K 1.0-1.5	Rich but should be composted before use.
Green manure (sunhemp)	Adds about 40-60 kg N per ha when grown well	Depends on biomass produced.

Note: nutrient percentages of organic materials vary greatly with source, age and handling. The ranges above are the commonly cited values; exact figures should be obtained from a laboratory test where precision is needed.

WORKED EXAMPLE - Converting one tonne of crop residue

- One tonne of dry crop residue contains roughly 5-10 kg nitrogen, 1-2 kg phosphorus and 10-15 kg potassium, depending on the crop.
- If burnt, almost all of this is lost and carbon is released as carbon dioxide.
- If composted, most of the nutrients are retained and returned to the soil, and about 0.2-0.3 tonnes of stable organic matter is added.
- If used as mulch, the nutrients are released slowly over the season, moisture is conserved and soil life is fed.
- Conclusion: recycling one tonne of residue is worth several hundred rupees of fertiliser and, more importantly, builds long-term soil health.

Mounish Raj · MouRa · Agri AI

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Key points to remember

- Biodiversity operates at genetic, species and ecosystem levels, plus functional biodiversity of useful organisms.
- Indigenous seeds are farmer-maintained land races adapted to local conditions; they give low-input performance, tolerance and seed sovereignty.
- Tamil Nadu has rich land races of paddy, millets, pulses, oilseeds and vegetables.
- Seed conservation uses community seed banks, on-farm growing, selection, Bijamrit treatment and careful storage.
- Farm waste is a resource: residues, dung, urine, weeds and kitchen waste are recycled.
- Recycling methods: mulching, composting (Indore, NADEP, vermicompost) and liquid preparations.
- Recycled materials supply nutrients and, more importantly, organic matter and soil life.

Short-answer questions with model answers

Q1. What are indigenous seeds and why are they important?

Model answer: Indigenous seeds are traditional varieties or land races developed and maintained by farmers over generations. They are important because they are adapted to local soil, climate and pests, perform well with low inputs, tolerate drought and pests better than many modern varieties, often have better taste and nutrition, carry genetic diversity for future breeding, and give the farmer independence from purchased seed.

Q2. Name any four methods of composting farm waste.

Model answer: The Indore method, the Bangalore method, vermicomposting and NADEP composting. Microbial waste decomposer cultures and liquid preparations such as Jivamrit are further ways of recycling farm waste.

Q3. Give two Tamil Nadu indigenous paddy varieties and state one benefit of using them.

Model answer: Kallurundai Samba and Karuppu Kavuni (black rice) are examples, along with Mappillai Samba and Seeraga Samba. A benefit is that they are adapted to local conditions and perform well under low-input natural farming, while also fetching a premium in niche markets for traditional rice.

Long-answer questions with model answer outlines

Q1. Explain the importance of biodiversity and indigenous seed conservation in natural farming.

Outline:

- Biodiversity: definition and levels (genetic, species, ecosystem, functional).
- Role of biodiversity in stability, pest control and pollination.
- Indigenous seeds: definition and advantages.
- Tamil Nadu examples of land races.
- Conservation methods: community seed banks, on-farm conservation, selection, treatment and storage.
- Conclusion: biodiversity and indigenous seed are the genetic base of natural farming.

Q2. Describe how farm wastes are recycled into inputs for soil enrichment on a natural farm.

Outline:

- Introduction: waste as a resource.
- Types of waste and their recycling methods (table).
- Composting methods: Indore, Bangalore, vermicompost, NADEP.
- Liquid preparations: Jivamrit, panchagavya.

- Nutrient value of recycled materials.
- Waste recycling flow diagram.
- Conclusion: recycling builds organic matter and closes the nutrient loop.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Biodiversity operates at genetic, species and ____ levels.

2 Traditional varieties maintained by farmers are called land ____.

3 A farmer-managed store of indigenous seed is a community ____ bank.

4 The Indore method is a method of ____.

5 Earthworms produce nutrient-rich ____.

6 Green manure crops such as sunhemp add ____ to the soil.

B. State whether true or false

1 Indigenous seeds are less tolerant of local pests than modern varieties. (False - often more tolerant)

2 Farm waste should be burnt to keep the field clean. (False - it should be recycled)

3 Vermicomposting uses earthworms. (True)

4 Community seed banks conserve indigenous varieties. (True)

C. Match the following

Column A	Column B
Karuppu Kavuni	Black rice land race
Sunhemp	Green manure crop
Vermicompost	Earthworm-produced manure
NADEP	Low-cost composting method
Mulching	Direct use of residue as cover

D. Answer in one word or one line

1 Name two Tamil Nadu indigenous paddy varieties.

2 Name three methods of composting.

3 What is a community seed bank?

4 Name two benefits of using indigenous seed.

Practical application in Tamil Nadu - biodiversity and seed
- Traditional paddy varieties such as Mappillai Samba and Karuppu Kavuni fetch premium prices.
- Millet land races of the southern districts are conserved by farmer groups and seed savers.
- Vermicompost units are widely promoted for recycling farm and kitchen waste.

Lecture 13. Water Conservation, Renewable Energy and Nutrient Management

Learning objectives

- Explain water conservation practices suited to natural farming.
- Describe renewable energy applications on the farm.
- Explain natural nutrient sources and nutrient management in natural farming.

13.1 Water conservation

Water is a limiting resource in most of Tamil Nadu, where rainfall is seasonal and often erratic. Natural farming conserves water both by storing it and by reducing losses, and by improving the soil's capacity to hold moisture.

13.1.1 Soil-based water conservation

- **Organic matter:** every one per cent increase in soil organic carbon raises the water-holding capacity of the soil substantially; mulch and manure build this.
- **Mulching:** a mulch layer cuts evaporation from the soil surface by a large fraction, so the same rainfall or irrigation lasts longer.
- **No tillage:** undisturbed soil has continuous pores and root channels that improve infiltration.
- **Cover crops:** live mulch protects the surface and reduces runoff.

13.1.2 Farm-level water conservation structures

Structure	Purpose
Farm pond	Stores runoff for supplementary irrigation and for fish and ducks.
Contour bund and trench	Slows runoff, increases infiltration on sloping land.
Water absorption trench	Recharges groundwater along the contour.
Percolation pond	Recharges groundwater and improves well yield.
Check dam	Slows stream flow and recharges the channel bed.
Field bund	Retains rainwater in the field and prevents soil loss.
Roof water harvesting	Collects rainwater for drinking and homestead use.
Recharge pit	Directs runoff into the ground near the source of use.

13.1.3 Efficient irrigation methods

- **Drip irrigation:** delivers water directly to the root zone, saving a large fraction of water compared with flooding.
- **Micro-sprinklers:** suited to close-grown crops and to sandy soils.
- **Alternate wetting and drying:** in paddy, allowing the field to dry partially between irrigations saves water without much yield loss.
- **Irrigation scheduling:** irrigate based on soil moisture and crop stage, not on a fixed calendar.
- **Mulch plus drip:** the most water-efficient combination, because evaporation loss is minimised.

13.2 Renewable energy on the farm

Natural farming reduces dependence on purchased energy and makes use of the sun, wind, animals and farm waste.

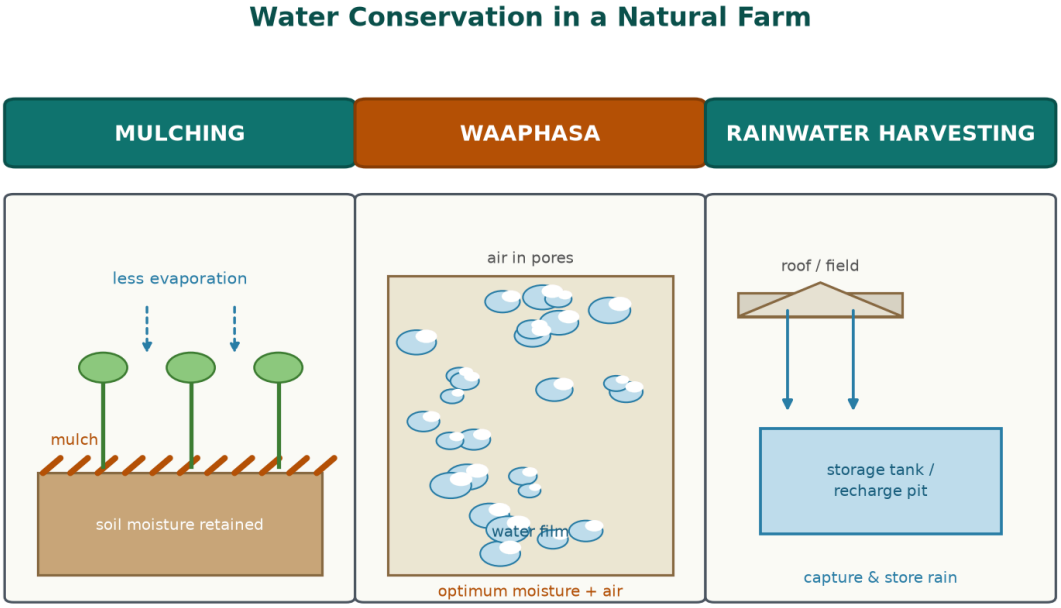


Figure 13.1 Water conservation in a natural farm

Mulching, Waaphasa soil moisture balance and rainwater harvesting working together.

Renewable energy source	Application on the farm
Solar photovoltaic	Water pumps, lighting, fencing, drying and charging batteries.
Solar thermal	Water heating and solar drying of produce.
Biogas from dung	Cooking gas and lighting; the digested slurry is a rich manure.
Wind (small)	Water pumping in windy locations.
Draught animal power	Ploughing, carting, water lifting and threshing.
Bullock / animal power	Reduces fossil fuel dependence for field operations.
Bio-energy from residues	Fuel and compost; no residue is wasted.
Human power	Hand tools and cycle-based pumps for small operations.

A biogas plant is particularly suited to a natural farm because it uses cattle dung, which is already the base of the input system, and it returns a nutrient-rich slurry that can be applied to crops.

13.3 Nutrient management in natural farming

Nutrient management means supplying the crop with the nutrients it needs, in the right balance and at the right time, from natural sources. In natural farming the approach is to feed the soil, not the crop, and to let soil biology deliver the nutrients.

13.3.1 Natural nutrient sources

Source	Main nutrients	How it is used
Farmyard manure / compost	N, P, K and micronutrients, plus organic matter	Broadcast or placed at sowing; base dressing.
Vermicompost	N, P, K, humus and microbes	Applied to nursery and field crops.
Jivamrit	Microbes, N, K	Soil drench or through irrigation.
Cattle urine	N, K	Diluted as a spray or in preparations.
Green manure (sunhemp, daincha)	N	Grown and incorporated or mulched before flowering.
Legume rotation	N	Fixes atmospheric nitrogen in root nodules.
Oil cakes (neem, groundnut, castor)	N, P and pest repellent action	Composted or applied as slurry.
Wood ash	K, Ca, P	Applied sparingly to potassium-demanding crops.
Rock phosphate and bone meal	P, Ca	Used in organic systems; sparingly in natural farming.
Panchagavya and botanical extracts	Growth promoters and micronutrients	Foliar spray at critical stages.

13.3.2 Principles of nutrient management

- **Feed the soil:** build organic matter and soil life; the crop is nourished indirectly.
- **Balance:** supply a mix of sources so that nitrogen, phosphorus, potassium and micronutrients are all available.
- **Timeliness:** apply at sowing, at critical growth stages and after stress.
- **Slow release:** organic sources release nutrients slowly, matching crop demand and reducing loss.
- **Recycling:** return all residues and manures to the soil to reduce nutrient export.
- **Monitor:** use soil tests and crop observation to adjust inputs.

WORKED EXAMPLE - Nutrient budget from on-farm inputs for one acre

- Compost: 2 tonnes per acre of good compost at roughly 0.5-1.5 per cent N gives about 10-30 kg nitrogen, plus phosphorus and potassium.
- Jivamrit: two or three applications per season supply a microbial boost and some readily available nitrogen and potassium.
- Cattle urine: 200 L diluted and sprayed supplies readily available nitrogen and potassium and acts as a mild pest repellent.
- Legume intercrop or green manure: a good sunhemp crop can add the equivalent of 40-60 kg nitrogen per hectare (about 16-24 kg per acre).
- Mulch: crop residue returning to the soil recycles several kilograms each of N, P and K per tonne.
- Total: on-farm sources can meet a large part of the crop's nitrogen and potassium needs; phosphorus is the nutrient most likely to need supplementation in the early years.
- Note: figures are indicative; a soil test is the reliable guide.

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Key points to remember

- Water is conserved by building soil organic matter, mulching, no tillage and cover crops.
- Farm-level structures include farm ponds, contour bunds and trenches, percolation ponds, check dams and recharge pits.
- Efficient irrigation uses drip, micro-sprinklers, alternate wetting and drying, and moisture-based scheduling.
- Renewable energy on the farm includes solar pumps and dryers, biogas from dung, wind, draught animals and hand tools.
- Biogas is especially suited to natural farms because it uses dung and returns a rich slurry.
- Natural nutrient sources include compost, vermicompost, Jivamrit, cattle urine, green manure, legumes, oil cakes and ash.
- Nutrient management principles: feed the soil, balance the sources, apply on time, use slow-release organics and recycle residues.

Short-answer questions with model answers

Q1. Name any five farm-level water conservation structures.

Model answer: Farm pond, contour bund and trench, percolation pond, check dam and recharge pit. Field bunds and roof water harvesting are further structures.

Q2. How is biogas suited to a natural farm?

Model answer: Biogas plants run on cattle dung, which is already the base of the natural farming input system. The plant provides cooking gas and lighting, reducing fuel cost, and the digested slurry that comes out is a nutrient-rich manure that can be applied to crops. Thus a waste product becomes both energy and fertiliser.

Q3. List four natural sources of nutrients in natural farming.

Model answer: Farmyard manure or compost, vermicompost, Jivamrit and green manure such as sunhemp or daincha. Cattle urine, oil cakes, wood ash, legume rotation and panchagavya are further sources.

Long-answer questions with model answer outlines

Q1. Explain water conservation and renewable energy practices suitable for a natural farm.

Outline:

- Water conservation: soil organic matter, mulching, no tillage, cover crops.
- Farm structures: farm pond, contour bunds, trenches, percolation ponds, check dams, recharge pits.

- Efficient irrigation: drip, micro-sprinkler, alternate wetting and drying, scheduling.
- Renewable energy: solar pump and dryer, biogas, wind, draught animals, hand tools.
- Biogas integration with dung and slurry.
- Conclusion: conservation of water and energy reduces cost and increases resilience.

Q2. Describe nutrient management in natural farming with reference to natural nutrient sources.

Outline:

- Definition and approach: feed the soil, not the crop.
- Natural nutrient sources with nutrients supplied (table).
- Principles: balance, timeliness, slow release, recycling, monitoring.
- Worked nutrient budget from on-farm inputs.
- Role of Jivamrit and mulch in nutrient release.
- Conclusion: soil biology is the delivery system for nutrients.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Mulching reduces ____ from the soil surface.

2 A ____ pond stores runoff for supplementary irrigation.

3 Drip irrigation delivers water to the ____ zone.

4 A ____ plant uses cattle dung to produce cooking gas and rich slurry.

5 Green manure crops such as ____ add nitrogen to the soil.

6 Every one per cent increase in soil organic carbon raises water-holding ____.

B. State whether true or false

1 Tillage improves the soil's water-holding capacity. (False - organic matter does)

2 Biogas slurry is a nutrient-rich manure. (True)

3 Alternate wetting and drying saves water in paddy. (True)

4 Organic nutrient sources release nutrients very quickly. (False - they release them slowly)

C. Match the following

Column A	Column B
Farm pond	Stores runoff
Percolation pond	Recharges groundwater
Biogas	Energy from cattle dung
Vermicompost	Nutrient-rich manure
Jivamrit	Microbial soil inoculum

D. Answer in one word or one line

1 Name three farm-level water conservation structures.

2 Name two renewable energy applications on a farm.

3 Name three natural sources of nutrients.

4 What is alternate wetting and drying?

Practical application in Tamil Nadu - water and energy

- Drip and micro-sprinkler irrigation are promoted with subsidy under state schemes.
- Farm ponds and recharge structures are built under watershed development programmes.
- Solar pumps are widely used to reduce electricity and diesel cost.

Lecture 14. Insect, Pest, Disease and Weed Management in Natural Farming

Learning objectives

- Explain the principles of ecological pest management.
- Describe botanical formulations used in natural farming.
- Explain biodiversity-based prevention and non-chemical weed management.

14.1 The natural farming attitude to pests

In natural farming a pest is not an enemy to be destroyed but a symptom of imbalance. A healthy, diverse, well-nourished crop in a living soil rarely suffers serious pest damage, because natural enemies keep pests in check and because the plant's own defences are strong. The aim is therefore prevention through a healthy ecosystem, and only then control by ecological means. A spray is a last resort, not a routine.

14.2 Principles of ecological pest management

- **Prevention first:** a healthy crop in a diverse system is the best protection.
- **Tolerate the harmless:** a small pest population feeds the natural enemies and does not reduce yield; do not spray at the first sight of an insect.
- **Use the action threshold:** act only when pest numbers cross the level at which economic damage begins.
- **Conserve natural enemies:** avoid broad-spectrum poisons and provide habitat for predators and parasitoids.
- **Use diversity:** mixed cropping, rotation, trap crops and border rows reduce pest build-up.
- **Prefer the least disruptive method:** cultural and mechanical methods before botanicals.
- **Treat the cause:** correct soil, nutrition and water problems that make plants vulnerable.

14.3 Categories of pest management methods

Category	Methods	Examples in natural farming
Cultural	Practices that prevent pest build-up	Timely sowing, crop rotation, mixed cropping, sanitation, removal of infested parts, summer ploughing where permitted, flooding in paddy.
Mechanical and physical	Direct removal or exclusion	Hand-picking of larvae and egg masses, light traps, yellow and blue sticky traps, pheromone traps, bird perches, netting.
Biological	Use of living organisms	Conservation and release of predators and parasitoids (ladybird beetles, Chrysoperla, Trichogramma), entomopathogens (Beauveria, Metarhizium, Bacillus thuringiensis, NPV).
Botanical	Plant-based preparations	Neem, nochi, erukku, adathoda, garlic, chilli, tobacco, tulsi, pungam.
Habitat / ecological engineering	Designing the environment	Flowering plants on bunds, banker plants, trap crops, beetle banks, refuge patches.
Biological control of disease	Antagonistic microbes	Trichoderma, Pseudomonas fluorescens, Bacillus subtilis for soil-borne disease.

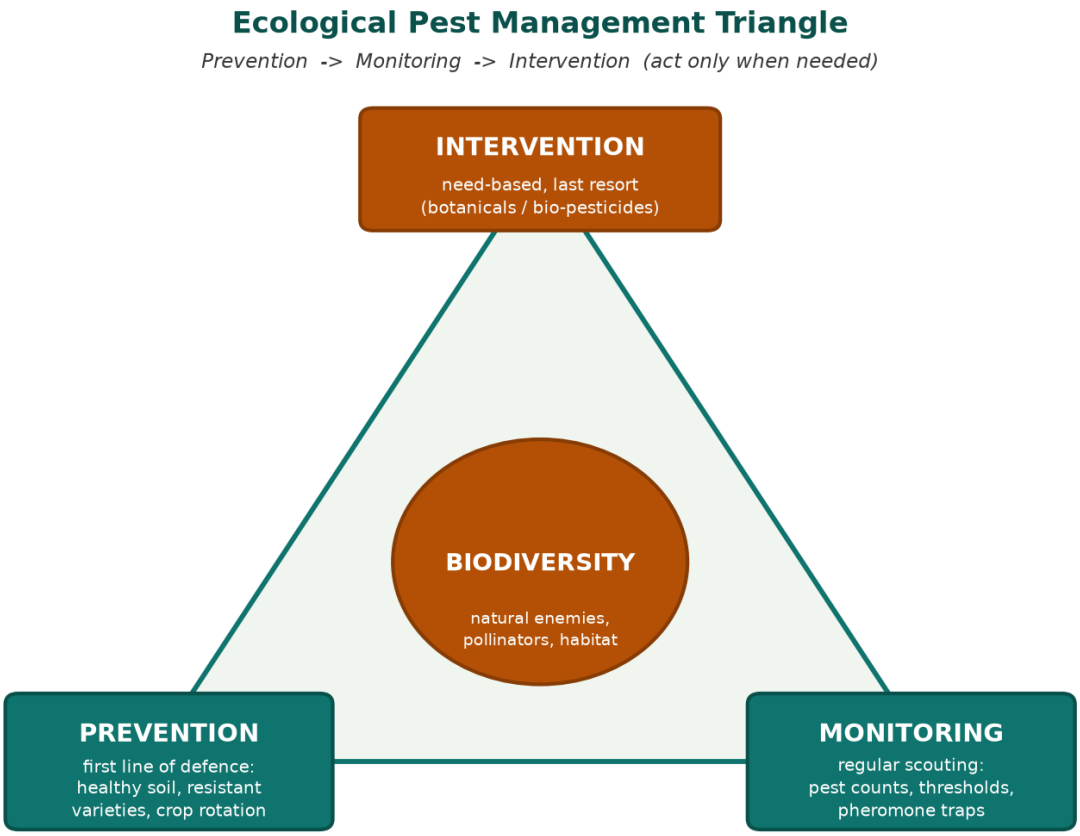


Figure 14.1 Ecological pest management triangle

Prevention, monitoring and intervention arranged around farm biodiversity.

14.4 Botanical formulations

Botanical preparations are made from locally available plants. They are less harmful than synthetic pesticides but must still be used carefully, because even plant extracts can harm beneficial insects if applied indiscriminately.

Plant	Active principle / action	Typical use
Neem (Azadirachta indica)	Azadirachtin - antifeedant, repellent, growth regulator	Seed kernel extract or neem oil spray for sucking pests and larvae.
Nochi (Vitex negundo)	Repellent and insecticidal compounds	Leaf extract spray for caterpillars and sucking pests.
Eruku / Erukku (Calotropis gigantea)	Latex compounds, repellent	Leaf extract in combination sprays.
Adathoda (Adhatoda vasica)	Alkaloids with insecticidal action	Leaf extract spray for larvae.
Garlic and onion	Sulphur compounds, repellent	Bulb extract spray against sucking pests and as a fungicide.
Chilli	Capsaicin, irritant	Combined with garlic and soap for a repellent spray.
Tobacco	Nicotine - insecticidal	Decoction, used sparingly; harmful to humans and bees, so use with care.
Tulsi (Ocimum)	Aromatic repellent	Leaf extract spray.
Pongam (Pongamia pinnata)	Karanjin - insecticidal	Seed oil spray.

A common combination spray is 'neem-garlic-chilli-soap': an extract of neem kernel, garlic, chilli and a little soap solution to help the spray stick. It is used against sucking pests and small caterpillars.

14.5 Disease management

- **Healthy soil:** a biologically active soil suppresses soil-borne pathogens; this is the foundation of disease control.
- **Antagonists:** Trichoderma and Pseudomonas fluorescens applied to seed and soil compete with and suppress pathogens.
- **Resistant varieties:** indigenous and locally adapted varieties tolerate disease better.
- **Cultural practice:** crop rotation, sanitation, spacing for air movement and avoiding excess moisture reduce disease.
- **Botanical and mineral sprays:** neem, garlic and dilute milk or buttermilk sprays for some fungal diseases; lime and ash for certain problems.
- **Seed treatment:** Bijamrit and warm-water treatment reduce seed-borne disease.

14.6 Weed management without herbicides

Weeds are managed by making the crop more competitive and by covering the soil, rather than by killing plants with chemicals.

Method	How it works
Mulching	A thick mulch layer blocks light and prevents weed germination.
Live mulch and cover crops	Smother weeds and occupy the space.
Competitive crops and close spacing	The crop shades out weeds.
Crop rotation	Changes the weed flora and prevents a single weed from dominating.

Method	How it works
Timely hand weeding	Remove weeds young, before they set seed.
Stale seedbed and solarisation	Allow weeds to germinate, then destroy them before sowing.
Green manuring with smother crops	Dense crops such as sunhemp suppress weeds.
Animals	Ducks and poultry graze young weeds and insects in some systems.

In Fukuoka's system, weeds are managed by a clover ground cover, by timing and by brief flooding in paddy, rather than by tillage or herbicide.

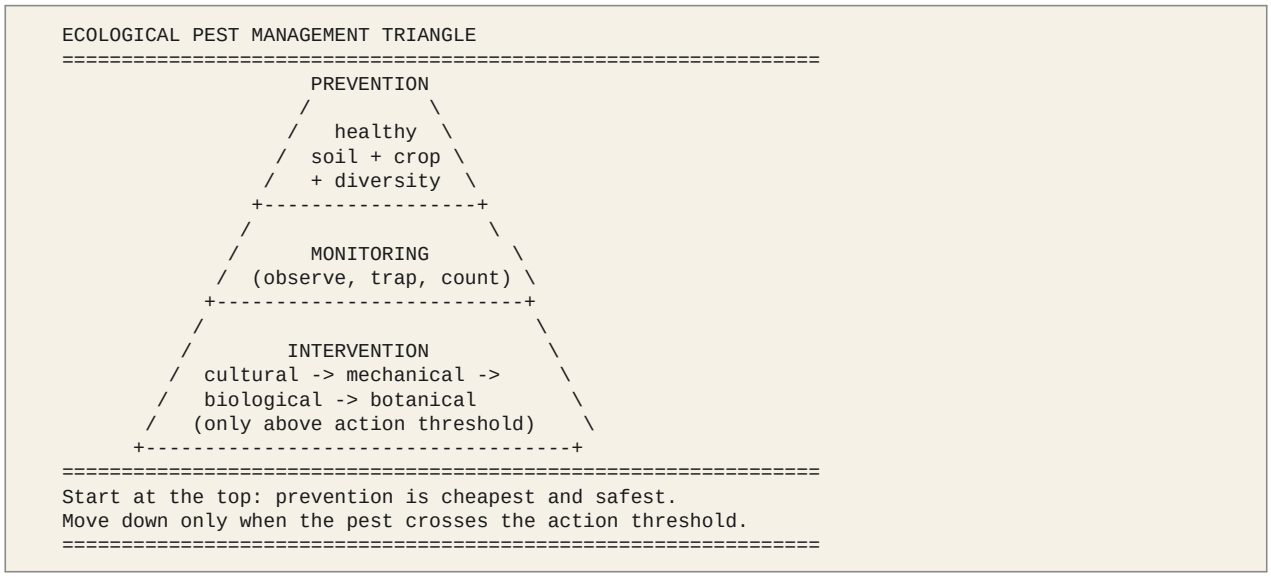


Fig. 14.1 The ecological pest management triangle - prevention first, intervention last.

WORKED EXAMPLE - Managing fruit borer in tomato by ecological means

- Prevention: grow tomato with a border of marigold (trap for nematodes and nectar for enemies), rotate away from the previous solanaceous crop, and keep the soil covered with mulch.
- Monitoring: set pheromone traps and yellow sticky traps; count moths and adults twice a week.
- Cultural: remove and destroy infested fruits and fallen fruit; keep the field clean.
- Biological: conserve ladybird beetles and spiders; spray Bacillus thuringiensis only against early-instar larvae.
- Botanical: if the count crosses the action threshold, spray neem kernel extract at the recommended dilution, in the evening to protect bees.
- Result: no synthetic pesticide is used, and the pest is kept below the economic injury level.

EXAM FOCUS

Key points to remember

- In natural farming a pest is a symptom of imbalance, so prevention comes first and spraying is a last resort.
- Principles: prevention, tolerance, action threshold, conserving natural enemies, diversity and treating the cause.
- Method categories: cultural, mechanical and physical, biological, botanical, habitat design and biological disease control.
- Botanical plants: neem, nochi, erukku, adathoda, garlic, chilli, tobacco, tulsi and pongam.
- Disease control rests on healthy soil, antagonists such as Trichoderma and Pseudomonas, resistant varieties and cultural practice.
- Weeds are managed by mulching, live mulch, competition, rotation, timely hand weeding and smother crops.

Mounish Raj · MouRa · Agri AI

- The pest management triangle: prevention, then monitoring, then intervention only above threshold.

Short-answer questions with model answers

Q1. State the four principles of ecological pest management.

Model answer: Prevention through a healthy, diverse crop in a living soil; tolerance of harmless pest levels; action only when the pest crosses the action threshold; and conservation of natural enemies by avoiding broad-spectrum poisons and providing habitat. Diversity and correction of the underlying cause are part of the same approach.

Q2. Name any four plants used for botanical pest control and one action of each.

Model answer: Neem (azadirachtin - antifeedant and growth regulator), nochi (repellent and insecticidal against caterpillars), garlic (sulphur compounds - repellent and fungicidal) and pongam (karanjin - insecticidal). Eruku, adathoda, chilli and tobacco are further examples.

Q3. How are weeds managed in natural farming without herbicides?

Model answer: By mulching to block light, using live mulch and cover crops to smother weeds, growing competitive crops and close spacing to shade them out, rotating crops to change the weed flora, removing weeds young by hand, using stale seedbed and solarisation, growing smother crops such as sunhemp, and in some systems using ducks or poultry to graze young weeds.

Long-answer questions with model answer outlines

Q1. Explain ecological pest management in natural farming with a suitable diagram.

Outline:

- Attitude: pest as a symptom of imbalance; prevention first.
- Principles of ecological pest management.
- Method categories with examples (cultural, mechanical, biological, botanical, habitat).
- The pest management triangle diagram.
- Botanical formulations and their use.
- Conclusion: a healthy ecosystem is the best pesticide.

Q2. Describe botanical formulations, disease management and weed management in natural farming.

Outline:

- Botanical plants, active principles and uses (table).
- Combination spray example and precautions for bees.
- Disease management: healthy soil, antagonists, resistant varieties, cultural practice, seed treatment.
- Weed management: mulching, live mulch, competition, rotation, hand weeding, smother crops.
- Fukuoka's approach to weeds.
- Conclusion: all three rely on ecology rather than chemicals.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 In natural farming a pest is treated as a symptom of ____.

2 Control is taken only when the pest crosses the action ____.

3 Azadirachtin is the active compound in ____.

4 Trichoderma is used for biological control of ____.

5 A thick ____ layer suppresses weeds by blocking light.

6 Broad-spectrum poisons kill natural ____.

B. State whether true or false

1 Natural farming sprays pesticide on a fixed calendar. (False - only above the action threshold)

2 Botanical sprays are completely harmless to bees. (False - they can harm bees, so spray in the evening)

3 Neem acts as an antifeedant and growth regulator. (True)

4 Diversity helps reduce pest build-up. (True)

C. Match the following

Column A	Column B
Neem	Azadirachtin
Nochi	Repellent and insecticidal
Trichoderma	Disease control
Bacillus thuringiensis	Bacterial control of larvae
Marigold	Trap crop for nematodes

D. Answer in one word or one line

1 Name three principles of ecological pest management.

2 Name four botanical pest-control plants.

3 How is disease controlled in natural farming?

4 Name two non-chemical methods of weed control.

Practical application in Tamil Nadu - pest management
- Neem-based sprays and panchagavya are widely used by organic farmers.
- Pheromone and sticky traps are used for monitoring in vegetable and fruit crops.
- Border rows of marigold and castor are common in cotton and vegetable fields.

Lecture 15. Mechanization and Farm Operations in Natural Farming

Learning objectives

- Explain the principles that guide mechanisation in natural farming.
- Describe suitable tools, machines and low-cost technologies.
- Show how farm operations are carried out with minimum soil and ecosystem disturbance.

15.1 The place of mechanisation in natural farming

Mechanisation means using tools and machines to carry out farm operations. In natural farming, mechanisation is welcome when it reduces drudgery and cost without harming the soil or the ecosystem. It is avoided when it disturbs the soil, compacts it or destroys soil life. The guiding rule is: use the machine that does the job with the least disturbance, and prefer the smallest machine that will do it.

15.2 Principles of mechanisation in natural farming

- **Minimum soil disturbance:** avoid the mouldboard plough and heavy discing; prefer no-till seed drills, dibblers and surface tools.
- **Minimum compaction:** use light machines, low tyre pressure and controlled traffic, and avoid working wet soil.
- **Low cost:** prefer small, locally made and repairable equipment over large, expensive machines.
- **Appropriate scale:** match the machine to the size of the holding and the availability of labour.
- **Energy efficiency:** prefer human, animal, solar and small-engine power where suitable.
- **Ecosystem protection:** avoid machines and practices that kill soil life, harm beneficial insects or damage habitat.
- **Timeliness:** mechanisation should allow operations at the right time, which is often more important than speed.

15.3 Tools and implements by operation

Operation	Suitable tools and machines	Note on natural farming
Sowing without tillage	No-till seed drill, zero-till drill, dibbler, jab planter, seed balls	Places seed in the soil with minimal disturbance; seed can be sown into mulch.
Nursery and transplanting	Tray nursery, mat nursery, manual and semi-automatic transplanters	Reduces labour and root damage.
Weeding	Hand hoe, wheel hoe, cono-weeder, brush cutter (surface only)	Shallow surface weeding only; never deep cultivation.
Mulch management	Straw spreader, hand tools, bale chopper	Spreads mulch evenly without disturbing soil.
Spraying botanicals	Knapsack sprayer, battery sprayer, boom sprayer	Use low-drift nozzles; spray in the evening to protect bees.
Irrigation	Drip and micro-sprinkler systems, solar pump, hose reel	Water-efficient and root-zone targeted.
Harvesting	Sickles, reapers, small combine for large farms	Keep stubble as mulch; avoid burning.
Threshing and winnowing	Pedal thresher, small power thresher, winnower	Retains straw for mulch and fodder.
Transport and haulage	Bullock cart, small tractor trailer, cycle cart	Low-cost, low-compaction options.
Post-harvest processing	Cleaner, grader, solar dryer, mill, oil expeller	Adds value on the farm.

15.4 Low-cost and appropriate technologies

- **Seed ball making:** a simple hand or drum method to coat seed in clay and compost for broadcasting.
- **Dibbler and jab planter:** hand tools that place seed and fertiliser without ploughing.
- **Cono-weeder:** a light, hand-pulled weeder that works on the surface between rows.
- **Battery knapsack sprayer:** cheaper and cleaner than engine sprayers for small farms.
- **Solar pump:** reduces diesel and electricity cost for irrigation.
- **Bullock-drawn implements:** low-cost, low-compaction tools for small and medium farms.
- **Cycle-based pumps and carts:** low-cost human-powered equipment.
- **Farm-made compost and Jivamrit units:** drums, pits and simple stirrers for input preparation.
- **Solar dryer and improved storage:** reduce post-harvest losses without fuel.

15.5 Farm operations calendar

Operations in a natural farm follow the crop calendar but with the natural farming approach at each step.

Stage	Natural farming operation
Land preparation	No ploughing. Clear only what is necessary; retain mulch and residue; repair bunds and water structures.
Seed selection and treatment	Select own or indigenous seed; treat with Bijamrit.
Sowing	Sow with a no-till drill or dibbler; sow into mulch; follow a mixed or intercropped plan.
Input application	Apply Jivamrit to the soil; spray botanical or growth preparations at critical stages.
Mulching	Maintain a continuous mulch layer; add residue after each harvest.

Stage	Natural farming operation
Weeding	Shallow surface weeding only, before weeds set seed.
Irrigation	Irrigate on the basis of soil moisture and crop stage; use drip or micro-sprinkler.
Pest and disease management	Monitor; act only above threshold; use ecological methods.
Harvesting	Harvest at the right maturity; keep residues for mulch and fodder.
Post-harvest	Clean, grade, process and store; add value where possible.

15.6 Precautions

- Never use a mouldboard plough or heavy disc harrow, which invert and pulverise the soil.
- Do not work or drive on wet soil, which causes severe compaction.
- Keep machinery weight low and traffic in fixed paths to protect the rest of the field.
- Do not use herbicides; weed by surface tools, mulch and competition.
- Spray botanicals in the evening to avoid harming bees and other pollinators.
- Service and calibrate sprayers to avoid over-application.

WORKED EXAMPLE - Cost of tools for a one-hectare natural farm

- Basic hand tools (spade, hoe, sickle, dibbler, cono-weeder): low one-time cost, long life.
- Battery knapsack sprayer: moderate one-time cost, low running cost.
- Drip system for part of the holding: higher one-time cost, but large water and labour savings; often supported by subsidy.
- Solar pump where the water source allows: higher one-time cost, but near-zero running cost and no diesel.
- Jivamrit drums and compost pit: very low cost, built from local materials.
- Rule: the total machinery investment should be recovered within a few seasons from reduced input cost and labour saving.
- Note: exact prices change with the market and with government subsidy schemes; obtain current rates locally.

EXAM FOCUS

Key points to remember

- Mechanisation is welcome in natural farming when it reduces drudgery and cost without disturbing the soil or ecosystem.
- Principles: minimum soil disturbance, minimum compaction, low cost, appropriate scale, energy efficiency, ecosystem protection and timeliness.
- Avoid the mouldboard plough and heavy discing; prefer no-till drills, dibblers and surface weeders.
- Low-cost technologies: seed balls, dibbler, cono-weeder, battery sprayer, solar pump, bullock implements.
- Farm operations follow the crop calendar but with no tillage, permanent mulch and ecological pest management.
- Do not work wet soil; keep machinery light and traffic in fixed paths.
- Spray botanicals in the evening to protect pollinators.

Short-answer questions with model answers

Q1. Why is the mouldboard plough avoided in natural farming?

Model answer: The mouldboard plough inverts and pulverises the soil, destroys soil structure and aggregates, breaks root channels and fungal networks, oxidises organic matter and buries residue that should stay on the

surface as mulch. This harms soil life and increases erosion, so it is avoided. No-till drills, dibblers and surface tools are used instead.

Q2. Name any four low-cost tools or technologies suitable for natural farming.

Model answer: Seed-ball making, the dibbler or jab planter, the cono-weeder and the battery knapsack sprayer. The solar pump, bullock-drawn implements and farm-made Jivamrit units are further examples.

Q3. State two precautions to be observed while mechanising a natural farm.

Model answer: Never work or drive on wet soil, because it causes severe compaction; and keep machinery light with traffic confined to fixed paths, so that the rest of the field and its soil life are protected.

Long-answer questions with model answer outlines

Q1. Explain the principles of mechanisation in natural farming and describe suitable tools for the main farm operations.

Outline:

- Definition and place of mechanisation.
- Principles: minimum disturbance, minimum compaction, low cost, appropriate scale, energy efficiency, ecosystem protection, timeliness.
- Tools by operation (table): sowing, weeding, spraying, irrigation, harvesting, threshing, transport, processing.
- Low-cost and appropriate technologies.
- Precautions.
- Conclusion: mechanisation should serve the soil, not damage it.

Q2. Describe the farm operations calendar of a natural farm and the machinery suited to each stage.

Outline:

- Land preparation without tillage; bund and water structure repair.
- Seed selection, Bijamrit treatment and no-till sowing.
- Input application and mulching.
- Surface weeding and irrigation.
- Ecological pest management and monitoring.
- Harvesting, threshing, post-harvest and value addition.
- Conclusion: each operation is matched to a low-disturbance tool.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 The ____ plough is avoided in natural farming.

2 A ____ planter places seed without ploughing.

3 A ____ weeder is a light, hand-pulled weeder.

4 A ____ pump reduces irrigation cost using solar energy.

5 Machinery should not be driven on ____ soil, to avoid compaction.

6 Botanical sprays should be applied in the ____ to protect bees.

B. State whether true or false

1 Heavy machinery is ideal for natural farming. (False - light, low-disturbance machinery is preferred)

2 No-till drills place seed with minimal soil disturbance. (True)

3 Working wet soil causes compaction. (True)

4 Herbicides are part of natural farming. (False - they are not used)

C. Match the following

Column A	Column B
No-till drill	Sowing without ploughing
Cono-weeder	Surface weeding
Dibbler	Hand tool for placing seed
Solar pump	Energy-efficient irrigation
Battery sprayer	Low-cost spraying

D. Answer in one word or one line

1 Why is the mouldboard plough avoided?

2 Name three low-cost tools for natural farming.

3 State two precautions in mechanisation.

4 How is weeding done without herbicides?

Practical application in Tamil Nadu - mechanisation

- Custom hiring centres provide small machinery and implements to farmers on hire.
- Solar pumps and drip systems are supported under state and central irrigation schemes.
- Cono-weeders and battery sprayers are widely used on smallholdings.

Lecture 16. Processing, Labelling and Economic Viability of Natural Farming

Learning objectives

- Describe post-harvest handling and value addition for natural farming produce.
- Explain packaging and labelling requirements and their market value.
- Assess the economic viability of natural farming with worked examples.

16.1 Why processing and value addition matter

A natural farmer who sells only raw grain at the farm gate captures a small part of the final price the consumer pays. By cleaning, grading, processing, packaging and labelling, the farmer moves up the value chain and keeps a larger share. Value addition is therefore central to the economic viability of natural farming.

16.2 Post-harvest handling

- **Timely harvest:** harvest at correct maturity to get the best quality and storability.
- **Cleaning:** remove stones, chaff, damaged grain and foreign matter.
- **Grading:** separate produce by size, colour and quality to get better prices.
- **Drying:** dry to safe moisture to prevent mould and insect damage; solar dryers are ideal.
- **Storage:** store in clean, dry, well-ventilated conditions; use neem leaves, ash or airtight containers to control insects without chemicals.
- **Reducing losses:** good handling can cut post-harvest losses substantially, which is a direct gain to the farmer.

16.3 Value addition

Crop / produce	Value-addition options
Paddy	Traditional rice (hand-pounded, unpolished), rice flour, puffed rice, rice bran oil.
Millets	Millet flour, ragi malt, millet cookies, noodles, ready mixes.
Pulses	Dal, papad, flour, sprouted and roasted snacks.
Oilseeds	Cold-pressed oil, oil cake as feed and manure.
Fruits (mango, banana)	Pulp, jam, pickle, chips, dried fruit.
Vegetables	Dried vegetables, pickles, powders, sauces.
Coconut	Copra, virgin coconut oil, coconut milk, coir, shell charcoal.
Spices	Cleaned, graded and ground spice powders.
Milk	Curd, ghee, paneer, value-added dairy products.
Honey and beeswax	Filtered honey, beeswax products.

16.4 Packaging

- Packaging protects the produce, prevents contamination and carries the brand message.
- Use food-grade, moisture-proof and tamper-evident packaging.
- Choose eco-friendly packaging - cloth, paper, jute, glass or recyclable material - to match the natural image.

- Include the product name, net weight, ingredients, farmer or group name, contact, batch number, date of packing and best-before date.
- Attractive, honest labelling raises perceived value and supports a premium price.

16.5 Labelling

A label is the identity of a product. For natural and organic produce, the label also carries the certification mark and the traceability information that builds consumer trust.

Label element	Purpose
Product name and description	Tells the buyer what the product is.
Net quantity and weight	Mandatory legal information.
Ingredients and nutrition	For processed foods; builds trust.
Farmer / group / brand name	Creates identity and traceability.
Certification mark (e.g. PGS-India / Jaivik Bharat)	Assures the buyer of organic or natural status.
Batch number and date of packing	Enables traceability and recall if needed.
Best-before or expiry date	Quality and safety information.
Contact details and QR code	Links the buyer to the farm for trust and repeat sales.
FSSAI licence number (for processed foods)	Mandatory for food business operators.

16.6 Economic viability

Economic viability means the farm can earn enough to cover its costs and give the family a reasonable income over the long term. Natural farming improves viability mainly by reducing cost and by opening premium markets, even where yield is similar.

16.6.1 Sources of higher income

- Lower input cost (fertiliser and pesticide replaced by on-farm preparations).
- Price premium for chemical-free and certified produce.
- Value addition through processing and packaging.
- Direct marketing to consumers, reducing middlemen.
- Diversified income from livestock, honey, mushrooms and trees.
- Reduced post-harvest losses through better handling.

WORKED EXAMPLE - Cost comparison - natural farming vs conventional farming for one acre of paddy

- Conventional paddy (one acre, one season):

- Seed Rs. 600; fertiliser Rs. 6,000; pesticide Rs. 2,500; land preparation (tractor) Rs. 3,000; labour Rs. 10,000; irrigation Rs. 2,000; harvesting and threshing Rs. 4,000. Total input cost about Rs. 28,100.

- Natural farming paddy (one acre, one season):

- Seed Rs. 400 (own/indigenous); fertiliser Rs. 0 (Jivamrit from farm); pesticide Rs. 300 (botanical materials); land preparation Rs. 500 (no tillage, only bund repair); labour Rs. 11,500 (more labour for mulching and input preparation); irrigation Rs. 1,500; harvesting and threshing Rs. 4,000. Total input cost about Rs. 18,200.

- Difference: natural farming costs about Rs. 9,900 less per acre per season.

- If the natural produce also gets a price premium of 10-20 per cent, the net advantage widens further.

- Note: figures are illustrative for teaching; actual values vary with location, season and prices.

WORKED EXAMPLE - Partial and complete nutrient and financial budget for a natural farm (one acre)

- PARTIAL BUDGET - nutrients supplied from on-farm sources:
- Compost 2 tonnes: about 10-30 kg N, plus P and K.
- Jivamrit (2 applications): microbial boost plus a few kg of readily available N and K.
- Legume intercrop / green manure: equivalent of about 16-24 kg N per acre.
- Residue mulch returned: several kg each of N, P and K.
- PARTIAL BUDGET - financial:
- Income: grain (Rs. 40,000) + straw/fodder value (Rs. 3,000) + intercrop (Rs. 4,000) = about Rs. 47,000.
- Variable cost: seed Rs. 400 + labour Rs. 11,500 + botanical materials Rs. 300 + irrigation Rs. 1,500 = about Rs. 13,700 (excluding harvest, which is common to both systems).
- Gross margin: about Rs. 33,300.
- COMPLETE BUDGET - include fixed and family costs:
- Add land rent or opportunity cost, depreciation of tools, and imputed value of family labour.
- Net return after these: still positive, and higher than the conventional system because the input cost is lower.
- Note: a complete budget is essential for a correct comparison; a partial budget alone can overstate the advantage.



Fig. 16.1 The value chain from farm to certified market.

16.7 Collective marketing and farmer producer organisations

Individual small farmers cannot usually meet the volume, quality consistency or paperwork that buyers and exporters demand. Farmer Producer Organisations (FPOs) and self-help groups solve this by aggregating produce, arranging certification, processing and marketing collectively, and giving farmers bargaining power.

WORKED EXAMPLE - Economics of a chemical-free produce value chain

- Take one tonne of paddy valued at Rs. 22,000 as raw grain at the farm gate.
- If milled and sold as hand-pounded traditional rice at a premium, the same grain can return substantially more, after allowing for milling cost and losses.
- If processed further into rice flour or puffed rice and packed under a brand, the return rises again, though with additional cost and effort.
- The key point: value addition and direct or certified marketing capture a larger share of the consumer's rupee than selling raw grain.
- Note: the exact multiples depend on the product, the market and the cost of processing; the principle is that each step of processing and each step closer to the consumer increases the farmer's share.

EXAM FOCUS

Key points to remember

- Value addition lets the farmer capture a larger share of the consumer price.
- Post-harvest steps: timely harvest, cleaning, grading, drying, safe storage and loss reduction.
- Value-addition options exist for paddy, millets, pulses, oilseeds, fruits, vegetables, coconut, spices and milk.
- Packaging protects and identifies the produce; eco-friendly packaging matches the natural image.
- Labels carry product details, weight, farmer or brand name, certification mark, batch number and dates.
- Economic viability improves through lower cost, price premium, value addition and direct marketing.
- FPOs and self-help groups help small farmers with volume, certification and collective marketing.

Short-answer questions with model answers

Q1. Why is value addition important for a natural farmer?

Model answer: Because selling raw produce at the farm gate captures only a small part of the final price. By cleaning, grading, processing, packaging and labelling, the farmer moves up the value chain and keeps a larger share of the consumer's rupee. Value addition also reduces dependence on middlemen and creates local employment.

Q2. List any six elements that must appear on the label of a packaged natural produce.

Model answer: Product name and description, net quantity or weight, ingredients or nutrition information, farmer or group or brand name, certification mark such as PGS-India, and batch number with date of packing. Best-before date, contact details and the FSSAI licence number for processed foods are further requirements.

Q3. How does natural farming improve the economic viability of a farm?

Model answer: Mainly by reducing the cost of cultivation, because fertiliser and pesticide are replaced by on-farm preparations; and by opening premium markets for chemical-free and certified produce. Value addition, direct marketing, diversified income from livestock and trees, and reduced post-harvest losses add to viability.

Long-answer questions with model answer outlines

Q1. Explain processing, packaging and labelling practices that enhance the market value of natural farming produce.

Outline:

- Importance of value addition and moving up the value chain.
- Post-harvest handling: harvest, cleaning, grading, drying, storage, loss reduction.
- Value-addition options for major crops (table).
- Packaging: protection, food-grade material, eco-friendly choice, required information.
- Labelling: elements and their purpose; certification mark and traceability.

- Marketing channels and the role of FPOs.
- Conclusion: processing and labelling convert a commodity into a premium product.

Q2. Assess the economic viability of natural farming with a cost comparison and a nutrient and financial budget for one crop.

Outline:

- Concept of economic viability.
- Cost comparison: natural vs conventional paddy, item by item.
- Nutrient budget from on-farm inputs (compost, Jivamrit, legumes, mulch).
- Partial budget: income and variable cost, gross margin.
- Complete budget: adding fixed cost, land and family labour.
- Value chain and premium; role of FPOs.
- Conclusion: lower cost and premium markets make natural farming viable.

Quick Revision - Objective Type Questions

A. Fill in the blanks

1 Cleaning, grading and ____ are important post-harvest operations.

2 A ____ dryer uses solar energy to dry produce.

3 Value addition moves the farmer up the ____ chain.

4 An ____ is a collective of farmers for marketing and certification.

5 The certification mark for Indian organic produce is ____ Bharat.

6 FSSAI licence is required for processed ____.

B. State whether true or false

1 Selling raw grain at the farm gate earns the farmer the largest share of the consumer price. (False - value addition earns more)

2 Good packaging protects the produce and carries the brand. (True)

3 Labelling includes the certification mark and batch number. (True)

4 Natural farming has no effect on post-harvest losses. (False - good handling reduces them)

C. Match the following

Column A	Column B
Value addition	Processing to raise market value
FPO	Collective marketing and certification
Solar dryer	Fuel-free drying
Label	Product identity and traceability
Premium price	Reward for chemical-free produce

D. Answer in one word or one line

1 Name four post-harvest operations.

2 Name three value-added products from millets.

3 Name four elements of a product label.

4 What is an FPO?

Practical application in Tamil Nadu - value chains

- Traditional rice varieties are hand-pounded and sold at a premium in the state.
- Millet-based value-added products are produced by women's self-help groups.
- Cold-pressed oil units add value to groundnut and coconut.

Lecture 17. Certification, Standards, Export Potential, Government Initiatives and Entrepreneurship

Learning objectives

- Explain certification systems for organic and natural produce, including PGS and third-party certification.
- Describe quality assurance mechanisms and export potential.
- List government initiatives, NGO programmes and farmer case studies.
- Identify entrepreneurship opportunities in natural and chemical-free agriculture.

17.1 Why certification is needed

A consumer cannot tell by looking whether produce was grown without chemicals. Certification is an independent assurance, backed by inspection and record-keeping, that the produce meets a defined standard. It builds consumer trust and gives the farmer access to premium markets and exports. For natural farming, certification is still developing, and several systems are used.

17.2 Types of certification

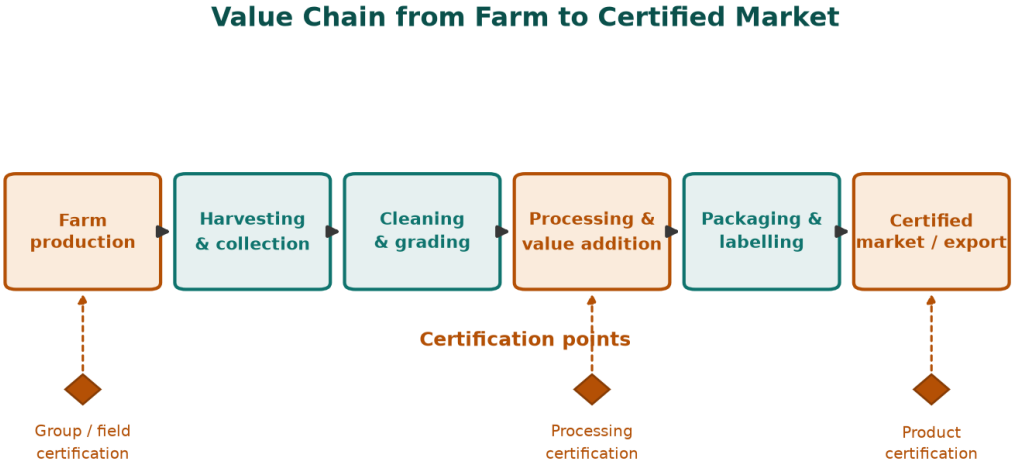
System	How it works	Suitability
Participatory Guarantee System (PGS-India)	Peer certification by a group of farmers, with a local group, records, peer inspection and a regional council. Low cost, farmer-driven.	Small and marginal farmers, local and domestic markets, direct sales.
Third-party certification (NPOP / NOP / EU)	Inspection and certification by an accredited certification body against a written standard.	Export markets and large buyers who demand accredited certification.
Self-declaration / trust-based	Farmer declares chemical-free practice; trust built through direct relationship with consumers.	Direct marketing, community-supported agriculture, farmers' markets.
Group / cluster certification	A group of farmers certified together, sharing the cost of inspection and records.	FPOs and clusters seeking scale.

17.3 PGS vs third-party certification

Basis	PGS-India (participatory)	Third-party certification
Who certifies	The farmers themselves, through peer review	An independent accredited certification body
Cost	Low; shared among the group	High; per-farm and per-inspection fees
Approach	Trust, transparency and local participation	Verification against a formal written standard
Inspection	Peer inspection by neighbouring farmers	External inspector
Documentation	Simple group records and a farm diary	Detailed records and internal control system
Market	Domestic, local and direct markets	Export and premium institutional buyers
Suitability	Small and marginal farmers, groups	Large farms, exporters, contract buyers
Certification mark	PGS-India / Jaivik Bharat mark	NPOP (India Organic) or the buyer-country mark (NOP, EU)
Strength	Affordable, inclusive, builds community	Widely recognised, opens export markets
Limitation	Less recognised internationally	Expensive and paperwork-heavy for small farms

17.4 Quality assurance and traceability

- **Farm records:** a farm diary of inputs, operations and yields is the base of any certification.
- **Internal control system:** for groups, an internal system to check member compliance before external inspection.



Natural produce certified at field, processing and product stages; traceability maintained throughout

Figure 17.1 Value chain from farm to certified market

Stages from farm production to certified market, with certification points marked.

- **Traceability:** batch numbers and records that link a packet back to the field and the farmer.
- **Residue testing:** laboratory testing for pesticide residues where required by buyers.
- **Conversion period:** a defined period (commonly two to three years for perennial crops and shorter for annuals under organic standards) during which the land is managed organically before certification.
- **Certification mark:** the logo or mark that assures the buyer, e.g. PGS-India or the India Organic mark.

Note: the exact length of the conversion period and the detailed standards differ between PGS and third-party systems and are revised from time to time; the general principle of a conversion period should be stated rather than a fixed number.

17.5 Export potential

World demand for organic and chemical-free food is growing, and India is a significant exporter of organic produce. Natural farming produce, once certified, can enter this market.

Aspect	Position
Main export products	Organic spices, tea, coffee, basmati rice, pulses, oilseeds, fruits, nuts, honey and processed foods.
Key markets	European Union, United States, United Kingdom, Canada, Japan, Australia and the Gulf countries.
Requirements	Accredited certification (NPOP and the buyer-country standard), traceability, residue-free testing and consistent quality.
Advantages for India	Diverse agro-climatic zones, traditional knowledge, low-input smallholder base and a large domestic market.
Challenges	High certification cost, small and scattered holdings, logistics, quality consistency and documentation.
Way forward	Cluster and group certification, FPOs, contract farming with exporters and government support for certification.

17.6 Government initiatives

Both the central government and the states support organic and natural farming through programmes, missions and certification support. The names below are the standard national programmes; scheme details, subsidy amounts and coverage change with each budget and should be confirmed from current official documents.

Programme / initiative	Purpose (general description)
Paramparagat Krishi Vikas Yojana (PKVY)	Promotes organic and traditional farming through cluster-based support for inputs, certification and marketing.
PGS-India	Provides a low-cost, farmer-participatory organic certification system and the Jaivik Bharat mark.
Mission Organic Value Chain Development for North Eastern Region (MOVCDNER)	Develops the organic value chain in the north-eastern states.
National Mission on Natural Farming / natural farming initiatives	Promotes natural farming through demonstration, training and support to farmers.
National Project on Organic Farming (NPOF)	Supports organic inputs, quality control and certification infrastructure.
Rashtriya Krishi Vikas Yojana (RKVY)	State-level support for organic and natural farming projects.
Andhra Pradesh Community Managed Natural Farming (APCNF)	A large state programme that promotes natural farming through community institutions; a widely cited model.

Programme / initiative	Purpose (general description)
Tamil Nadu organic and natural farming initiatives	State agriculture department support for organic and natural farming, certification support through the Tamil Nadu Organic Certification Department, and farmer outreach through the Uzhavan platform.
State seed and input support schemes	Support for indigenous seed, bio-inputs and organic input production units.

Note on the Uzhavan platform: it is the Tamil Nadu government's agriculture information and service platform, used for farmer registration, scheme information, subsidy applications and advisory services. The specific schemes offered on it change over time.

17.7 NGO and community initiatives

- Non-governmental organisations that train farmers in ZBNF, organic and natural farming methods.
- Community seed banks and seed savers' groups that conserve indigenous varieties.
- Farmer producer organisations that aggregate produce and arrange certification and marketing.
- Self-help groups, especially of women, involved in input production, processing and marketing.
- Participatory guarantee groups formed under PGS-India for peer certification.
- Farmer field schools and demonstration plots that spread knowledge farmer to farmer.

17.8 Case studies

The case studies below are presented in general terms, based on widely reported experience. They are meant to show the pattern of success, not to state precise figures, which vary by farm and year.

Case	What was done	Reported outcome (general)
Andhra Pradesh community-managed natural farming	Village-level community institutions supported large numbers of farmers to adopt natural farming methods.	Reduced input cost, improved soil and reported increases in net income and resilience in many villages.
Subhash Palekar's ZBNF movement	Farmer-to-farmer training across states in the four pillars.	Widespread adoption, particularly among smallholders and in rain-fed areas.
Tamil Nadu traditional paddy farmers	Cultivation of indigenous rice varieties under natural methods.	Premium prices in niche markets and conservation of land races.
Organic spice and plantation growers	Certification through PGS or third-party systems and direct or export marketing.	Access to premium markets and higher realisation per unit.
Women's self-help group enterprises	Collective production of value-added products such as millet mixes and pickles.	Additional household income and greater participation of women.

17.9 Entrepreneurship opportunities

Natural and chemical-free agriculture creates business opportunities beyond the farm gate. These are the avenues a graduate can pursue or advise farmers on.

Opportunity	Description
Natural input production	Producing and selling Jivamrit, Bijamrit, vermicompost, panchagavya and bio-inputs.
Desi cow based enterprises	Dairy, dung and urine products, and a desi-cow-centred input unit.
Organic and natural produce trading	Aggregating, grading and selling chemical-free produce.
Value-added food processing	Millet mixes, flours, oils, pickles, dried products and ready-to-cook foods.
Direct marketing and e-commerce	Farm-to-consumer sales, subscription boxes and online platforms.
Certification and advisory services	Helping farmers prepare records and obtain PGS or third-party certification.

Opportunity	Description
Nursery and seed enterprise	Producing and selling indigenous seed and planting material.
Agri-tourism and farm experience	Hosting visitors for natural farming experience and education.
Training and knowledge services	Conducting farmer training and demonstration.
Beekeeping and honey	Pollination services and honey production.
Mushroom and millet enterprises	Using crop residues and millets for high-value products.

17.10 Standards and quality control

- Standards define what may and may not be used and how records must be kept.
- For organic produce, the National Programme for Organic Production (NPOP) sets the Indian standard, and PGS-India provides the participatory route.
- The India Organic (NPOP) mark and the Jaivik Bharat mark identify certified produce.
- Quality control includes residue testing, input verification and inspection.
- For natural farming produce, standards are still developing; many farmers rely on PGS, group certification or direct consumer trust.
- FSSAI regulations apply to processed and packaged foods.

WORKED EXAMPLE - Economics of a certified chemical-free produce value chain

- Raw produce sold at the farm gate earns the base price.

- Certified produce, sold to an organic store or buyer, earns a premium over the base price because certification assures quality.

- Value-added, branded and packaged produce, sold directly or online, earns a further premium, after allowing for processing and packaging cost.

- For a farmer group, the cost of PGS certification is shared and low, so the premium is largely retained.

- For an exporter, third-party certification is costly but opens higher-value markets, so the net gain depends on volume.

- Conclusion: certification and value addition together turn a low-margin commodity into a higher-margin branded product.

EXAM FOCUS

Key points to remember

- Certification is independent assurance that produce meets a defined standard; it builds trust and opens premium and export markets.
- PGS-India is participatory, low-cost and peer-certified; third-party certification is external, costly and required for export.
- Quality assurance rests on farm records, an internal control system, traceability, residue testing and a conversion period.
- India exports organic spices, tea, coffee, rice, pulses, oilseeds, fruits and honey, mainly to the EU, USA, UK, Canada, Japan and the Gulf.
- Government programmes include PKVY, PGS-India, MOVCDNER, the National Mission on Natural Farming, NPOF and RKVY, plus state programmes such as APCNF and Tamil Nadu initiatives.
- Entrepreneurship opportunities include input production, value-added processing, direct marketing, certification services and agri-tourism.
- Standards for organic produce are set by NPOP and PGS-India; natural farming standards are still developing.

Short-answer questions with model answers

Q1. Distinguish PGS-India from third-party certification.

Model answer: PGS-India is a participatory system in which farmers themselves, through peer inspection and group records, certify their produce; it is low-cost and suited to small farmers and domestic markets. Third-party certification is carried out by an independent accredited body against a written standard; it is costly and paperwork-heavy but is widely recognised and required for export markets.

Q2. Name any five government programmes that support organic or natural farming.

Model answer: Paramparagat Krishi Vikas Yojana (PKVY), PGS-India, Mission Organic Value Chain Development for North Eastern Region (MOVCDNER), National Project on Organic Farming (NPOF) and Rashtriya Krishi Vikas Yojana (RKVY). Andhra Pradesh Community Managed Natural Farming and Tamil Nadu's organic and natural farming initiatives are state examples.

Q3. List any four entrepreneurship opportunities in natural farming.

Model answer: Natural input production (Jivamrit, vermicompost, bio-inputs); value-added food processing (millet mixes, flours, oils); direct marketing and e-commerce of chemical-free produce; and certification and advisory services. Nursery and seed enterprises, agri-tourism and beekeeping are further opportunities.

Long-answer questions with model answer outlines

Q1. Explain the certification systems for natural and organic produce and compare PGS with third-party certification.

Outline:

- Why certification is needed.
- Types: PGS-India, third-party, self-declaration, group certification.
- Comparison table on certifier, cost, inspection, documentation, market, suitability, mark.
- Quality assurance: records, internal control, traceability, residue testing, conversion period.
- Conclusion: choice of system depends on market and scale.

Q2. Discuss the government initiatives, case studies and entrepreneurship opportunities in natural farming.

Outline:

- Government initiatives: PKVY, PGS-India, MOVCDNER, National Mission on Natural Farming, NPOF, RKVY.
- State programmes: APCNF and Tamil Nadu initiatives, Uzhavan platform.
- NGO and community initiatives: training, seed banks, FPOs, SHGs, participatory guarantee groups.
- Case studies: APCNF, ZBNF movement, Tamil Nadu traditional rice, spice growers, women's groups.
- Entrepreneurship: input production, processing, direct marketing, certification services, agri-tourism, beekeeping.
- Conclusion: policy, community action and enterprise together scale natural farming.

Q3. Describe the export potential of natural farming produce and the requirements for export.

Outline:

- Growing global demand for organic and chemical-free food.
- Main export products and key markets.
- Requirements: accredited certification, traceability, residue-free testing, quality consistency.
- Advantages for India: agro-climatic diversity, traditional knowledge, smallholder base.
- Challenges: certification cost, scattered holdings, logistics, documentation.
- Way forward: cluster certification, FPOs, contract farming, government support.

- Conclusion: certification is the key to the export market.

Appendix A. Practical Exercises and Field Assignments

These exercises are designed for the practical and field-work component of the course. Each gives an aim, the materials needed, the procedure and what to record. Students should maintain a practical record book.

Exercise 1. Preparation of Bijamrit and seed treatment

Aim: To prepare Bijamrit and treat seed before sowing.

Materials: Desi cow dung (5 kg), desi cow urine (5 L), lime (50-100 g), a handful of farm soil, jaggery (optional), seed, water, a mixing vessel.

Procedure: Mix the dung and urine in water to a thin slurry; add lime, farm soil and jaggery; soak the seed briefly; dry in shade and sow the same day.

Record: Record the quantity of seed treated, the appearance of the slurry and the germination percentage compared with untreated seed.

Exercise 2. Preparation of Jivamrit

Aim: To prepare Jivamrit and observe fermentation.

Materials: Desi cow dung (10 kg), desi cow urine (10 L), jaggery (2 kg), pulse flour (2 kg), a handful of farm soil, 200 L water, a drum and a stirrer.

Procedure: Dissolve jaggery in water; add dung, urine, pulse flour and soil; stir twice daily; ferment in shade for two to three days.

Record: Record the smell and appearance each day and the volume prepared. Note the changes that show the culture is ready.

Exercise 3. Preparation of a botanical pest-repellent spray

Aim: To prepare a neem-garlic-chilli spray and test it on a crop.

Materials: Neem kernel or leaves, garlic, chilli, soap, water, a knapsack sprayer.

Procedure: Grind and soak the plant material overnight; strain; add a little soap solution; dilute and spray in the evening.

Record: Record the pest observed, the dilution used and the change in pest numbers after a few days.

Exercise 4. Composting by the Indore method

Aim: To prepare compost from farm waste.

Materials: Crop residues, cattle dung, urine, water, soil, a pit or heap area.

Procedure: Layer residues, dung and soil, moisten, and turn periodically over several weeks until the material is dark and crumbly.

Record: Record the weight of materials used, the temperature and the time taken for maturity.

Exercise 5. Vermicomposting

Aim: To prepare vermicompost from organic waste.

Materials: Earthworms, cattle dung, crop waste, a vermibed or tank, water.

Procedure: Half-fill the bed with dung and waste; introduce earthworms; keep moist and shaded; harvest the compost after it matures and separate the worms.

Record: Record the input weight, the compost weight and the time taken.

Exercise 6. Mulching and soil moisture observation

Aim: To study the effect of mulching on soil moisture and temperature.

Materials: Straw or crop residue, a small plot, a soil thermometer or moisture meter.

Procedure: Mulch one plot and leave a control plot bare; measure soil moisture and temperature at intervals.

Record: Record the difference in moisture and temperature between mulched and unmulched plots.

Exercise 7. Identification of natural enemies and pests

Aim: To identify common pests and their natural enemies in the field.

Materials: A hand lens, a sweep net, a notebook, the field.

Procedure: Collect and observe insects; classify each as pest or natural enemy; note the crop and the stage.

Record: Record the names and numbers observed and the predator-prey relationship.

Exercise 8. Setting up traps and monitoring

Aim: To monitor pest populations using traps.

Materials: Yellow and blue sticky traps, pheromone traps, pegs.

Procedure: Place traps at the recommended height and spacing; count the insects caught twice a week.

Record: Record the counts and decide whether the action threshold has been crossed.

Exercise 9. Seed collection and treatment of indigenous varieties

Aim: To collect, clean and store indigenous seed.

Materials: Indigenous seed, neem leaves or ash, storage containers.

Procedure: Collect seed from healthy plants; clean and dry it; treat it with Bijamrit or cow urine; store with neem leaves or ash.

Record: Record the variety name, the source and the germination percentage after storage.

Exercise 10. Preparing a farm layout plan

Aim: To design a natural farm layout for a given holding.

Materials: A map of the holding, drawing materials.

Procedure: Mark the main field, home garden, cattle shed, fodder plot, farm pond, input unit and boundary trees.

Record: Record the area allotted to each component and the reason for the arrangement.

Exercise 11. Cost of cultivation comparison

Aim: To compare the cost of cultivation of a crop under natural and conventional methods.

Materials: Records of inputs, labour and yields; current market prices.

Procedure: List every input and operation for each method; total the cost; compute the net return.

Record: Record the difference in cost and net return per acre.

Exercise 12. Soil sampling and organic carbon estimation

Aim: To collect a representative soil sample and estimate organic carbon.

Materials: An auger or spade, a sample bag, a laboratory or field kit.

Procedure: Collect samples from several spots, mix, quarter and label; test for organic carbon and other parameters.

Record: Record the soil test values and interpret them for natural farming.

Exercise 13. Water harvesting structure visit

Aim: To study a farm pond, percolation pond or recharge structure.

Materials: A notebook, measuring tape.

Procedure: Visit the structure; measure its size and capacity; note how runoff is directed into it.

Record: Record the capacity, the catchment and the use of the stored water.

Exercise 14. Value addition and labelling exercise

Aim: To process a farm product and design a label.

Materials: A farm product, processing equipment, packaging material.

Procedure: Clean, grade and process the product; pack it; design a label with all required information.

Record: Record the cost of processing, the packaging cost and the expected selling price.

Exercise 15. Survey of a certified or natural farm

Aim: To survey a natural or certified farm and document its practices.

Materials: A questionnaire, a notebook.

Procedure: Visit the farm; record the practices, inputs, crops, certification status, cost and income.

Record: Prepare a short report on the farm's practices and its economics.

Appendix B. Model End-Semester Question Paper

This model paper follows the usual pattern of a final theory examination. Marks and duration are indicative.

Model end-semester question paper (indicative)

- Time: 2.5 hours. Maximum marks: 100. Answer all questions in Part A, any four in Part B and any two in Part C.

- PART A - Objective and very short answer (20 x 1 = 20 marks). Covering all units.

- PART B - Short answer (4 x 5 = 20 marks):

- 1. Differentiate natural farming from organic farming. 2. Explain the composition and function of Jivamrit. 3. Write short notes on Korean Natural Farming. 4. Describe the integrated crop-tree-animal nutrient cycle. 5. Explain ecological pest management. 6. Write short notes on PGS-India certification.

- PART C - Long answer (2 x 15 = 30 marks):

- 7. Describe the history and evolution of natural farming from ancient Indian agriculture to the present. 8. Explain the four pillars of natural farming with their composition and function. 9. Discuss the design of a natural farm with a suitable layout and its nutrient cycle. 10. Explain water conservation, renewable energy and nutrient management in natural farming.

- PART D - Application and case-based (2 x 15 = 30 marks):

- 11. Prepare a cost comparison and a nutrient budget for one acre of a crop under natural and conventional farming. 12. Discuss the certification systems, government initiatives and entrepreneurship opportunities in natural farming. 13. Design a plan to convert a two-hectare chemically farmed holding into a certified natural farm.

Blueprint of the paper

Unit	Objective	Short	Long	Application
Heritage and history (1-2)	3	1	1	-
Concept, principles, pillars (3-5)	5	2	1	-
Scope, schools, design (6-8)	4	1	1	1
Ecology, integration, biodiversity (10-12)	4	1	1	1
Water, energy, pest, mechanisation (13-15)	2	1	-	-
Processing, certification, enterprise (16-17)	2	1	1	1

Appendix C. Master Revision Tables

These are the comparison tables that appear most often in examinations, gathered in one place for quick revision.

C.1 Natural farming vs organic farming vs conventional farming

Basis	Natural	Organic	Conventional
Chemical fertiliser and pesticide	Not used	Not used	Used heavily
Tillage	None or minimum	Permitted	Intensive
External organic inputs	Minimal	Permitted and often purchased	Not the main input
Soil cover	Always mulched	Encouraged	Often bare
Source of nutrition	Soil biology and on-farm recycling	Organic amendments	Synthetic NPK
Input cost	Very low	Moderate to high	High
Certification	Community or PGS where applicable	PGS or NPOP	Not applicable

C.2 The four pillars

Pillar	Composition	Function
Bijamrit	Desi cow dung, cow urine, lime, farm soil, water	Seed treatment; microbial coat; better germination and protection
Jivamrit	Desi cow dung, cow urine, jaggery, pulse flour, farm soil, water	Living microbial culture; decomposition; nutrient mobilisation
Mulching	Crop residues, straw, leaves or live cover	Moisture, temperature, weed control, soil life, organic matter
Waaphasa	Achieved by mulching and judicious irrigation	Balance of air and moisture in soil pores

C.3 Schools of natural farming

School	Founder / origin	Signature practice
Fukuoka natural farming	Masanobu Fukuoka, Japan	Seed balls, clover cover, no tillage, minimal intervention
ZBNF	Subhash Palekar, India	Four pillars; desi cow based; farmer training
Rishi Krishi	Ancient Indian texts	Amritpani, Amritjal, panchagavya, spiritual calendar
Korean Natural Farming	Cho Han-Kyu, South Korea	Indigenous Micro-Organisms, FPJ, FFJ, OHN, LAB

C.4 Certification systems

Basis	PGS-India	Third-party certification
Who certifies	Farmers, by peer review	Independent accredited body
Cost	Low, shared	High
Market	Domestic and direct	Export and premium institutional
Documentation	Simple group records	Detailed records and internal control
Mark	PGS-India / Jaivik Bharat	NPOP (India Organic) or buyer-country mark

C.5 Key government programmes

Programme	Purpose
PKVY	Cluster support for organic and traditional farming
PGS-India	Low-cost participatory organic certification
MOVCDNER	Organic value chain in the north-eastern states
National Mission on Natural Farming	Promotion of natural farming through demonstration and training
NPOF	Organic inputs, quality control and certification infrastructure
RKVY	State-level organic and natural farming projects
APCNF	State programme on community-managed natural farming

Appendix D. Abbreviations

Abbreviation	Full form
APCNF	Andhra Pradesh Community Managed Natural Farming
CA	Conservation Agriculture
FPO	Farmer Producer Organisation
FFJ	Fermented Fruit Juice
FPJ	Fermented Plant Juice
FSSAI	Food Safety and Standards Authority of India
IFS	Integrated Farming System
IMO	Indigenous Micro-Organisms
IPM	Integrated Pest Management
KNF	Korean Natural Farming
LAB	Lactic Acid Bacteria
MOVCDNER	Mission Organic Value Chain Development for North Eastern Region
NPOF	National Project on Organic Farming
NPOP	National Programme for Organic Production
PGS	Participatory Guarantee System
PKVY	Paramparagat Krishi Vikas Yojana
RKVY	Rashtriya Krishi Vikas Yojana
SDG	Sustainable Development Goal
SHG	Self-Help Group
SOC	Soil Organic Carbon
TNAU	Tamil Nadu Agricultural University
ZBNF	Zero Budget Natural Farming

Glossary of Key Terms

The terms below are the ones most likely to appear in the examination. Learn the definition of each in one or two lines.

Term	Meaning
Natural farming	A chemical-free, low-input farming system that works with nature, avoids soil disturbance and relies on soil biology for nutrition and on ecological methods for pest control.
Zero Budget Natural Farming (ZBNF)	A natural farming system developed by Subhash Palekar, based on four pillars and designed to bring the external cash cost of inputs close to zero.
Bijamrit	A seed treatment preparation of desi cow dung, cow urine, lime and farm soil, used to coat seed with beneficial microbes.
Jivamrit	A fermented microbial culture of desi cow dung, cow urine, jaggery, pulse flour and farm soil, applied to the soil.
Mulching	Covering the soil surface with crop residues, straw, leaves or a live cover crop to conserve moisture and suppress weeds.
Waaphasa / Whapasa	The balanced state of air and moisture in soil pores, ideal for roots and aerobic microorganisms.
Desi cow	An indigenous cow breed, valued in natural farming for the microbial richness of its dung and urine.

Term	Meaning
Panchagavya	A preparation of five cow products (dung, urine, milk, curd and ghee) used as a growth promoter.
Amritpani / Amritjal	Liquid preparations of cow dung, cow urine, jaggery and gram flour used in Rishi Krishi.
Green manure	A crop grown and then incorporated or mulched to add organic matter and nitrogen to the soil.
Cover crop	A crop grown mainly to cover and protect the soil and to provide live mulch.
Vermicompost	Nutrient-rich compost produced by earthworms from organic waste.
Farmyard manure	Decomposed mixture of cattle dung, urine, litter and feed waste used as organic manure.
Humus	The stable, dark organic fraction of soil formed by the decomposition of organic matter.
Soil organic carbon	The carbon held in soil organic matter; a key indicator of soil health.
Rhizobium	A bacterium that fixes atmospheric nitrogen in the root nodules of legumes.
Mycorrhiza	A beneficial association between a fungus and plant roots that improves nutrient and water uptake.
Azadirachtin	The main active compound in neem, acting as an antifeedant, repellent and growth regulator.
Trichoderma	A beneficial fungus used to control soil-borne plant pathogens.
Pseudomonas fluorescens	A beneficial bacterium used for biological control of disease and plant growth promotion.
Indigenous Micro-Organisms (IMO)	Locally collected and multiplied microorganisms, central to Korean Natural Farming.
Ecological engineering	Designing the farm habitat to support natural enemies and pollinators, so that pests are controlled naturally.
Trap crop	A crop grown to attract pests away from the main crop, which is then destroyed with the pest.
Banker plant	A plant that hosts alternative prey, allowing natural enemies to build up before the pest arrives.
Action threshold	The pest population level at which control measures become economically justified.
Integrated farming system	A system combining crops, livestock, trees and other enterprises so that the waste of one becomes the input of another.
Land race	A traditional crop variety developed and maintained by farmers over generations, adapted to local conditions.
Community seed bank	A farmer-managed store of indigenous seed for conservation and exchange.
Mulch	The material spread on the soil surface as a cover.
PGS-India	Participatory Guarantee System, a low-cost, peer-based organic certification system of India.
Third-party certification	Certification by an independent accredited body against a written organic standard, used mainly for export.
NPOP	National Programme for Organic Production, which sets the Indian organic standard and the India Organic mark.
Jaivik Bharat	The Indian organic certification mark.
PKVY	Paramparagat Krishi Vikas Yojana, a central programme promoting organic and traditional farming in clusters.
APCNF	Andhra Pradesh Community Managed Natural Farming, a large state programme promoting natural farming.
Farmer Producer Organisation (FPO)	A collective of farmers that aggregates produce, arranges certification and markets jointly.
Value chain	The sequence of steps from production to the final consumer, at each of which value is added.
Conversion period	The period during which land must be managed organically before produce can be certified.
Biodiversity	The variety of life at genetic, species and ecosystem levels.
Ecological balance	The stable relationship among the living and non-living parts of an ecosystem.

Term	Meaning
Renewable energy	Energy from sources that are naturally replenished, such as the sun, wind and biomass.
Waaphasa-based irrigation	Irrigation managed to maintain the soil's air-moisture balance rather than to saturate it.

Mounish Raj · MouRa · Agri AI

Reference List and Suggested Reading

The following standard texts, government documents and institutional materials are the sources for this course. Students should consult the current editions and the latest official documents, since programme details change over time.

Books on natural and organic farming

- Fukuoka, M. *The One-Straw Revolution: An Introduction to Natural Farming.* (English translation, New York Review Books).
- Fukuoka, M. *The Natural Way of Farming: The Theory and Practice of Green Philosophy.*
- Palekar, S. *The Philosophy of Spiritual Farming* (ZBNF series, Zero Budget Natural Farming).
- Palekar, S. *Zero Budget Natural Farming* - booklets on the four pillars and crop-wise practice.
- Howard, A. *An Agricultural Testament.*
- Howard, A. and Wad, Y.D. *The Waste Products of Agriculture: Their Utilization as Humus.*
- Balfour, E.B. *The Living Soil.*
- Lampkin, N. *Organic Farming.*
- Sharma, A.K. *A Handbook of Organic Farming.*
- Cho Han-Kyu. *Korean Natural Farming* - material on Indigenous Micro-Organisms and natural inputs.

Ancient and traditional Indian sources

- Krishi-Parashara (attributed to Parashara) - ancient Indian treatise on agriculture.
- Kautilya's Arthashastra - sections on agriculture, irrigation and land administration.
- Surapala's Vrikshayurveda - treatise on trees and plant health.
- Brihat Samhita and Agni Purana - sections on rainfall and agriculture.
- Sangam Tamil literature - references to dryland farming, tanks and cattle.

ICAR, TNAU and institutional e-course material

- ICAR e-course material on organic farming and natural farming.
- TNAU e-course and lecture material on organic farming, sustainable agriculture and natural farming.
- TNAU Agritech Portal - technology and package-of-practice pages on organic and natural farming.
- National Centre of Organic and Natural Farming (NCONF) - technical and training material.
- ICAR-Indian Institute of Farming Systems Research - integrated farming system material.

Government programmes and documents

- Paramparagat Krishi Vikas Yojana (PKVY) - operational guidelines.
- PGS-India - programme guidelines and certification manual.
- National Programme for Organic Production (NPOP) - standards document.

- Mission Organic Value Chain Development for North Eastern Region (MOVCDNER) - guidelines.
- National Project on Organic Farming (NPOF) - scheme documents.
- Andhra Pradesh Community Managed Natural Farming (APCNF) - programme reports.
- Tamil Nadu Department of Agriculture - organic and natural farming scheme orders and the Uzhavan platform.
- National Mission on Natural Farming - scheme documents and guidelines.

Suggested reading for examinations

- Revise the comparison tables of every lecture - they are the most examinable content.
- Learn the four pillars and the ZBNF preparations with their ingredients and quantities.
- Learn the schools of natural farming with their founders and signature practices.
- Learn the certification systems, government programmes and entrepreneurship opportunities.
- Practise the worked examples so that you can reproduce a cost or nutrient budget.
- Draw the text diagrams - pillars, principles wheel, farm layout, nutrient cycle, pest triangle, value chain - from memory.

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End Note

This study material follows the 2025 syllabus of AGR 202 - Principles and Practices of Natural Farming for the Third Semester of B.Sc. (Hons) Agriculture at Tamil Nadu Agricultural University. It covers all seventeen lectures of the course in the order given in the syllabus. Quantitative figures in the worked examples are illustrative teaching values; where a figure is uncertain, a standard range has been given rather than a false precision. Scheme names, subsidy amounts and certification details change over time and should be confirmed from current official documents.